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The energy-absorbing behaviour of fibre reinforced honeycombs for battery pack structures --Manuscript Draft--

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Abstract:	This paper presents an experimental and numerical investigation on the energy absorption characteristics of lightweight battery packs fabricated from short or continuous carbon fibre (SCF/CCF) reinforced honeycomb composites. The thermoplastic short and continuous carbon fibre reinforced composite structures were produced via 3D printing. The thermosetting continuous carbon fibre reinforced composite (CCF/Epoxy) structures were manufactured through vacuum bag moulding (VBM). These two types of structures were then compared to evaluate their energy absorption performances and failure mechanisms under axial loading conditions. Test results indicate that the honeycomb-structured battery pack made of 3D-printed continuous carbon fibre-reinforced thermoplastic (CCF/PLA) exhibits a higher compressive strength than CCF/Epoxy. For 1-cell configuration, CCF/PLA gives 36.53 MPa vs. 26.75 MPa for CCF/Epoxy; for 8-cell configuration, the values are 18.60 MPa and 15.95 MPa, respectively. Additionally, the specific energy absorption (SEA) values for CCF/PLA are 18.20 J/g (1-cell) and 24.10 J/g (8-cell). In contrast, thermosetting CCF/Epoxy composites demonstrate a superior SEA performance, with values of 31.22 J/g (1-cell) and 38.24 J/g (8-cell). Subsequently, a finite element (FE) model was developed to simulate the mechanical behaviour of the composite specimens subjected to axial loading. It shows a good agreement between numerical predictions and experimental results. Furthermore, the synergistic mechanical matching relationship between honeycomb structures with varying heights and battery cell configurations was predicted through numerical simulations. Quasi-static experimental and numerical simulation results provide a design guidance to improve the load bearing capacity.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlight

- Evaluate the energy absorption capacity of honeycomb composite structures for lightweight battery packs.
- CCF/Epoxy composites show the highest SEA (38.25 J/g), outperforming 3D-printed CCF/PLA (20.07 J/g) and SCF/PLA (18.48 J/g).
- Reveal the influence of structural height on the mechanical performance and protection efficiency of honeycomb battery packs.
- Provide mechanical-structural synergy for both high energy absorption and favorable energy density (up to 129.53 Wh/Kg for CCF/Epoxy).

The energy-absorbing behaviour of fibre reinforced honeycombs for battery pack structures

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Abstract

This paper presents an experimental and numerical investigation on the energy absorption characteristics of lightweight battery packs fabricated from short or continuous carbon fibre (SCF/CCF) reinforced honeycomb composites. The thermoplastic short and continuous carbon fibre reinforced composite structures were produced via 3D printing. The thermosetting continuous carbon fibre reinforced composite (CCF/Epoxy) structures were manufactured through vacuum bag moulding (VBM). These two types of structures were then compared to evaluate their energy absorption performances and failure mechanisms under axial loading conditions. Test results indicate that the honeycomb-structured battery pack made of 3D-printed continuous carbon fibre-reinforced thermoplastic (CCF/PLA) exhibits a higher compressive strength than CCF/Epoxy. For 1-cell configuration, CCF/PLA gives 36.53 MPa vs. 26.75 MPa for CCF/Epoxy; for 8-cell configuration, the values are 18.60 MPa and 15.95 MPa, respectively. Additionally, the specific energy absorption (SEA) values for CCF/PLA are 18.20 J/g (1-cell) and 24.10 J/g (8-cell). In contrast, thermosetting CCF/Epoxy composites demonstrate a superior SEA performance, with values of 31.22 J/g (1-cell) and 38.24 J/g (8-cell). Subsequently, a finite element (FE) model was developed to simulate the mechanical behaviour of the composite specimens subjected to axial loading. It shows a good agreement between numerical predictions and experimental results. Furthermore, the synergistic mechanical matching relationship between honeycomb structures with varying heights and battery cell configurations was predicted through numerical simulations. Quasi-static experimental and numerical simulation

1 28 results provide a design guidance to improve the load bearing capacity.

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6 30 **Keywords:** 3D-printed; Energy absorption; Honeycomb structures; Battery pack; Load bearing capacity.

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1. Introduction

Batteries serve as the heart of electric aircraft and are essential for hybrid/electric propulsion. However, current battery technology limits electric aircraft to small airframes with very limited range, making battery pack weight a critical constraint^[1]. Recent advances in electric aircraft largely rely on lightweight structural materials and improved power systems^[2-4]. Reducing battery pack weight lowers energy consumption and allows more batteries to be installed without increasing drone mass, which is vital for enhancing electric aircraft performance.^[5, 6]. Despite its importance, research on aerospace battery systems remains limited, whereas significant progress in EV battery pack design has provided valuable insights into structural concepts and material selection^[7-10]. Commercial battery packs are primarily made of aluminum, steel, or carbon fibre reinforced plastic (CFRP)^[11-13]. Compared with aluminum, a CFRP cell shell is about 40 % lighter^[14-16] and the thermal conductivity is about 200 times lower, which can better protect the battery from cold and heat.

Numerous studies have focused on lightweight battery packs^[17-20]. For automotive applications, a high-strength, crash-resistant enclosure must address safety issues such as internal fire or explosion^[21, 22]. Over the past two decades, modelling of the battery pack has advanced toward more efficient management systems^[23, 24]. Pana et al.^[25] experimentally and numerically investigated a battery pack with a high-strength steel base and a CFRP outer shell, reporting a 6.6% weight reduction compared with an all-steel enclosure. Liang^[26] and Greve et al.^[27] used finite element simulations to evaluate the impact performance of battery and pack. Cellular and honeycomb structures have gained increasing attention to battery protection, due to their high specific strength, excellent energy absorption, and design flexibility^[28-31]. Unlich et al.^[32] proposed that the overall performance of hexagonal battery packs is superior to other test geometries. Sinz et al.^[33] designed a replaceable battery pack for electric buses and recommended adding beams beneath it to reduce intrusion and acceleration peaks for better protection. Other studies have explored composite materials in battery systems. Pan et al.^[34] studied the impact

resistance of battery packs through computer simulations. Wang et al. [35] investigated a novel battery pack system consisting of a non-modular battery pack (CTP) and two tubular structures with negative Poisson's ratio (NPR). Han et al. [36] tested carbon/glass hybrid composite honeycomb enclosures under flexural and impact loads, with detailed damage analysis. Dong et al. [37] focused on structural battery composites, examining their out-of-plane compressive behaviour and electrochemical stability under compressive stress of 10 MPa. Samir et al. [38] fabricated non-prismatic reinforced honeycomb (NPRH) sandwich structures via fused filament fabrication and conducted multi-directional quasi-static compression tests, reporting high specific energy absorption.

Limited research exists on the structural design of de-modular battery packs for improved protection. In electric aircraft, conventional wing fuel tanks could be replaced by battery packs integrated into honeycomb sandwich wing panels using unitized packaging technology (UPT). This enables both lightweight design and enhanced safety. To fill this gap, the current study carried out fabrication of thermoplastic short-carbon-fibre and continuous-carbon-fibre honeycomb structures via 3D printing and vacuum bag molding. The corresponding quasi-static crush energy absorption was evaluated through experimental and numerical investigations. Furthermore, the load-displacement responses were analysed to assess mechanical interaction and deformation compatibility between the battery and the protective structure.

2. Experimental procedure

2.1 Specimen preparation

2.1.1 3D printed short carbon fibre honeycomb structures

All structures were manufactured according to Fused Deposition Modeling (FDM) method, as shown in Figure 1(a). For modeling specimen's geometry, Creality Slicer Software was used, then model was exported as STL file from the software and imported to the 3D printing software for printing preparation. ENDER-3MAX 3D

printer was used to print SCF/PLA (Carbon Fibre/Polylactic Acid) specimens. This printer was selected because of its simplicity and ease of use. The diameter of SCF/PLA printing wire is 1.75 mm, and the printing adopts standard wire feeding method, which is fed into the printing head through (Polytetrafluoroethylene) PTFE tube. All the printing parameters and dimensions of 3D printed specimens are presented in Table 1. The actual fibre mass (%) in SCF/PLA filaments is 8-10%, the fibre volume fraction ranging from 5.5% to 7.0% is exhibited by these composites.

Table 1. Short carbon fibre-reinforced polylactic acid composite 3D printing parameters.

Printing parameter	Value	Units
Filament diameter	1.75	mm
Nozzle diameter	0.4	mm
Extrusion/ Nozzle temperature	220	°C
Nozzle	60	°C
Layer height	0.2	mm
infill	100	%
Extrusion speed	150	mm/s
Infill pattern	Tetrahedral	-
Outer shell	0.6	mm

2.1.2 3D printed continuous carbon fibre honeycomb structures

As shown in Figure 1(a), continuous PLA wire and carbon fibre bundle mix in the guide tube through heating the PLA resin to the melting temperature. Both materials are subsequently delivered into the heating zone, where the molten PLA impregnates the carbon fibre bundle in-situ. The resulting impregnated composite filament is then extruded through the nozzle and deposited to form the printed structure. The heat sink at the entrance of guide tube is to ensure the solid state of PLA wire. To uniformly mix carbon fibre and PLA, a conical nozzle is employed. During the printing process, the already extruded carbon fibre reinforced PLA filaments stick on the panel and provide persistent traction load to pull the carbon fibre clad by PLA out of the nozzle. The printing spacing was set to 1 mm, a printing speed to 150 mm/min, a layer height to 0.3 mm, and a nozzle temperature of 220 °C. The fibre volume fraction of the structure in this study was determined to be 17 %. The printing path of the 3D-

printed CCF/PLA composite is presented in Figure 1(a). The as-printed material exhibits a good impregnation quality, with uniform resin distribution around the fibres and no significant fibre exposure. Due to weakened fibre fixation after resin melting, pull-back phenomena are observed (approximately 20% angular deviation at corners and 45% at path reversals), which are compensated by reserved path design.

2.1.3 Continuous carbon fibre reinforced thermosetting honeycomb structures

Thermosetting continuous carbon fibre reinforced CCF/Epoxy composites are prepared by vacuum bag moulding (VBM) method, as shown in Figure 1(b). The composite material system used in this study was a unidirectional T700/USN12500 CFRP prepreg carbon/epoxy tape (Guangwei Co., Ltd, China) with a nominal cured ply thickness of 0.125 mm and a 12 K tow. The symmetric quasi-angle lay-up $[45/-45/0/-45/45]_s$ was adopted to balance axial stiffness and shear-dominated progressive failure, since $\pm 45^\circ$ plies induce progressive damage and enhance energy absorption capacity under crushing loads^[39]. Following the manufacturer's guidelines, a vacuum pressure of approximately 0.1 MPa was applied, and the initial tow spacing (0.2-1.0 mm) was reduced to approximately 0.05-0.5 mm after consolidation. Also, the curing cycle was performed with a temperature ramping rate of 2 °C/min to 80 °C, maintained for 60 minutes, followed by a second temperature rise to 120 °C, which was held for 90 minutes. Under this process, the fibre content of the CCF/Epoxy composite material can reach approximately 50%.

Honeycomb structure was prepared using nylon moulds consisting of hexagonal blocks, which were made by 3D printing. Here, a total of 8 hexagonal-shaped nylon blocks, with a height of 100 mm and a face-to-face dimension of 22 mm, were inserted into the mould. The manufacturing processes for the honeycomb can be divided into five steps, Firstly, the nylon moulds, with a thermal deformation temperature of 180 °C, were manufactured using the selective laser sintering technique. Unidirectional carbon fibre prepreg tapes were sectioned according

to designated fibre orientations to obtain the required lay-up configurations for subsequent manufacturing. In the second stage, five layers of prepreg tape were wrapped around the surface of each nylon mould. In the third stage, all the moulds were assembled and then five layers of prepreg tape were wrapped around the surface of the assembly.

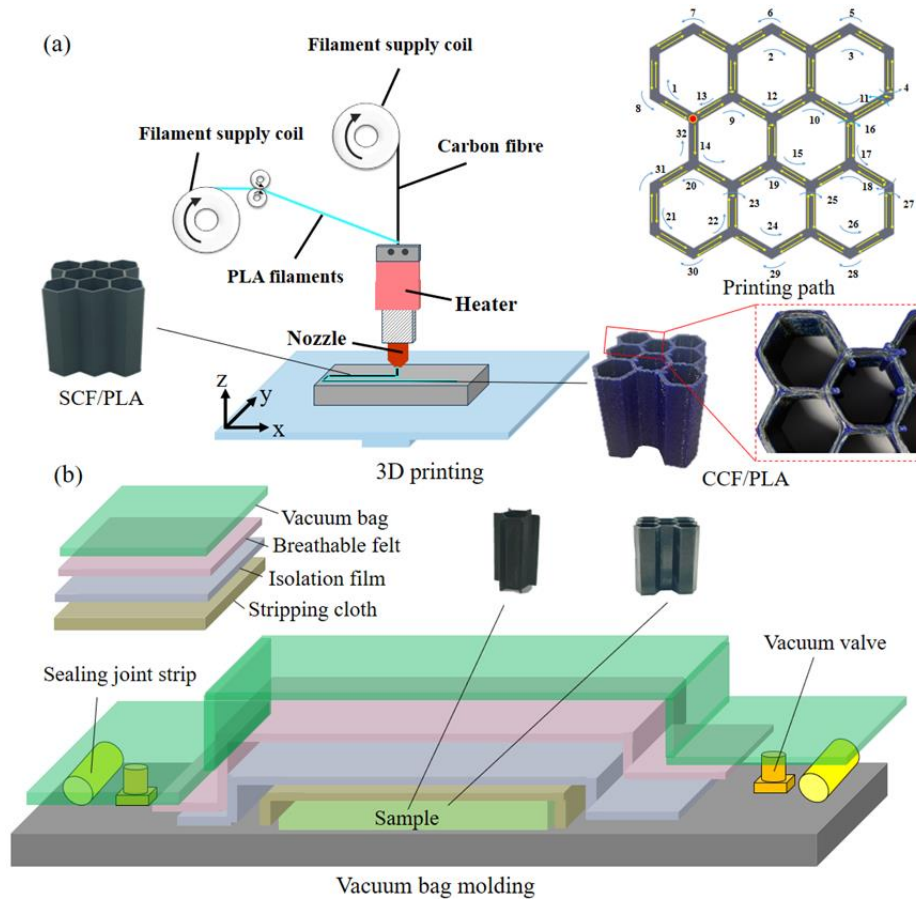


Figure 1. (a) Schematic of the 3D printing process of short/continuous carbon fibre thermoplastic composite, (b) preparation process of continuous carbon fibre thermosetting composite honeycomb structure by vacuum bag moulding method.

As a result, thicknesses of all parts of the honeycomb were the same, equivalent to ten layers of prepreg. Fourthly, the preform was then vacuum bagged in a polymer film and placed in an oven to cure. Finally, following completion of the cure cycle, the moulds were removed from the assembly, with the final product being produced.

Lithium battery (produced by Shenzhen Kyoto Electronic Technology Co., LTD.) with a voltage of 3.7 V and capacity of 4800 mA (h) was embedded into the honeycomb. For safety reasons, the residual power of the

cylindrical battery used in the test was less than 25 % of the total capacity, and the bases were superimposed up and down to avoid sliding during the battery crushing test.

Three-dimensional internal images were acquired via X-ray computed tomography (CT), allowing the void content (%) to be quantified. The measured void contents are 2.7% for SCF/PLA (Figure 2a), 3.46% for CCF/PLA (Figure 2b) and 1.32% for CCF/epoxy (Figure 2c). This quantitatively confirms that the filament-based additive manufacturing introduces a higher interfacial porosity compared with vacuum-assisted curing, primarily due to limited inter-bead fusion and fibre impregnation.

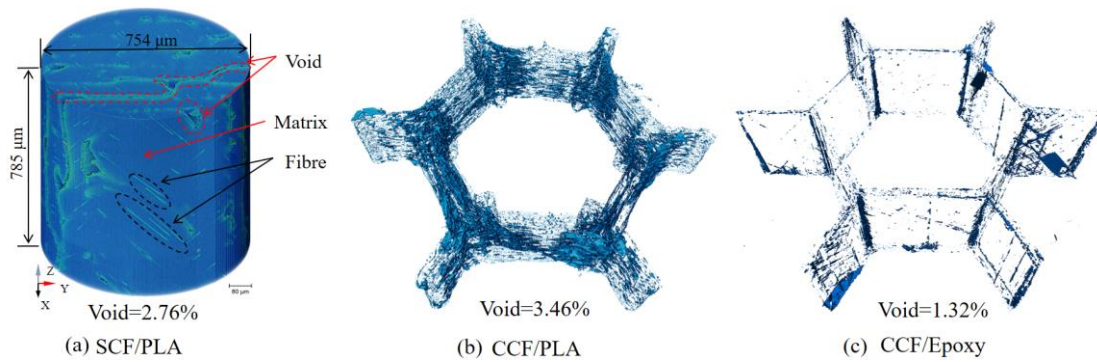


Figure 2. Comparison of void content (%) in honeycomb composite structures manufactured by different processes: (a) 3D-printed SCF/PLA, (b) 3D-printed CCF/PLA, and (c) vacuum bag cured CCF/epoxy, as determined by X-ray computed tomography (CT).

2.2. Quasi-static mechanical tests

The compression strength and energy-absorbing characteristics of the carbon fibre honeycomb structure and cylindrical battery were evaluated at a crosshead displacement rate of 5 mm/min using an MTS universal testing machine (ETM504C) with a load capacity of 200 kN. The two ends of the battery were connected to the voltage acquisition instrument (SIRIUS MINI 4-channel vibration/noise analyzer). The cylindrical battery used was standard cylindrical battery 21700 (Panasonic). The mechanical testing machine used for carbon fibre honeycomb structure and cylindrical battery testing are shown in Figure 3. All quasi-static crushing tests of the lithium-ion cells were conducted in a well-ventilated and open testing environment equipped with remote loading control and

a fire blanket for emergency protection. The state of charge (SOC) of all tested cells was maintained at $(20 \pm 5\%)$. The recorded temperature values represent localised surface heating during deformation rather than a full thermal runaway event.

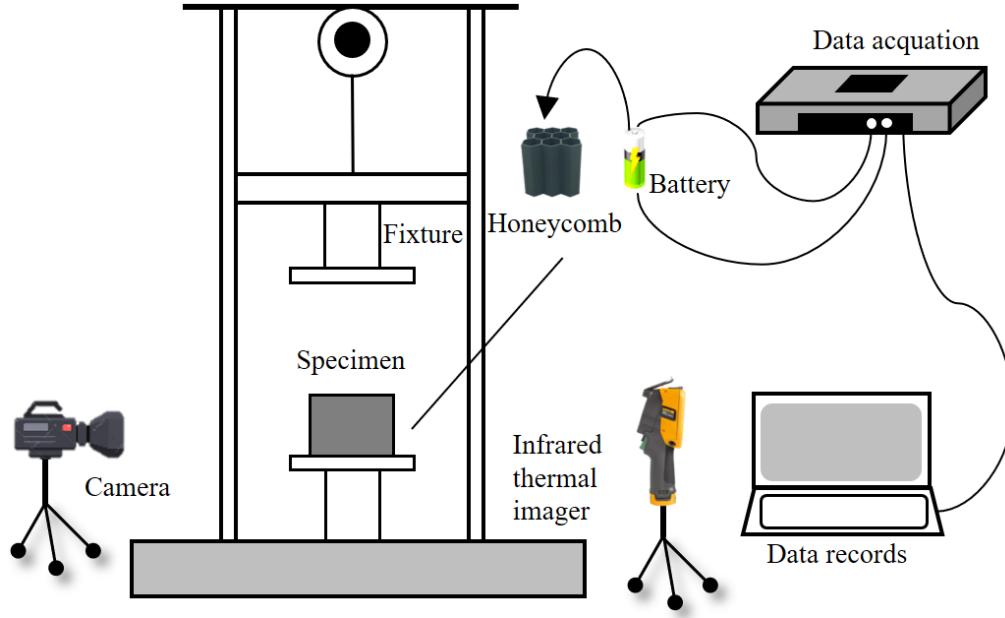


Figure 3. Schematic of the quasi-static compression test setup.

The energy absorbed by the specimen during crushing is equal to the area beneath the load-displacement curve, which can be calculated by Eq. (1):

$$EA = \int_0^l F(x)dz \quad (1)$$

where x is the displacement corresponding to the applied load, ' l ' represents the effective crushing distance and F is an applied load.

Specific energy absorption (SEA) is used to evaluate energy absorbing efficiency, accounting for the lightweight requirement in the design of a good energy absorber. This can be calculated by dividing the measured value of EA by the mass of the damaged part (m), according to Eqs. (2) and (3).

$$SEA = \frac{\int_0^l F(x)dz}{m} \quad (2)$$

$$m = M \cdot \frac{l}{H} \quad (3)$$

where ‘M’ and ‘H’, as indicated before, correspond to the total mass and height of each specimen before test, respectively.

Table 2. Summary of geometries of the honeycomb structure investigated in this study.

Sample	Mass (g)	Length (mm)	Wide (mm)	Height (mm)	Thicknes s (mm)	Loading rate (mm·min ⁻¹)
SCF/PLA-1Cell-a	20.10	41.46	36.41	71.70	2.03±0.11	5.00
SCF/PLA-1Cell-b	20.40	41.51	36.62	71.75	2.04±0.08	5.00
SCF/PLA-1Cell-c	20.20	41.49	36.54	71.72	2.04±0.10	5.00
SCF/PLA-1Cell-d	20.15	41.48	36.63	71.74	2.02±0.08	5.00
SCF/PLA-1Cell-e	20.22	41.52	36.53	71.73	2.03±0.10	5.00
SCF/PLA-8Cell-a	77.30	73.52	71.38	73.50	2.09±0.09	5.00
SCF/PLA-8Cell-b	77.50	73.90	72.34	73.51	2.05±0.11	5.00
SCF/PLA-8Cell-c	77.40	73.87	72.21	73.45	2.07±0.12	5.00
CCF/PLA-1Cell-a	31.60	44.34	38.96	72.07	3.35±0.11	5.00
CCF/PLA-1Cell-b	31.70	44.35	38.89	72.10	3.36±0.13	5.00
CCF/PLA-1Cell-c	31.50	44.34	38.88	72.09	3.36±0.12	5.00
CCF/PLA-8Cell-a	115.00	77.19	73.49	71.72	3.02±0.11	5.00
CCF/PLA-8Cell-b	115.40	77.15	73.70	71.83	3.05±0.10	5.00
CCF/PLA-8Cell-c	115.20	77.16	73.74	71.80	3.03±0.09	5.00
CCF/Epoxy-1Cell-a	20.00	44.24	41.65	75.41	1.36±0.05	5.00
CCF/Epoxy-1Cell-b	20.00	43.04	42.96	73.85	1.55±0.03	5.00
CCF/Epoxy-1Cell-c	20.00	44.04	43.85	76.21	1.32±0.04	5.00
CCF/Epoxy-8Cell-a	77.50	72.29	68.75	75.16	1.75±0.04	5.00
CCF/Epoxy-8Cell-b	73.50	73.26	69.49	74.62	1.45±0.03	5.00
CCF/Epoxy-8Cell-c	77.20	73.16	68.48	76.01	1.55±0.02	5.00

Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated up to

the onset of densification in the load-displacement curve to ensure a consistent comparison. Three repeated tests were carried out for each test condition to obtain an average value, the geometries of all structures are shown in Table 2.

3. Numerical modelling

3.1. Modelling of honeycomb structure and battery

3.1.1 Modelling of 3D printed short carbon fibre reinforced composite honeycomb structure

A high-fidelity FE model consisting of the 1-Cell and 8-Cell honeycomb structure was constructed in Abaqus/Explicit, as shown in Figures 4(a), (a1).

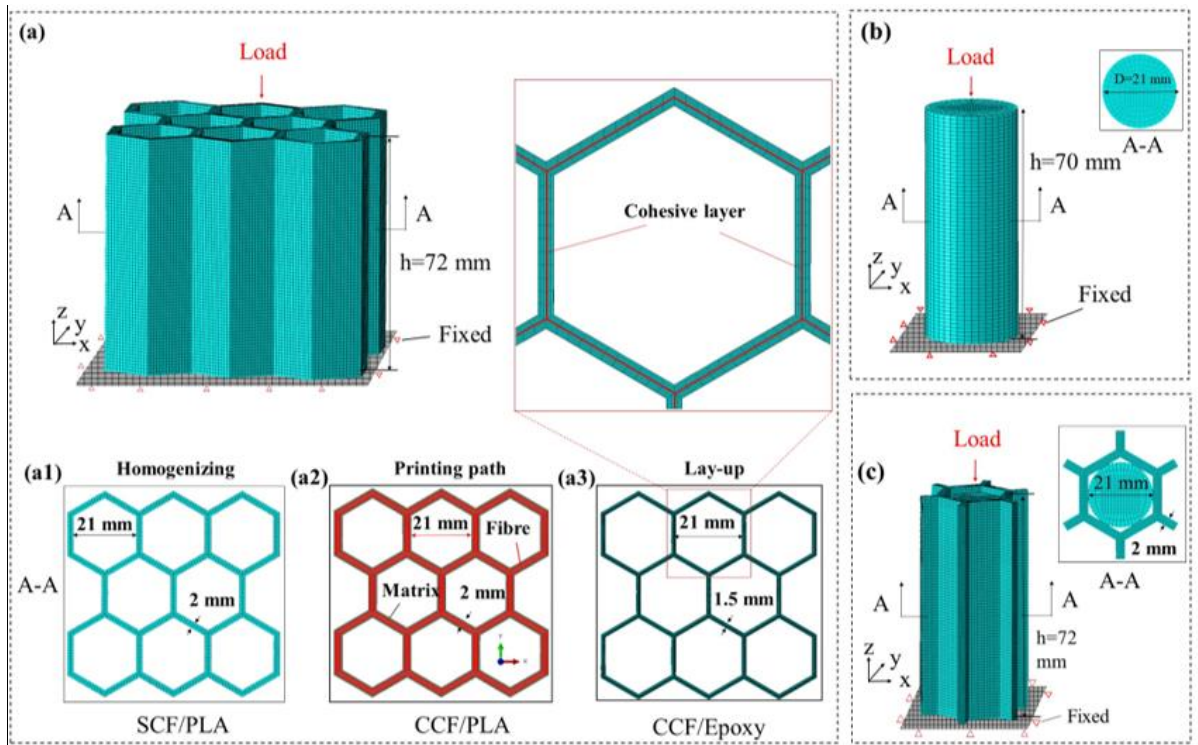


Figure 4. (a) Full-scale FE model of honeycomb of different manufactures (a1, 3D printing thermoplastic short carbon fibre (SCF/PLA) composites, a2, 3D printing continuous carbon fibre (CCF/PLA) composites, a3, vacuum bag moulding method (CCF/Epoxy) composites), (b) the geometry and mesh generation of the FE model for the cylindrical battery.

The height is 72 mm, the inner diameter is 21 mm, and the honeycomb thickness is 2 mm. The top view of

the model is also exhibited. The honeycomb was meshed using 8-noded solid elements with reduced integration (C3D8R) to model the out-of-plane mechanical behaviour. Based on mesh sensitivity studies (Appendix A), a mesh size of 1 mm × 1 mm was adopted, as shown in Figure 4(a1). The interaction properties were applied to the face-to-face contact to feature the tangential behaviour with a friction penalty of 0.2 ^[40] and hard normal contact to prevent self-penetration of elements. The material properties are listed in Table 3. Among them, the stress triaxiality of -0.33 was calculated to represent the compressive shear dominated stress state during honeycomb wall buckling under quasi-static compression.

Table 3. Properties of short carbon fibre composite honeycomb ^[40].

Parameter		Value	Units
Density		1.35	g/cm ³
Young's Modulus		2294	MPa
Poisson's Ratio		0.3	/
Ductile damage (Damage evolution)	Fracture strain	1.45	%
	Stress triaxiality	-0.33	/
	Strain rate	0.000656	s ⁻¹
	Failure displacement	0.14143	mm
Shear damage (Damage evolution)	Fracture strain	1.35	%
	Shear stress ratio	1.6	/
	Strain rate	0.00416	s ⁻¹
	Failure displacement	1.1343	mm

3.1.2 Modelling of 3D printed continuous carbon fibre reinforced composite honeycomb structure

The finite element model of 3D printed continuous carbon fibre reinforced composite honeycomb structure is established using the commercial finite element software ABAQUS/Explicit, to simulate the compressive behaviour of the honeycomb structure. The established FE models of 1-cell and 8-cell honeycombs are shown in Figures 4(a) and (a2), and their finite element models with structural geometric parameters are established by combining the printing path, material properties of matrix and continuous carbon fibre, as shown in Table 4. When establishing the FE model, PLA is defined as an isotropic material, and its elastic and plastic parameters are obtained from the previous experimental study ^[41].

To improve computational efficiency, a simplified numerical model was adopted, in which the curvature of real fibres, intermittent deposition, and interlayer defects were neglected. Also, the printed continuous fibre composite was treated using homogeneous anisotropic material properties ^[42], where their stiffness and strength characteristics along three main directions were incorporated through homogenised orthotropic properties.

Table 4. Basic mechanical parameters of continuous carbon fibre bindle and PLA ^[41].

Parameter	Tensile strength/MPa	Tensile modulus/GPa	Density/g·cm ⁻³
Fibre bindle	4100	240	1.78
PLA filament	62.63	3.20	1.24

Consequently, this modeling framework is suitable for predicting the overall structural response and crushing trends, but is not intended for a detailed description of micro-scale failure mechanisms. The matrix and fibre bundle parts are also meshed by eight-noded hexahedral linear element with reduced integration (C3D8R). The same contact properties as described in the previous section are used for both tangential and normal interaction. Face-to-face contact is also applied between the fibre bundle and matrix parts. A displacement control boundary condition is applied on the top rigid surface. All degrees of freedom (Translational and rotational ones) on the bottom rigid surface and rotational degrees of freedom on the top rigid surface are constrained (Figure 4a).

3.1.3 Modelling of vacuum bag mould continuous carbon fibre reinforced composite honeycomb structure

The simulations were conducted by using the explicit solver in Abaqus/Explicit. The mesh, geometry and boundary conditions of the FE model are shown in Figures 4(a) and 4(a3). Composite samples have a layup of [45/−45/0/−45/45]_s and dimensions of the inner diameter of 21 mm, the height of 72 mm, and the cell thickness of 1.5 mm. A user-defined material model (VUMAT) for fabric reinforced composite materials, namely “ABQ_PLY_FABRIC”, was adopted to define the intralaminar material. The continuum shells were individually meshed using 8-noded quadrilateral stress/displacement continuum shell elements with reduced integration (SC8R)

^[43]. Each shell represents one fibre ply with a thickness of 0.15 mm. A regular mesh size of 1 mm × 1 mm was employed after a mesh sensitivity study (Appendix A). There are 66528 elements for 1-cell and 144000 elements for 8-cell structures used in the numerical analysis. A 45° chamfer triggering was modelled in order to ensure a stable and progressive collapse of the structure. The model was positioned between two steel plates that were modelled as discrete rigid shell plates. Based on the traction-separation law, the bonding behaviour was defined. As shown in Figure 4(a3), zero-thickness 8-node cohesive elements (COH3D8) were employed to model the inter-laminar interfaces of the thermosetting composite laminates (the red line). The bottom plate was fixed, whilst constant displacement loading was assigned to the top plate.

The elastic and strength properties adopted in the numerical simulations were provided by the material manufacturers and the remaining material properties were taken from Ref. ^[44]. The material properties used in this study are shown in Table 5.

Table 5. Mechanical properties of honeycomb composite ^[44].

Properties	Values	Units
Density	$\rho=1.7$	g/cm^3
Stiffness properties	$E_{11}=130$ $E_{22}=E_{33}=9$	GPa
Poisson's ratio	$V_{12+}=0.35$ $V_{12-}=0.3$	/
Material strengths	$X_{1+}=1954.1$ $X_{1-}=2000$ $Y_{2+}=90$ $Y_{2-}=724$ $S=119$	MPa
Critical fracture energy	$G_f^{1+}=280$ $G_f^{1-}=200$ $G_f^{2+}=19$ $G_f^{2-}=31.5$	kJ/m^2

During the fracture process, open-section composite structures may exhibit multiple failure modes, including intralaminar shear, axial cracking, and transverse delamination. These modes result in spreading or petal-like

damage patterns. To fully represent these failure modes, detailed numerical analysis methods capable of capturing them are required. Single-shell modeling approaches cannot capture failure modes ^[45]. The interlayer cohesive properties are shown in Table 6 ^[46].

Table 6. Inter-laminar properties defined for the composite cohesive element layer shell interfaces ^[46].

Properties	Values	Units
Cohesive strengths	$t_n=100, t_s=t_t=130$	MPa
Initial cohesive stiffness	$K_{nn}=5.4E^6$ $K_{ss}=K_{tt}=4.2E^6$	N/mm ³
Fracture energy	$G_{Ic}=0.554$ $G_{IIc}=G_{IIIc}=2.8116$	N/mm ³
Density	$\rho=1.7$	g/cm ³
Benzeggagh–Kenane exponent	$\eta=2$	/

3.1.4 Modelling of battery

A FE model, representing almost all the details of the battery cell, was established in Abaqus/explicit. Here, cylindrical Lithium-ion 21700 cells (GAIA, HP 602030 NCA-45 A/162 Wh) were chosen to be simulated. The main dimensions and components of the battery are shown in Figure 4(b). The battery shell is made of stainless steel ^[47]. The outer diameter and the axial length of the battery are 21 mm and 70 mm, respectively. The battery was normalised in this model, so that the model parameters correspond to the actual size of the battery studied. All mechanical properties of the battery are shown in Table 7 ^[48].

Table 7. Properties of cylindrical battery ^[48].

Properties	Values	Units
Young's Modulus	8467.56	MPa
Poisson's Ratio	0.3	/
Density	3.5	g/cm ³
Fracture Strain	0.35	/
Stress Triaxiality	-0.33	/
Strain Rate	0.0002	s ⁻¹
Displacement at Failure	10	mm

3.2 Modelling of honeycomb structure with battery

To facilitate the study of the effects of honeycomb structures on battery protection, predictions of compression protection using honeycomb structures of different heights are particularly important. The FE model needs to include both the honeycomb unit cell structure and the battery. Based on the models of the composite honeycomb structure and the battery, we have calculated the mechanical matching between the honeycomb structures with heights of 72, 74, 76, 78, and 80 mm and the battery, to predict and analyze the mechanical matching design between the protective structure and the battery.

4. The test results and discussion

In the initial part of this investigation, a quasi-static compression test with a displacement loading rate of 1 mm/min was carried out to study the mechanical properties of the cylindrical battery. Then, the strength and energy-absorbing characteristics of 1-cell honeycomb samples, based on 3D printing and vacuum bag moulding, were investigated through a series of quasi-static compression tests at a crosshead displacement rate of 5 mm/min. The results of these tests are presented and discussed as follows.

4.1 Results of quasi-static compression test of cylindrical battery

Based on the load-, voltage- and temperature-displacement curves in Figure 5, the battery response can be divided into three stages (Figure 5a). At Stage I (Mechanical deformation), the battery undergoes elastic and plastic deformations until structural buckling occurs, whilst voltage and temperature remain nearly constant. The increase on load contributes to the accumulation of progressive strain energy. At Stage II (Separator deformation), after buckling, the separator with a high failure strain begins to carry the load. The applied load gradually decreases as the separator deforms under continuous platen displacement, whilst the voltage remains around 3.4 V. This stage highlights stress redistribution and progressive damage within the cell. At Stage III (Short circuit and thermal response), once the separator reaches its mechanical limit, a cathode-anode short circuit occurs, leading to a rapid

voltage drop and temperature rise to 129 °C. The accelerated changes are attributed to localised shorting and Joule heating. Further crush results in the ultimate mechanical failure and complete voltage collapse. This stage illustrates the coupling among the structural degradation, electrochemical response and thermal effect.

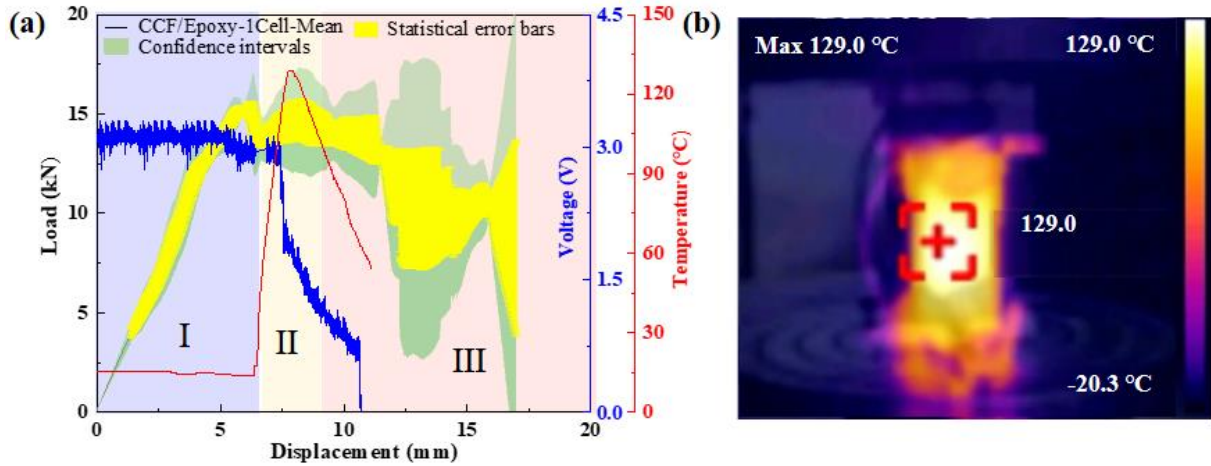


Figure 5. (a) The curves of load, voltage, and battery temperature versus time under quasi-static compression, and (b) the temperature results of the failure center.

4.2 Test results of honeycomb structures made of fibre composites with different manufacturing processes

The compressive response of the three honeycomb structures is illustrated in Figure 6, which depicts both the load-displacement curves and the corresponding progressive failure modes. The densification points are also marked on the curves. To ensure statistical robustness, the curves were analysed to derive the mean, variance, and confidence intervals, as well as the boundary intervals for the total energy absorption. For SCF/PLA honeycombs, both 1-cell (Figure 6a) and 8-cell (Figure 6b) configurations exhibit a rapid increase in load to peak values (8.16 kN and 39.81 kN), followed by a gradual load decrease and a relatively stable crushing plateau. This behaviour is governed by progressive cell-wall buckling and matrix-dominated failure. Damage initiates at stress concentration regions and propagates through matrix cracking and interlayer debonding, whilst the load was subsequently sustained by wall folding and frictional contact between fractured segments. The dominant failure modes are

therefore matrix cracking, wall buckling, and delamination (Figures 7a1, a2), which are the damage characteristics of composite cellular solids under axial loading ^[49]. The SEA values are 12.88 J/g for the 1-cell and 10.32 J/g for the 8-cell one.

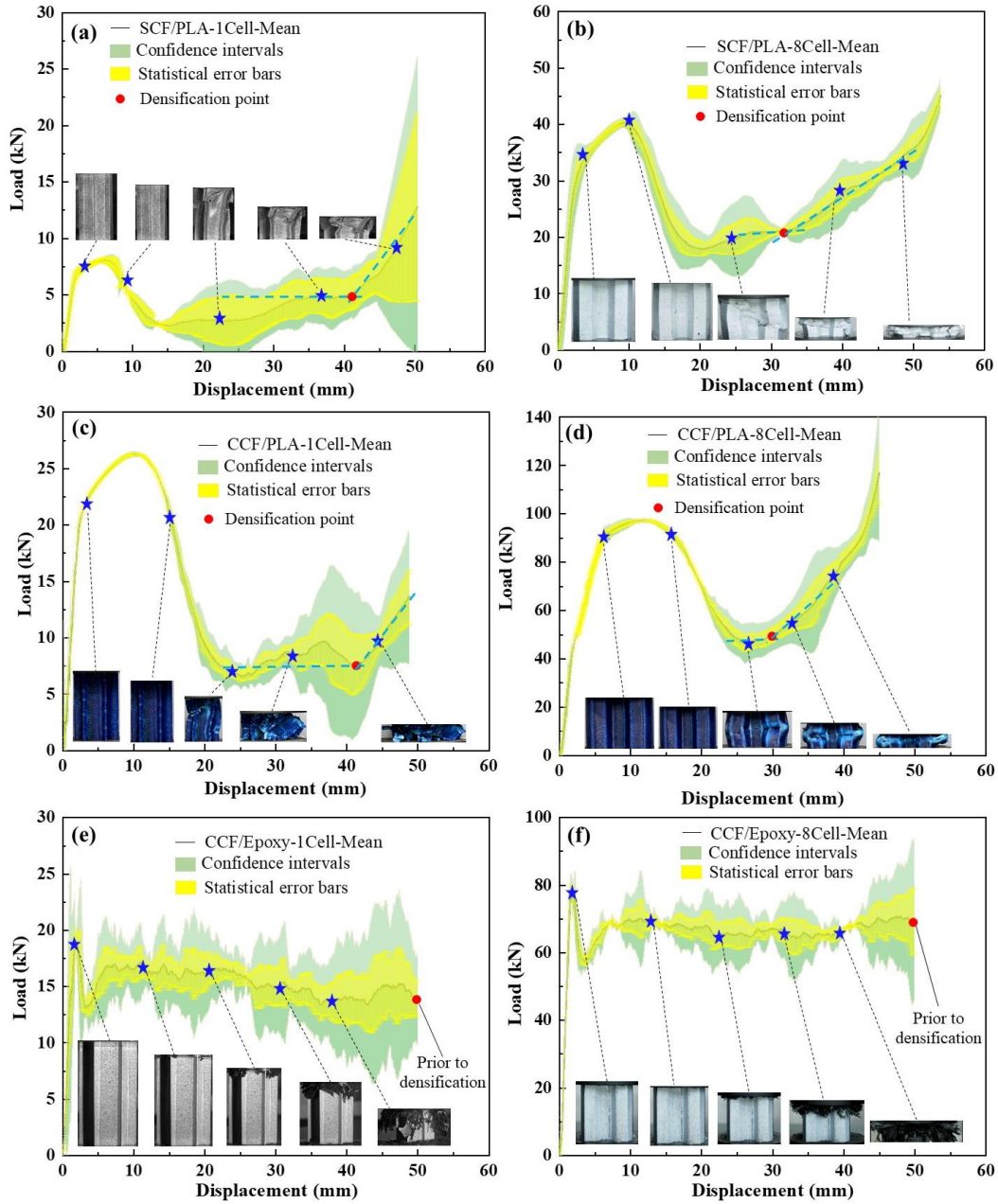


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static crushing tests on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

For the CCF/PLA honeycombs, the peak loads were measured as 25.29 kN for 1-cell (Figure 6c) and 97.68

kN for 8-cell samples (Figure 6d). A more pronounced post-peak load drop is observed. This sharp reduction is mechanistically associated with interface-controlled damage in additively manufactured continuous fibre composites, where limited interlayer fusion and in-situ impregnation promote early delamination and fibre-matrix debonding. After the initial drop of load, a secondary plateau is maintained through progressive fibre fracture, wall buckling, and interfacial sliding. The dominant failure mechanisms include fibre fracture, matrix cracking, and delamination (Figures 7b1, b2), consistent with reported post-peak instability features in continuous fibre AM composites.^[50] The 1-cell and 8-cell honeycombs exhibit SEA values of 18.48 J/g and 20.07 J/g, respectively.

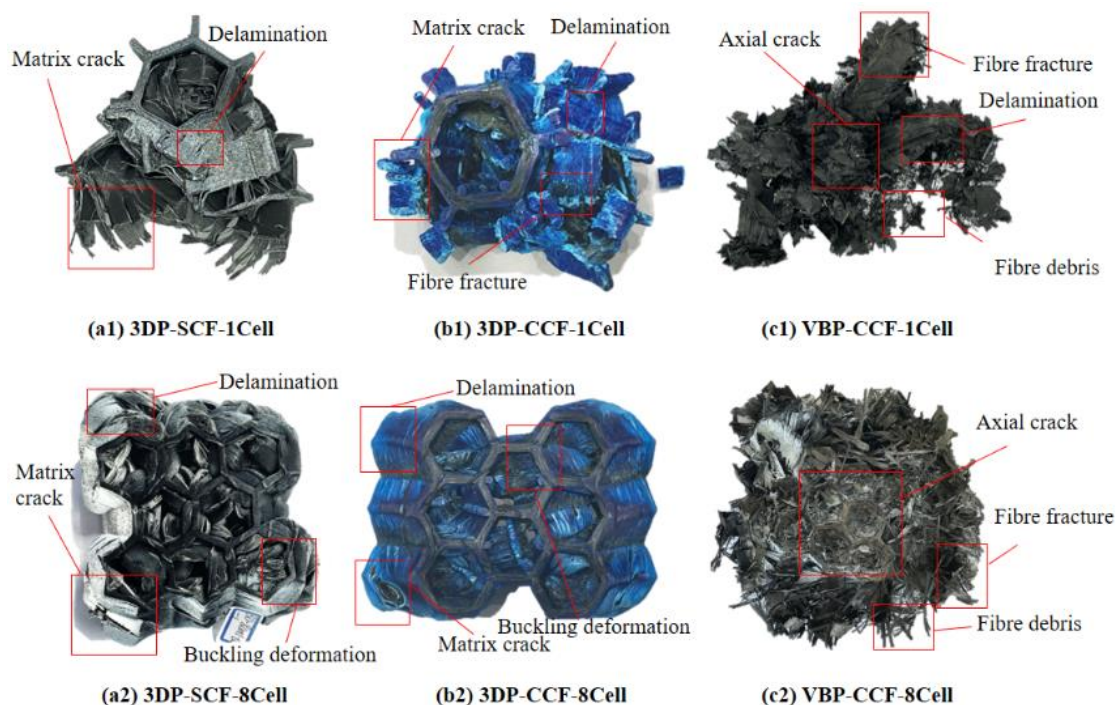


Figure 7. Failure modes of 1-cell (a1 for SCF/PLA, b1 for CCF/PLA, c1 for CCF/Eopxy) and 8-cell (a2 for SCF/PLA, b2 for CCF/PLA, c2 for CCF/Eopxy) honeycomb structures made of different materials after quasi-static compression.

Similar staged crushing behaviour is also observed in vacuum-bag-molded CCF/Epoxy honeycombs, but with a more stable post-peak response (Figures 6e, f). The 1-cell and 8-cell structures reach peak loads of 20.33 kN and 80.18 kN, respectively, followed by sustained plateau loads prior to densification. Failure initiates with matrix cracking and delamination, followed by wall sliding and progressive lamina fragmentation (Figures 7c1, c2).

Interfacial sliding and friction between fractured walls provide a sustained energy dissipation mechanism via friction, which delays collapse locally and directly contributes to the superior SEA (43.27 J/g for the 1-cell and 38.25 J/g for the 8-cell honeycomb) ^[51]. This contrasts with abrupt failure modes (e.g., sudden matrix/fibre fracture) of the previous two materials (SCF/PLA and CCF/PLA).

Table 8. Energy absorption characteristics of honeycomb structures made of 3D-printed short carbon fibre, 3D-printed continuous carbon fibre and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composites, respectively.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
SCF/PLA-1Cell-a	20.10	8.02		11.58		13.62	
SCF/PLA-1Cell-b	20.40	7.97		11.51		10.51	
SCF/PLA-1Cell-c	20.20	8.08	8.02	11.66	11.58	13.01	12.88
SCF/PLA-1Cell-d	20.15	8.10		11.70		13.58	
SCF/PLA-1Cell-e	20.22	7.98		11.50		13.67	
Variance	/	0.032		0.069		1.45	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	
CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

All data on the peak load, compressive strength, energy absorption and specific energy absorption of the

honeycomb structures are listed in Table 8. For 8-cell honeycomb structures made of different materials, it is observed that the 3D-printed CCF/PLA composites provide the highest compressive strength (18.61 MPa), surpassing those of CCF/Epoxy (15.95 MPa) and SCF/PLA (7.58 MPa).

Due to severe fragmentation after crush, the CCF/Epoxy specimens could not be examined by CT scanning. Therefore, post-crush CT analyses were performed on the 3D-printed SCF/PLA and CCF/PLA honeycomb structures. Figure 8(a) presents the overall 3D CT reconstruction of the crushed SCF/PLA honeycomb, which is consistent with the macroscopic observations and reveals interlayer delamination and matrix cracking. Figure 8(a1) shows sectional views from different orientations, which more distinctly illustrate the internal damage features, including interfacial separation, local buckling and fracture.

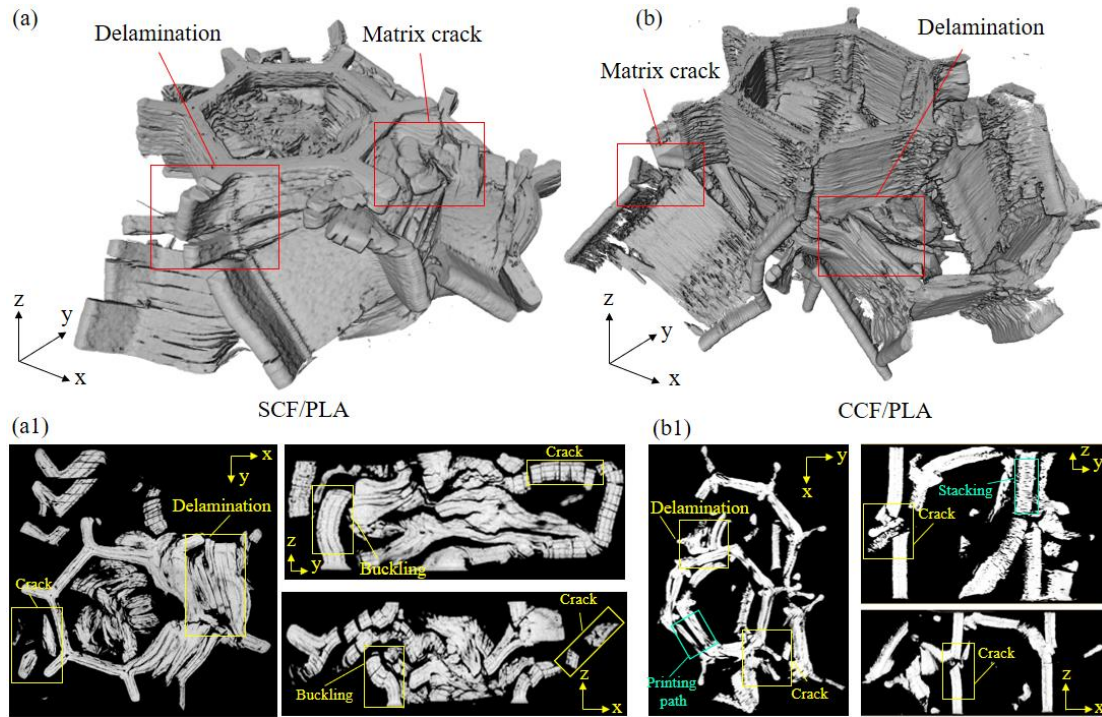


Figure 8 CT results of SCF/PLA (a, a1) and CCF/PLA (b, b1) after quasi-static compression.

For the CCF/PLA composite honeycomb, Figure 8(b) provides the 3D CT reconstruction after compression crush, exhibiting failure characteristics as observed during the physical test. The sectional views in Figure 8(b1) further reveal the layer-by-layer deposition features inherent to the 3D printing process, as well as the fibre

deposition paths. In addition, internal damage modes, such as delamination and fracture, can be clearly identified within the structure. These CT results provide direct internal damage evidence, with supporting the observed macroscopic failure mechanisms.

As shown in Figure 9, the compressive performances of the 3D-printed SCF/PLA and CCF/PLA honeycomb structures, as well as the VBM-fabricated CCF/Epoxy honeycomb structure developed in this study, are compared with various competitive cellular architectures reported in the literature. These include conventional aluminum honeycombs [52], aluminum pomelo-inspired hierarchical hexagonal honeycombs (Al Pomelo-HHH) [53], aluminum hierarchical hexagonal honeycombs (Al HHH) [54]. Other advanced cellular structures include polymer tube-reinforced honeycombs [55], CFRP hexagonal honeycombs, and 3D CFRP honeycombs [56]. The results demonstrate that the CCF/Epoxy honeycomb structure developed in this study exhibits a significantly higher specific energy absorption (SEA) compared to most of the comparable cellular designs, indicating its superior energy absorption capability. However, its density is higher than CFRP hexagon honeycomb.

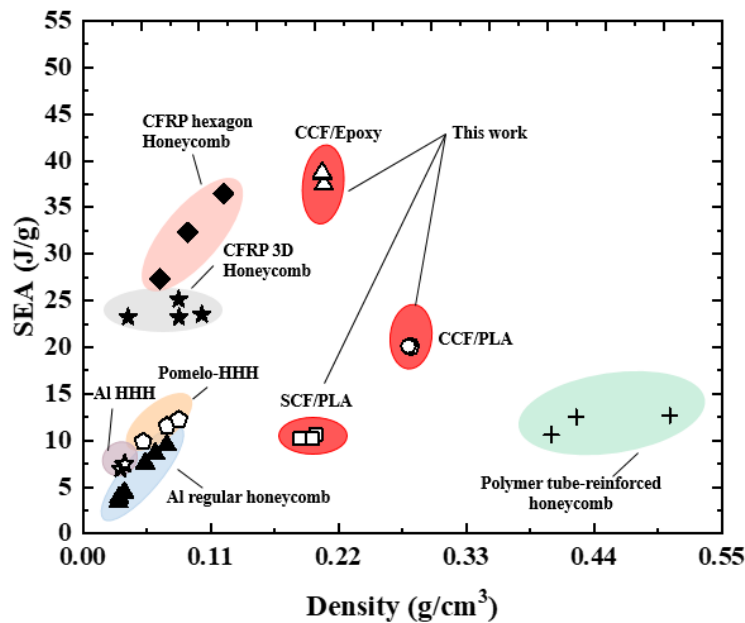


Figure 9. Comparison between the SEA of this work with other cellular structures: regular honeycomb [53] aluminium pomelo-inspired hierarchical hexagonal honeycomb (Al Pomelo-HHH) [54], aluminium hierarchical hexagonal honeycomb (Al HHH) [53] and other competing structures, including polymer tube-reinforced honeycombs [54], CFRP Hexagon and CFRP 3D Honeycomb [56].

In this study, the honeycomb structures made of three types of materials with embedded battery were tested under compressive loading. The experimental results are presented in Figure 10. All load-displacement curves were processed to obtain the mean values, standard deviations and confidence intervals. For the 3D honeycomb structures with embedded battery (Figures 10(a-c)), the structure failed prior to the battery, followed by combined load-bearing and subsequent failure together with the battery. The peak load of the SCF/PLA structure with embedded battery is approximately 18.5 kN (Figures 10(a, a1)). Subsequently, the battery also began to fail, with the temperature gradually rising to 70.7 °C. Similarly, for the CCF/PLA honeycomb structure with embedded battery (Figures 10(b, b1)), the overall structure underwent two elastic stages (0-2 mm, and 4.8-10 mm) before eventually entering the plastic stage, with a peak load of 33.2 kN. Ultimately, the structure and the battery experienced instability and failure together, with the temperature increasing to 59.5 °C. For the CCF/Epoxy structure with embedded battery, with the highest peak load of 35.2 kN (the compressive strength of 45.88 MPa). The structure exhibited a petal-like failure mode, leading to ultimately failure, with the temperature reaching 41 °C.

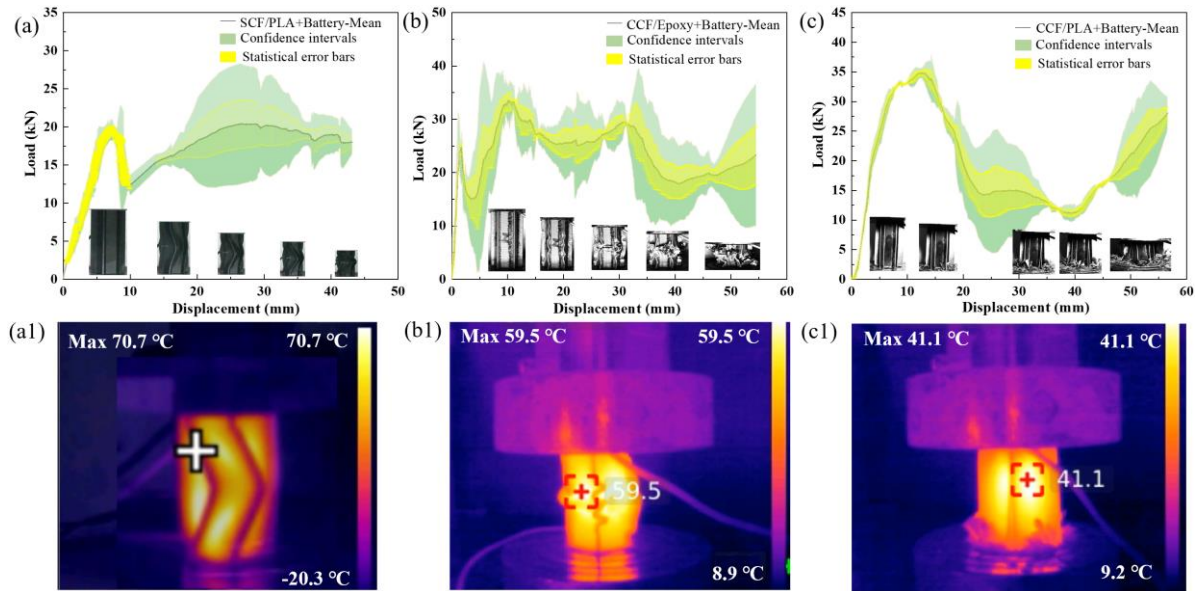


Figure 10. Quasi-static compression load–displacement responses, corresponding failure modes, and temperature evolution of 21700 cylindrical batteries embedded in honeycomb structures: (a, a1), SCF/PLA + battery, (b, b1) CCF/PLA + battery, (c, c1) CCF/epoxy + battery.

Comparative results indicate that CCF/PLA with battery achieves the highest compressive strength (50.44

MPa), while CCF/epoxy with battery provides the maximum specific energy absorption (11.4 J/g). Owing to the high material utilisation, design flexibility, and capability for manufacturing complex geometries, additive manufacturing presents significant advantages for structural applications. The findings offer useful references for material and process selection in engineering battery-integrated structures.

The 21700 lithium-ion cell used in this study has a nominal capacity of 5000 mAh, a rated voltage of 3.7 V, and a mass of 68 g, corresponding to an energy of approximately 18.6 Wh. Based on this value, the gravimetric energy density of the honeycomb-embedded battery system can be estimated for different material configurations. Specifically, the energy density is 128.28 Wh/kg for the SCF/PLA configuration, 101.64 Wh/kg for the CCF/PLA one, and 129.53 Wh/kg for the CCF/epoxy one. The gravimetric energy density reported here is a simplified structural-level estimate based only based on the battery cell and the honeycomb protective structure, without including other pack components such as BMS, interconnects, thermal management devices, or supporting frame. It is therefore used only for conceptual comparison. In addition, the SEA values are derived under axial crushing conditions only and are intended for relative comparison within this specific loading scenario, rather than representing full battery pack performance under general service conditions.

5. Verification of numerical models

5.1 The honeycomb structure

As shown in Figure 11, the finite element predictions are compared with the experimental results to evaluate the predictive capability of the proposed damage model. Overall, the simulations reproduce the load-displacement responses and the main failure characteristics observed in the experiments. For the SCF/PLA honeycomb structures (Figures 11a and b), the numerical results show good agreement with the experimental curves, with the error in peak load within 6 %. The model also captures the characteristic deformation stages, including the elastic response

(initial stiffness), peak load, progressive collapse and final densification. Also, micro-buckling is observed in both the experimental and simulation results, as shown in Figure 11a. These phenomena indicate that the progressive buckling and damage accumulation of the cell walls are reasonably simulated.

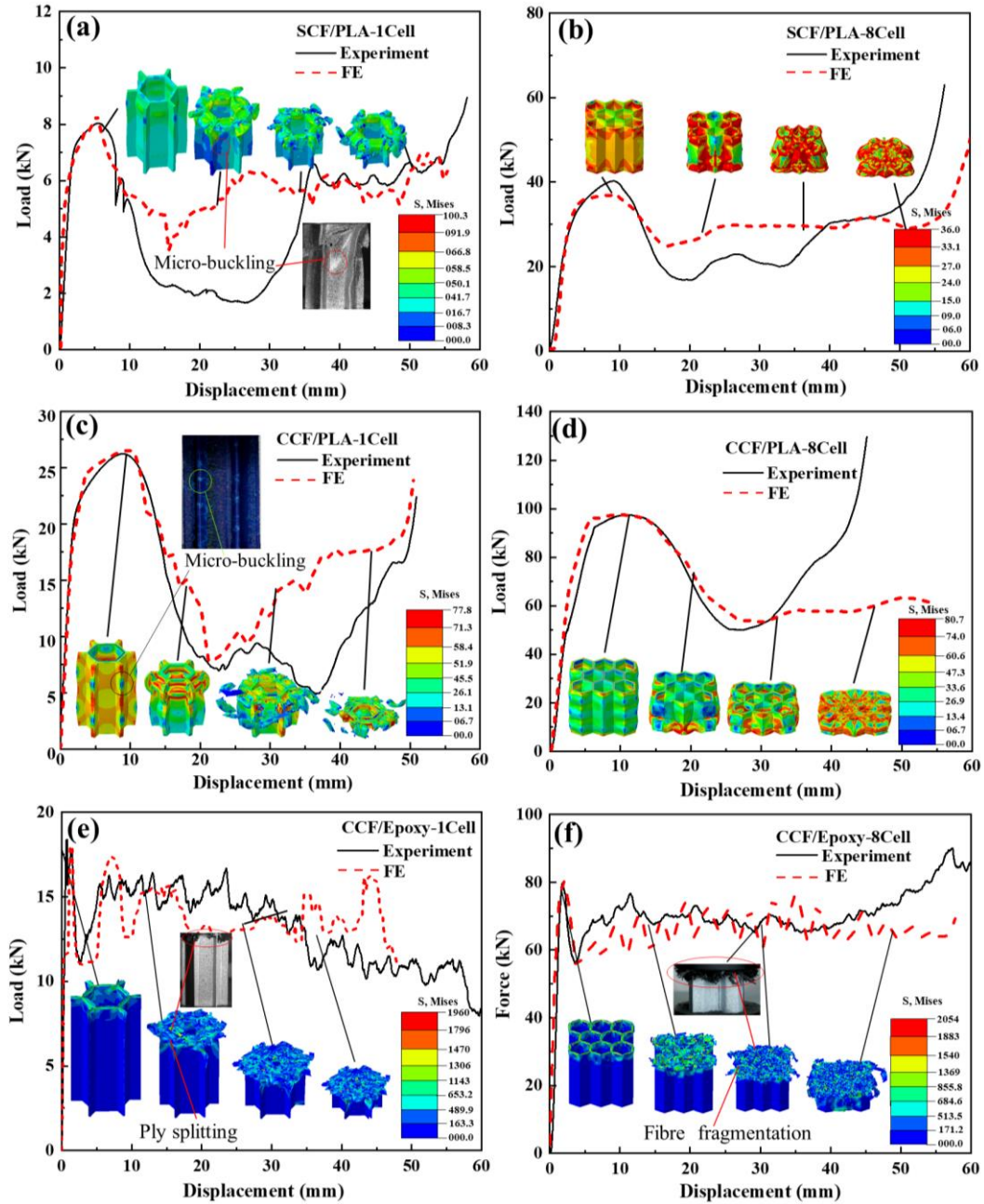


Figure 11. The load-displacement curves and predicted failure modes following crush tests on 1-cell (a) and 8-cell (b) honeycomb of 3D-printed SCF/PLA composite, 1-cell (c) and 8-cell (d) honeycomb of 3D-printed CCF/PLA composite and 1-cell (e) and 8-cell (f) honeycomb of vacuum bag moulding CCF/Epoxy composite. (the solid lines correspond to test data and the dashed lines to the FE predictions).

For the CCF/PLA honeycomb structures (Figures 11c and d), the predicted responses also agree well with the

experimental results. The discrepancies in peak load are between 5 and 9 %, whilst the differences on plateau load remain below 10 %. The micro-buckling phenomenon of the structure is captured in Figure 11(c), as well.

However, in the present model the honeycomb walls are represented using homogenised material properties. Fibre-matrix micromechanical interactions, discrete fibre fracture and layer-wise additive manufacturing defects are neglected. Consequently, the simulations exhibit fewer fibre fragments and more continuous damage patterns than those observed experimentally. Nevertheless, the model accurately reproduces the global structural response of the CCF/PLA honeycombs, including the initial stiffness, peak load, post-peak softening and overall crushing behaviour (Figures 11c and d). This indicates that the model is suitable for capturing the macroscopic deformation and progressive collapse behaviour of the integrated structure [57, 58].

For the CCF/epoxy honeycomb structures (Figures 11e and f), the simulations also reproduce the key deformation stages, including the elastic response, sudden load drop, and subsequent plateau region. The discrepancies between the predicted and experimental mechanical responses remain approximately 10-11 %. Figures 11(e) and (f) show the ply splitting and fibre fragmentation phenomenon of the structure, which are reasonably captured using the Abq_Ply_Fabric model. Overall, the FE models provide reliable predictions of the global mechanical response and dominant failure mechanisms of the battery protective system under crush compression.

5.2 The FE result of battery and battery with honeycomb structure

Figure 12 shows strong agreement between the experimental and simulated load-displacement curves for both the battery alone and the battery coupled with an SCF/PLA honeycomb structure. In Figure 12(a), the battery undergoes a linear elastic deformation, followed by a sharp load drop and steady plateau decline until the ultimate failure. Due to the simplifications inherent in the simulation model developed, the current numerical outputs

primarily indicate the agreement with the corresponding experimental results, in terms of overall profiles of load-displacement curves and damage evolutions of the structures. In numerical terms, the simulated peak load deviates from the experimental value by 20%, and the simulated energy absorption deviates by 5%. Damage patterns, including buckling and collapse, are consistent in both. Figure 12(b) shows that for the battery-structure system the simulation captures the buckling and failure of the honeycomb reasonably well, with almost identical initial stiffness and peak load. These results confirm the reliability of the simulation in predicting the mechanical response and failure behaviour of the battery and its integrated structure.

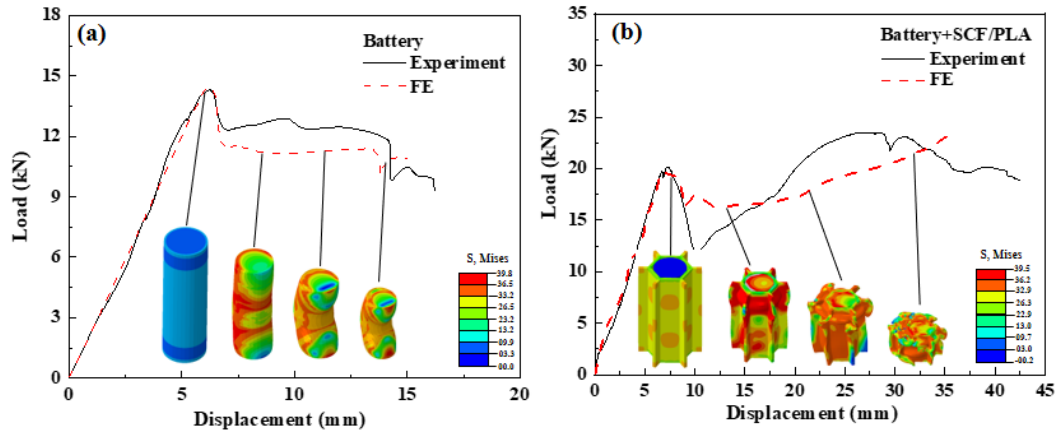


Figure 12. The load-displacement curves and the predicted failure modes following crush tests on a 21700 battery (a) and SCF/PLA composites (b) (the solid lines correspond to test data and the dashed lines to the FE predictions).

5.3 Result of the honeycomb-battery integrated structures with different heights

Finite element simulations were conducted to evaluate the mechanical interaction between the honeycomb structure and the cylindrical battery at different structural heights (72-80 mm), as shown in Figure 13. For the 3D-printed SCF/PLA honeycombs (Figure 13a), when the structural height of 72 mm is close to the battery height of 70 mm, the structure and battery deform cooperatively, resulting in the highest peak load of 23.08 kN. As the structural height increases, the protective structure collapses earlier than the battery, leading to a gradual reduction in peak load and energy absorption. Figure 13(c) further shows that the optimal structural matching height lies between 72-74 mm. in terms of peak load. Although the peak load at 74 mm is similar to that at 72 mm, the energy

absorption at 72 mm (Figure 13d) is about 3.2% higher, indicating a more effective structural matching condition.

A similar trend is observed for the CCF/PLA honeycombs (Figure 13b). At larger structural heights (e.g., 78-80 mm), premature structural failure reduces the load-sharing capability between the honeycomb and the battery, resulting in decreased peak load and energy absorption. Overall, the results indicate that a structural height of 72 mm, slightly higher than the battery height, provides the optimal mechanical compatibility and maximizes the load-bearing capacity and energy absorption of the integrated system.

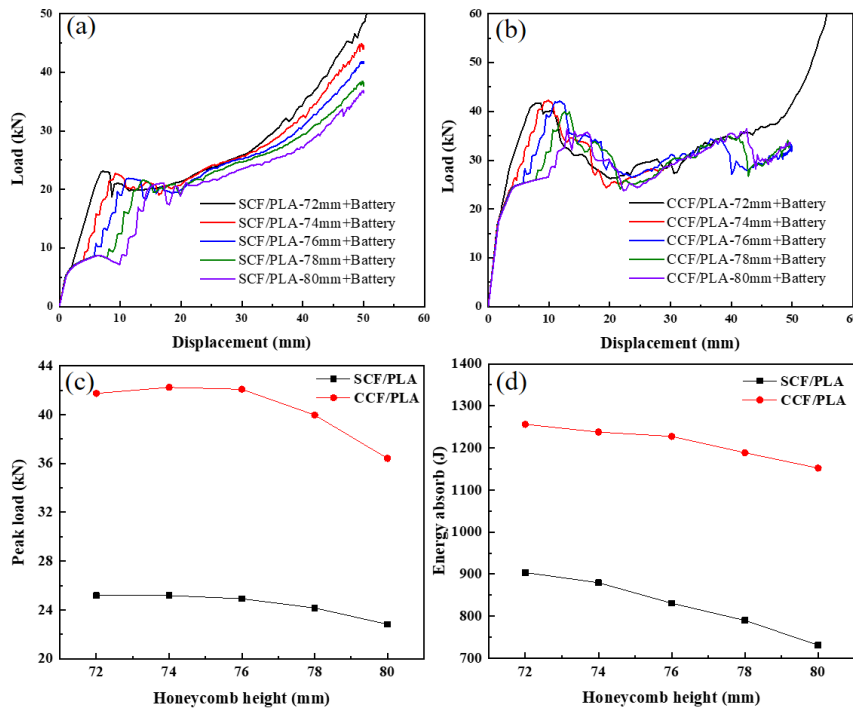


Figure 13. Mechanical responses of integrated honeycomb-battery systems with varying honeycomb heights: (a-b) load-displacement curves of 3D-printed SCF/PLA and CCF/PLA honeycombs, and (c-d) corresponding peak load and energy absorption.

To clarify the mechanical compatibility between the protective structure and the battery, a quantitative analysis was performed. As shown in Figure 14(a), the protective structure is designed to be slightly taller than the battery, allowing it to undergo elastic deformation and provide buffering before battery failure. The load-bearing enhancement of the integrated system therefore depends on the height difference between the two components.

As illustrated in Figure 14(b), the energy absorption behaviour during compression was evaluated using an analytical model. Based on experimental and simulation results, a semi-empirical relationship was established to

quantify the energy absorption of both the structure and the battery. With the fibre volume fraction and reinforcement distribution being kept constant across all specimens, the reinforcement phase density is not explicitly included as a variable. It should be noted that the present study does not aim to perform a global optimization for the battery-structure system. Instead, the novelty of this study is to identify the mechanical compatibility between the honeycomb structure and the cylindrical battery within the design range investigated. The empirical relations were calibrated using the results from four height configurations, whilst the remaining configurations were used as validation cases to assess the predictive capability of the proposed expressions. The resulting expressions are given as follows:

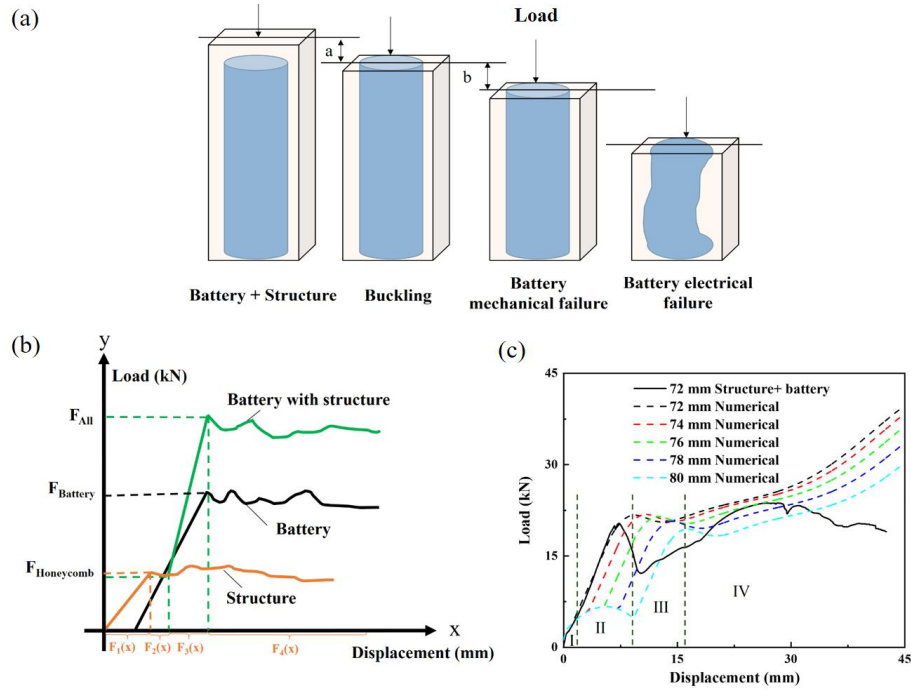


Figure 14. (a) Schematic diagram of structure and battery loading process, (b) Matching design with mechanics, (c) Theoretical calculation results at different heights.

$$E_{all} = \int_0^x F(x)dx, (F(x)=F_{all}) \quad (4)$$

$$F_1(x) = a * x + b \quad (0 \leq x \leq 1) \quad a=4.63 \quad b=0.013 \quad (5)$$

$$F_2(x) = a * x^2 + bx + c \quad (1 \leq x \leq 10 - db) \quad a=-0.1405, b=1.7577, c=3.1794 \quad (6)$$

$$F_3(x) = a * x + b \quad (10 - db \leq x \leq (14 - db) - (dely - dy) / 2.8) \quad a=2.8 \quad b=-21.05 \quad (7)$$

$$F_4(x) = a_1 * \exp(-((x - b_1) / c_1)^2) + a_2 * \exp(-((x - b_2) / c_2)^2) + a_3 * \exp(-((x - b_3) / c_3)^2)$$

$$(x \geq (14 - db) - (delty - dy) / 2.8) \quad a_1 = 62.089, \quad b_1 = 49.1409, \quad c_1 = 31.1435, \quad a_2 = -30.0051, \quad b_2 = 42.3597, \quad c_2 = 18.4386, \\ a_3 = 4.9067, \quad b_3 = 15.9977, \quad c_3 = 3.5338. \quad (8)$$

The y-direction translation dy is given by:

$$dy = -0.568 \cdot db^2 + 0.741 \cdot db + 0.0308 \quad (9)$$

The auxiliary x-direction translation $delty$ is given by:

$$delty = -0.1189 \cdot (8 - db)^2 + 1.4962 \cdot (8 - db) + 3.8481 \quad (10)$$

where db represents the height difference between the honeycomb and the battery.

As shown in the Figure14, $F_1(x)$ represents the expression for the elastic stage of the honeycomb structure, while $F_2(x)$ corresponds to its plastic stage. $F_3(x)$ denotes the expression for the combined elastic response of the structure and the battery, and $F_4(x)$ describes their combined plastic response.

6. Conclusions

In this study, honeycomb structure-battery packs have been fabricated using carbon fibre composite materials through 3D printing and vacuum bag moulding processes. In terms of specific energy absorption (SEA), the vacuum-bag-moulded CCF/epoxy honeycomb has exhibited the highest SEA (38.25 J/g), the 3D-printed CCF/PLA honeycomb has showed the highest compressive strength (18.61 MPa). The mechanical interaction between the honeycomb structure and the battery has been successfully analysed through experiments and finite element simulations. The results indicate that the honeycomb structure can provide initial protection through sacrificial deformation before the battery reaches its failure stage, followed by a coupled load-bearing response of the integrated system.

Using the validated FE model, parametric FE analyses have been conducted to evaluate the influence of structural height on the integrated performance. An optimal matching height is identified for both SCF/PLA and CCF/PLA honeycombs, with a balance being obtained between peak load and energy absorption. Overall, the

results highlight the importance of deformation compatibility and load-transfer mechanisms in governing the protective performance of honeycomb battery-pack structures. It provides a mechanics-based guidance for the design of integrated lightweight battery systems.

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Appendix.

Mechanical characterisation tests are carried out on SCF/PLA composite specimens manufactured under identical printing conditions. The composites exhibited a tensile strength of 32.40 MPa and a tensile modulus of 2.94 GPa. Under compressive loading, the measured strength and modulus are 47.63 MPa and 1.51 GPa, respectively. The corresponding flexural strength and flexural modulus are found to be 84.54 MPa and 6.32 GPa, whilst the shear strength and shear modulus reached 20.74 MPa and 0.30 GPa, respectively (Figure A1).

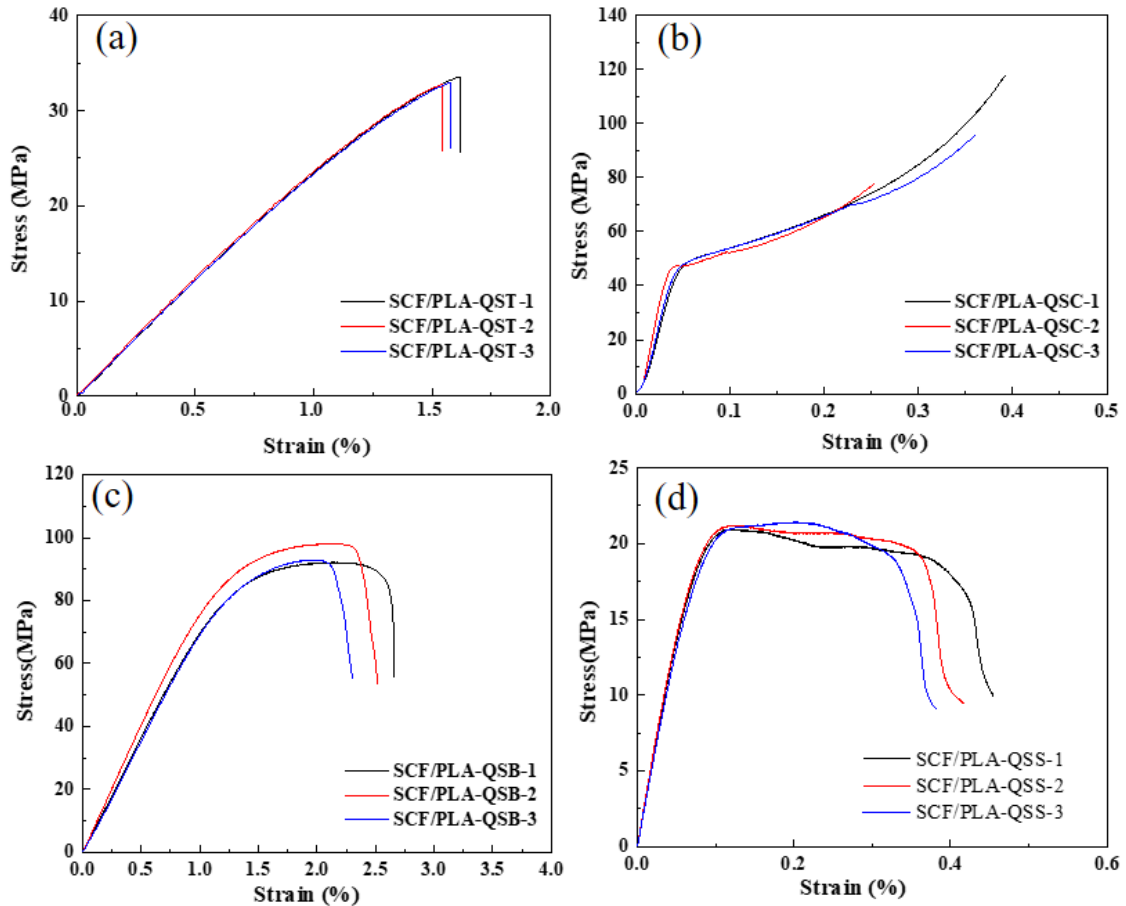


Figure A1. Mechanical test results of SCF/PLA in (a) tension, (b) compression, (c) bending and (d) shearing.

According to experimental measurements conducted on specimens fabricated in the identical printing conditions, the tensile strength and modulus of the CCF/PLA composites are determined as 333.46 MPa and 21.97 GPa, respectively. The measured flexural strength and modulus are 273.37 MPa and 16 GPa, respectively, whilst the shear strength is 32.47 MPa (Figure A2).

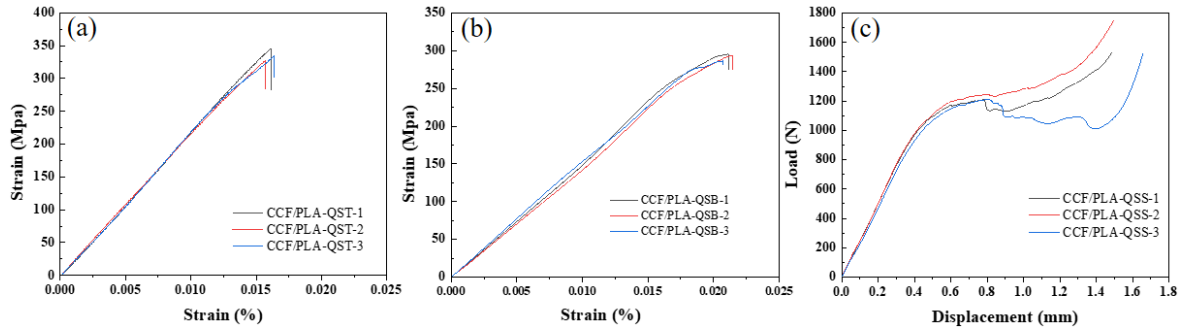


Figure A2. Mechanical test results of CCF/PLA in (a) tension, (b) bending and (c) shearing.

In order to prove the mesh independence of the proposed damage model, three mesh sizes (0.5, 1 and 1.5 mm) were used to investigate the mesh sensitivity in the honeycomb structure. As illustrated in Figure A3(b) (Appendix), the simulations with mesh sizes of 0.5 mm and 1 mm show good agreement with the experimental peak load, indicating that the numerical predictions are relatively insensitive to further mesh refinement. In contrast, the model with a mesh size of 1.5 mm significantly underestimates the peak load, as the coarse mesh cannot adequately capture the local deformation and damage evolution of the honeycomb cell walls.

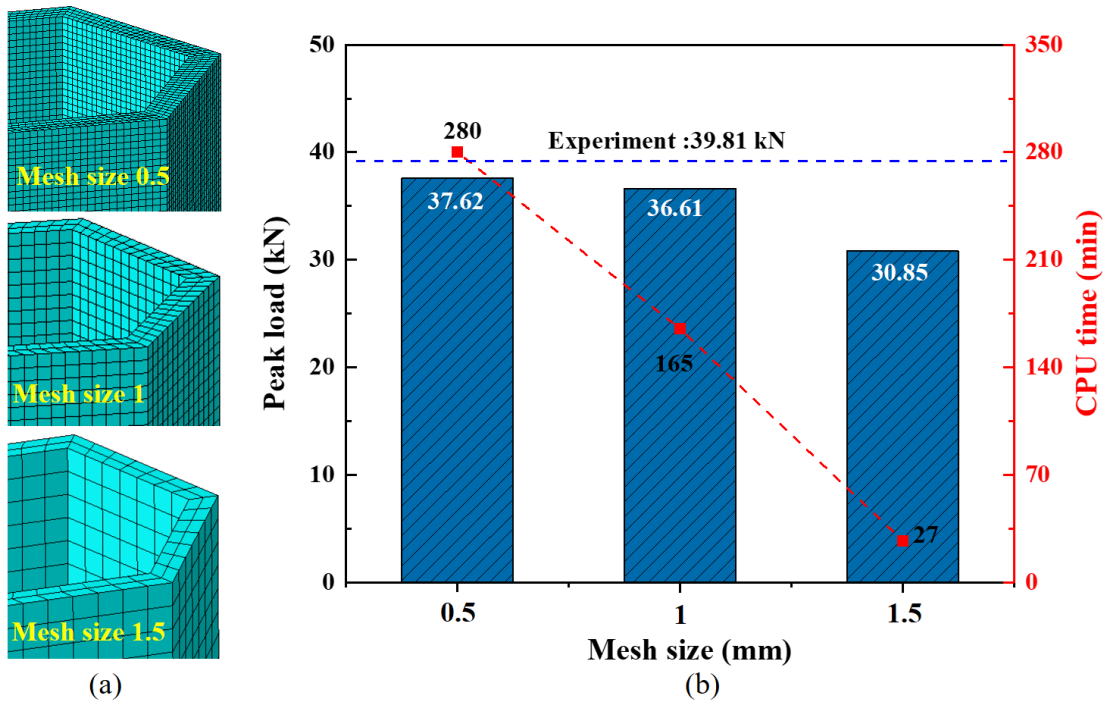


Figure A3. Mesh geometries in mesh sensitivity study: (a) specimens for mesh geometries and (b) corresponding peak load histories obtained from quasi-static compress test.

In addition, the computational efficiency was evaluated. The CPU time for mesh sizes of 0.5, 1 and 1.5 mm was approximately 280, 165 and 27 min, respectively. Considering both prediction accuracy and computational

1 648 efficiency, a mesh size of 1 mm was selected for the subsequent simulations. This sensitivity analysis demonstrates
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4 649 that the selected mesh size provides a reasonable balance between the computational cost and accuracy.
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The energy-absorbing behaviour of fibre reinforced honeycombs for battery pack structures

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Abstract

This paper presents an experimental and numerical investigation on the energy absorption characteristics of lightweight battery packs fabricated from short or continuous carbon fibre (SCF/CCF) reinforced honeycomb composites. The thermoplastic short and continuous carbon fibre reinforced composite structures were produced via 3D printing. The thermosetting continuous carbon fibre reinforced composite (CCF/Epoxy) structures were manufactured through vacuum bag moulding (VBM). These two types of structures were then compared to evaluate their energy absorption performances and failure mechanisms under axial loading conditions. Test results indicate that the honeycomb-structured battery pack made of 3D-printed continuous carbon fibre-reinforced thermoplastic (CCF/PLA) exhibits a higher compressive strength than CCF/Epoxy. For 1-cell configuration, CCF/PLA gives 36.53 MPa vs. 26.75 MPa for CCF/Epoxy; for 8-cell configuration, the values are 18.60 MPa and 15.95 MPa, respectively. Additionally, the specific energy absorption (SEA) values for CCF/PLA are 18.20 J/g (1-cell) and 24.10 J/g (8-cell). In contrast, thermosetting CCF/Epoxy composites demonstrate a superior SEA performance, with values of 31.22 J/g (1-cell) and 38.24 J/g (8-cell). Subsequently, a finite element (FE) model was developed to simulate the mechanical behaviour of the composite specimens subjected to axial loading. It shows a good agreement between numerical predictions and experimental results. Furthermore, the synergistic mechanical matching relationship between honeycomb structures with varying heights and battery cell configurations was predicted through numerical simulations. Quasi-static experimental and numerical simulation

1 28 results provide a design guidance to improve the load bearing capacity.

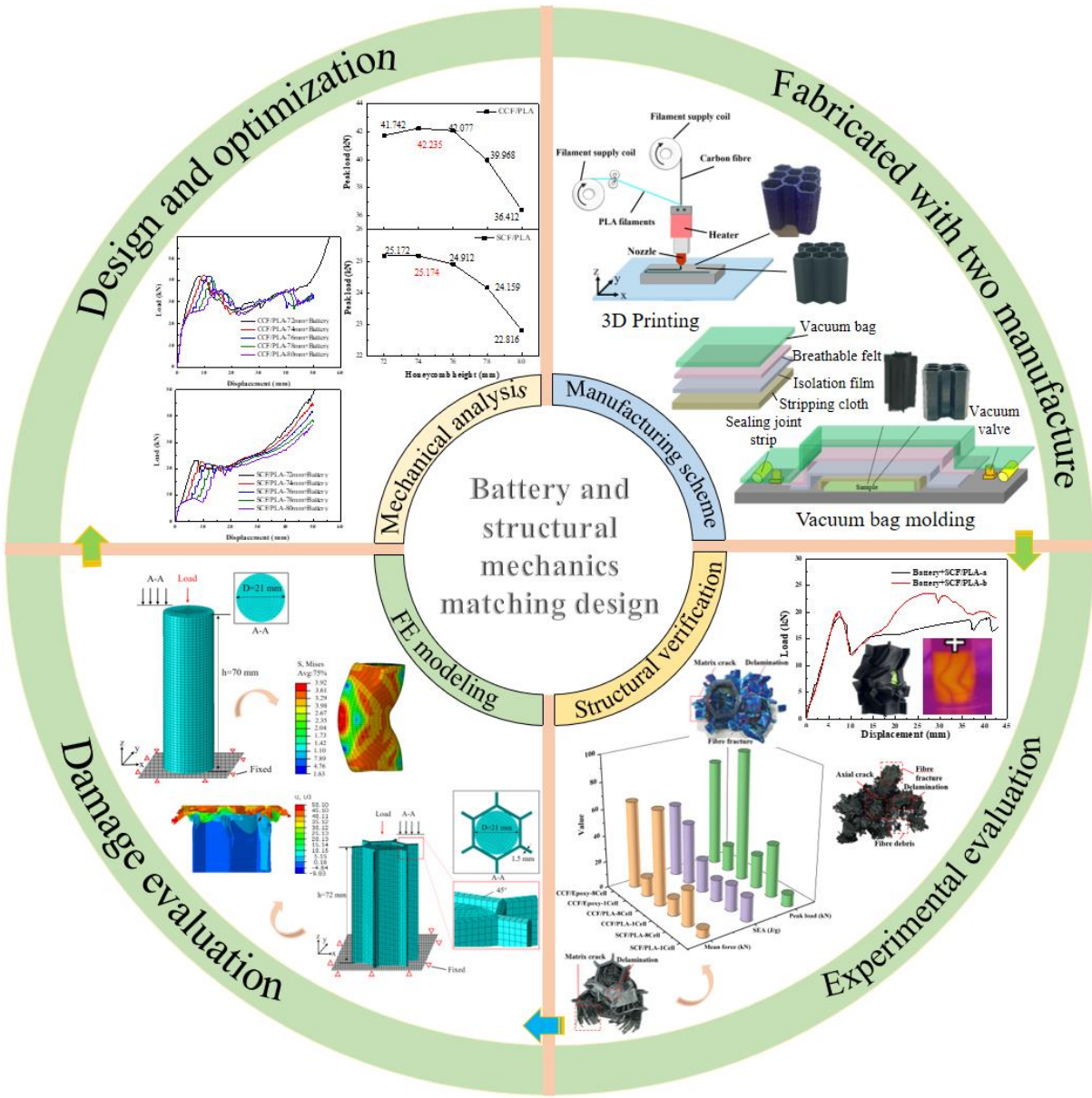
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6 30 **Keywords:** 3D-printed; Energy absorption; Honeycomb structures; Battery pack; Load bearing capacity.

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Graphical abstract



1. Introduction

Batteries serve as the heart of electric aircraft and are essential for hybrid/electric propulsion. However, current battery technology limits electric aircraft to small airframes with very limited range, making battery pack weight a critical constraint^[1]. Recent advances in electric aircraft largely rely on lightweight structural materials and improved power systems^[2-4]. Reducing battery pack weight lowers energy consumption and allows more batteries to be installed without increasing drone mass, which is vital for enhancing electric aircraft performance.^[5, 6]. Despite its importance, research on aerospace battery systems remains limited, whereas significant progress in EV battery pack design has provided valuable insights into structural concepts and material selection^[7-10]. Commercial battery packs are primarily made of aluminum, steel, or carbon fibre reinforced plastic (CFRP)^[11-13]. Compared with aluminum, a CFRP cell shell is about 40 % lighter^[14-16] and the thermal conductivity is about 200 times lower, which can better protect the battery from cold and heat.

Numerous studies have focused on lightweight battery packs^[17-20]. For automotive applications, a high-strength, crash-resistant enclosure must address safety issues such as internal fire or explosion^[21, 22]. Over the past two decades, modelling of the battery pack has advanced toward more efficient management systems^[23, 24]. Pana et al.^[25] experimentally and numerically investigated a battery pack with a high-strength steel base and a CFRP outer shell, reporting a 6.6% weight reduction compared with an all-steel enclosure. Liang^[26] and Greve et al.^[27] used finite element simulations to evaluate the impact performance of battery and pack. Cellular and honeycomb structures have gained increasing attention to battery protection, due to their high specific strength, excellent energy absorption, and design flexibility^[28-31]. Unlich et al.^[32] proposed that the overall performance of hexagonal battery packs is superior to other test geometries. Sinz et al.^[33] designed a replaceable battery pack for electric buses and recommended adding beams beneath it to reduce intrusion and acceleration peaks for better protection. Other studies have explored composite materials in battery systems. Pan et al.^[34] studied the impact

resistance of battery packs through computer simulations. Wang et al. [35] investigated a novel battery pack system consisting of a non-modular battery pack (CTP) and two tubular structures with negative Poisson's ratio (NPR). Han et al. [36] tested carbon/glass hybrid composite honeycomb enclosures under flexural and impact loads, with detailed damage analysis. Dong et al. [37] focused on structural battery composites, examining their out-of-plane compressive behaviour and electrochemical stability under compressive stress of 10 MPa. Samir et al. [38] fabricated non-prismatic reinforced honeycomb (NPRH) sandwich structures via fused filament fabrication and conducted multi-directional quasi-static compression tests, reporting high specific energy absorption.

Limited research exists on the structural design of de-modular battery packs for improved protection. In electric aircraft, conventional wing fuel tanks could be replaced by battery packs integrated into honeycomb sandwich wing panels using unitized packaging technology (UPT). This enables both lightweight design and enhanced safety. To fill this gap, the current study carried out fabrication of thermoplastic short-carbon-fibre and continuous-carbon-fibre honeycomb structures via 3D printing and vacuum bag molding. The corresponding quasi-static crush energy absorption was evaluated through experimental and numerical investigations. Furthermore, the load-displacement responses were analysed to assess mechanical interaction and deformation compatibility between the battery and the protective structure.

2. Experimental procedure

2.1 Specimen preparation

2.1.1 3D printed short carbon fibre honeycomb structures

All structures were manufactured according to Fused Deposition Modeling (FDM) method, as shown in Figure 1(a). For modeling specimen's geometry, Creality Slicer Software was used, then model was exported as STL file from the software and imported to the 3D printing software for printing preparation. ENDER-3MAX 3D

printer was used to print SCF/PLA (Carbon Fibre/Polylactic Acid) specimens. This printer was selected because of its simplicity and ease of use. The diameter of SCF/PLA printing wire is 1.75 mm, and the printing adopts standard wire feeding method, which is fed into the printing head through (Polytetrafluoroethylene) PTFE tube. All the printing parameters and dimensions of 3D printed specimens are presented in Table 1. The actual fibre mass (%) in SCF/PLA filaments is 8-10%, the fibre volume fraction ranging from 5.5% to 7.0% is exhibited by these composites.

Table 1. Short carbon fibre-reinforced polylactic acid composite 3D printing parameters.

Printing parameter	Value	Units
Filament diameter	1.75	mm
Nozzle diameter	0.4	mm
Extrusion/ Nozzle temperature	220	°C
Nozzle	60	°C
Layer height	0.2	mm
infill	100	%
Extrusion speed	150	mm/s
Infill pattern	Tetrahedral	-
Outer shell	0.6	mm

2.1.2 3D printed continuous carbon fibre honeycomb structures

As shown in Figure 1(a), continuous PLA wire and carbon fibre bundle mix in the guide tube through heating the PLA resin to the melting temperature. Both materials are subsequently delivered into the heating zone, where the molten PLA impregnates the carbon fibre bundle in-situ. The resulting impregnated composite filament is then extruded through the nozzle and deposited to form the printed structure. The heat sink at the entrance of guide tube is to ensure the solid state of PLA wire. To uniformly mix carbon fibre and PLA, a conical nozzle is employed. During the printing process, the already extruded carbon fibre reinforced PLA filaments stick on the panel and provide persistent traction load to pull the carbon fibre clad by PLA out of the nozzle. The printing spacing was set to 1 mm, a printing speed to 150 mm/min, a layer height to 0.3 mm, and a nozzle temperature of 220 °C. The fibre volume fraction of the structure in this study was determined to be 17 %. The printing path of the 3D-

printed CCF/PLA composite is presented in Figure 1(a). The as-printed material exhibits a good impregnation quality, with uniform resin distribution around the fibres and no significant fibre exposure. Due to weakened fibre fixation after resin melting, pull-back phenomena are observed (approximately 20% angular deviation at corners and 45% at path reversals), which are compensated by reserved path design.

2.1.3 Continuous carbon fibre reinforced thermosetting honeycomb structures

Thermosetting continuous carbon fibre reinforced CCF/Epoxy composites are prepared by vacuum bag moulding (VBM) method, as shown in Figure 1(b). The composite material system used in this study was a unidirectional T700/USN12500 CFRP prepreg carbon/epoxy tape (Guangwei Co., Ltd, China) with a nominal cured ply thickness of 0.125 mm and a 12 K tow. The symmetric quasi-angle lay-up $[45/-45/0/-45/45]_s$ was adopted to balance axial stiffness and shear-dominated progressive failure, since $\pm 45^\circ$ plies induce progressive damage and enhance energy absorption capacity under crushing loads^[39]. Following the manufacturer's guidelines, a vacuum pressure of approximately 0.1 MPa was applied, and the initial tow spacing (0.2-1.0 mm) was reduced to approximately 0.05-0.5 mm after consolidation. Also, the curing cycle was performed with a temperature ramping rate of 2 °C/min to 80 °C, maintained for 60 minutes, followed by a second temperature rise to 120 °C, which was held for 90 minutes. Under this process, the fibre content of the CCF/Epoxy composite material can reach approximately 50%.

Honeycomb structure was prepared using nylon moulds consisting of hexagonal blocks, which were made by 3D printing. Here, a total of 8 hexagonal-shaped nylon blocks, with a height of 100 mm and a face-to-face dimension of 22 mm, were inserted into the mould. The manufacturing processes for the honeycomb can be divided into five steps, Firstly, the nylon moulds, with a thermal deformation temperature of 180 °C, were manufactured using the selective laser sintering technique. Unidirectional carbon fibre prepreg tapes were sectioned according

to designated fibre orientations to obtain the required lay-up configurations for subsequent manufacturing. In the second stage, five layers of prepreg tape were wrapped around the surface of each nylon mould. In the third stage, all the moulds were assembled and then five layers of prepreg tape were wrapped around the surface of the assembly.

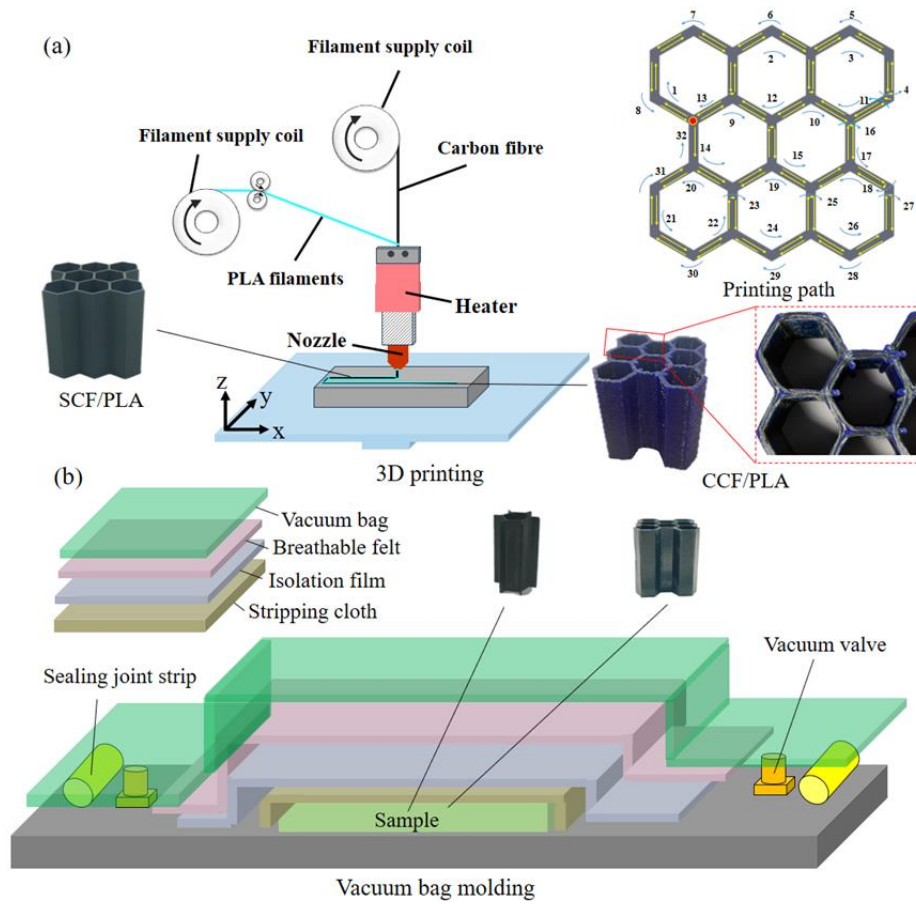


Figure 1. (a) Schematic of the 3D printing process of short/continuous carbon fibre thermoplastic composite, (b) preparation process of continuous carbon fibre thermosetting composite honeycomb structure by vacuum bag moulding method.

As a result, thicknesses of all parts of the honeycomb were the same, equivalent to ten layers of prepreg. Fourthly, the preform was then vacuum bagged in a polymer film and placed in an oven to cure. Finally, following completion of the cure cycle, the moulds were removed from the assembly, with the final product being produced. Lithium battery (produced by Shenzhen Kyoto Electronic Technology Co., LTD.) with a voltage of 3.7 V and capacity of 4800 mA (h) was embedded into the honeycomb. For safety reasons, the residual power of the

cylindrical battery used in the test was less than 25 % of the total capacity, and the bases were superimposed up and down to avoid sliding during the battery crushing test.

Three-dimensional internal images were acquired via X-ray computed tomography (CT), allowing the void content (%) to be quantified. The measured void contents are 2.7% for SCF/PLA (Figure 2a), 3.46% for CCF/PLA (Figure 2b) and 1.32% for CCF/epoxy (Figure 2c). This quantitatively confirms that the filament-based additive manufacturing introduces a higher interfacial porosity compared with vacuum-assisted curing, primarily due to limited inter-bead fusion and fibre impregnation.

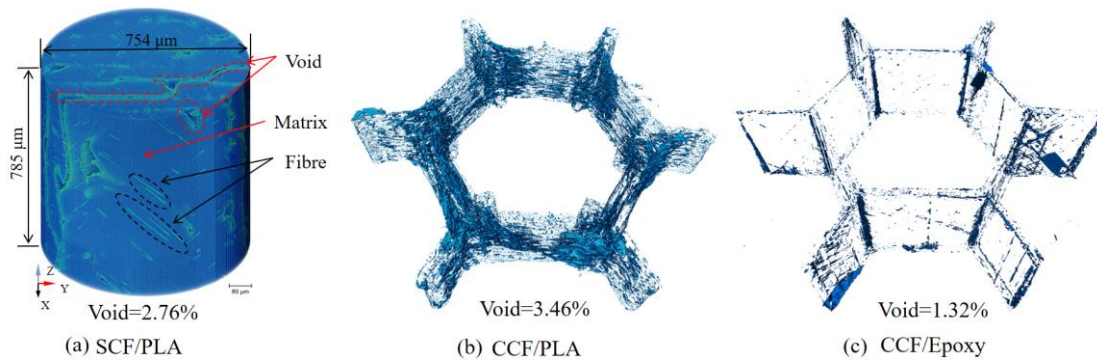


Figure 2. Comparison of void content (%) in honeycomb composite structures manufactured by different processes: (a) 3D-printed SCF/PLA, (b) 3D-printed CCF/PLA, and (c) vacuum bag cured CCF/epoxy, as determined by X-ray computed tomography (CT).

2.2. Quasi-static mechanical tests

The compression strength and energy-absorbing characteristics of the carbon fibre honeycomb structure and cylindrical battery were evaluated at a crosshead displacement rate of 5 mm/min using an MTS universal testing machine (ETM504C) with a load capacity of 200 kN. The two ends of the battery were connected to the voltage acquisition instrument (SIRIUS MINI 4-channel vibration/noise analyzer). The cylindrical battery used was standard cylindrical battery 21700 (Panasonic). The mechanical testing machine used for carbon fibre honeycomb structure and cylindrical battery testing are shown in Figure 3. All quasi-static crushing tests of the lithium-ion cells were conducted in a well-ventilated and open testing environment equipped with remote loading control and

a fire blanket for emergency protection. The state of charge (SOC) of all tested cells was maintained at $(20 \pm 5\%)$. The recorded temperature values represent localised surface heating during deformation rather than a full thermal runaway event.

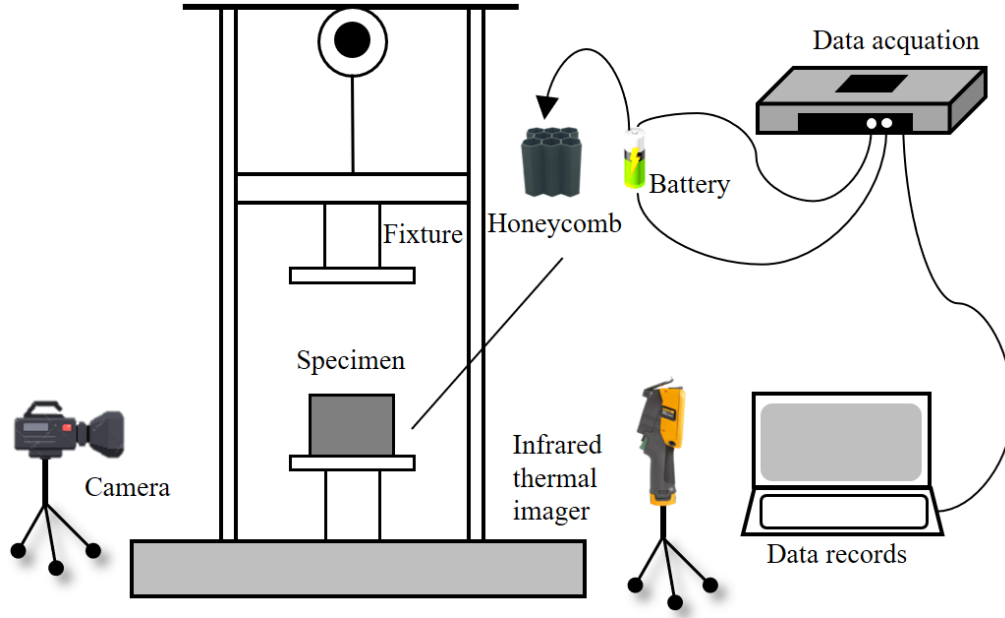


Figure 3. Schematic of the quasi-static compression test setup.

The energy absorbed by the specimen during crushing is equal to the area beneath the load-displacement curve, which can be calculated by Eq. (1):

$$EA = \int_0^l F(x)dz \quad (1)$$

where x is the displacement corresponding to the applied load, ' l ' represents the effective crushing distance and F is an applied load.

Specific energy absorption (SEA) is used to evaluate energy absorbing efficiency, accounting for the lightweight requirement in the design of a good energy absorber. This can be calculated by dividing the measured value of EA by the mass of the damaged part (m), according to Eqs. (2) and (3).

$$SEA = \frac{\int_0^l F(x)dz}{m} \quad (2)$$

$$m = M \cdot \frac{l}{H} \quad (3)$$

where ‘M’ and ‘H’, as indicated before, correspond to the total mass and height of each specimen before test, respectively.

Table 2. Summary of geometries of the honeycomb structure investigated in this study.

Sample	Mass (g)	Length (mm)	Wide (mm)	Height (mm)	Thicknes s (mm)	Loading rate (mm·min ⁻¹)
SCF/PLA-1Cell-a	20.10	41.46	36.41	71.70	2.03±0.11	5.00
SCF/PLA-1Cell-b	20.40	41.51	36.62	71.75	2.04±0.08	5.00
SCF/PLA-1Cell-c	20.20	41.49	36.54	71.72	2.04±0.10	5.00
SCF/PLA-1Cell-d	20.15	41.48	36.63	71.74	2.02±0.08	5.00
SCF/PLA-1Cell-e	20.22	41.52	36.53	71.73	2.03±0.10	5.00
SCF/PLA-8Cell-a	77.30	73.52	71.38	73.50	2.09±0.09	5.00
SCF/PLA-8Cell-b	77.50	73.90	72.34	73.51	2.05±0.11	5.00
SCF/PLA-8Cell-c	77.40	73.87	72.21	73.45	2.07±0.12	5.00
CCF/PLA-1Cell-a	31.60	44.34	38.96	72.07	3.35±0.11	5.00
CCF/PLA-1Cell-b	31.70	44.35	38.89	72.10	3.36±0.13	5.00
CCF/PLA-1Cell-c	31.50	44.34	38.88	72.09	3.36±0.12	5.00
CCF/PLA-8Cell-a	115.00	77.19	73.49	71.72	3.02±0.11	5.00
CCF/PLA-8Cell-b	115.40	77.15	73.70	71.83	3.05±0.10	5.00
CCF/PLA-8Cell-c	115.20	77.16	73.74	71.80	3.03±0.09	5.00
CCF/Epoxy-1Cell-a	20.00	44.24	41.65	75.41	1.36±0.05	5.00
CCF/Epoxy-1Cell-b	20.00	43.04	42.96	73.85	1.55±0.03	5.00
CCF/Epoxy-1Cell-c	20.00	44.04	43.85	76.21	1.32±0.04	5.00
CCF/Epoxy-8Cell-a	77.50	72.29	68.75	75.16	1.75±0.04	5.00
CCF/Epoxy-8Cell-b	73.50	73.26	69.49	74.62	1.45±0.03	5.00
CCF/Epoxy-8Cell-c	77.20	73.16	68.48	76.01	1.55±0.02	5.00

Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated up to

the onset of densification in the load-displacement curve to ensure a consistent comparison. Three repeated tests were carried out for each test condition to obtain an average value, the geometries of all structures are shown in Table 2.

3. Numerical modelling

3.1. Modelling of honeycomb structure and battery

3.1.1 Modelling of 3D printed short carbon fibre reinforced composite honeycomb structure

A high-fidelity FE model consisting of the 1-Cell and 8-Cell honeycomb structure was constructed in Abaqus/Explicit, as shown in Figures 4(a), (a1).

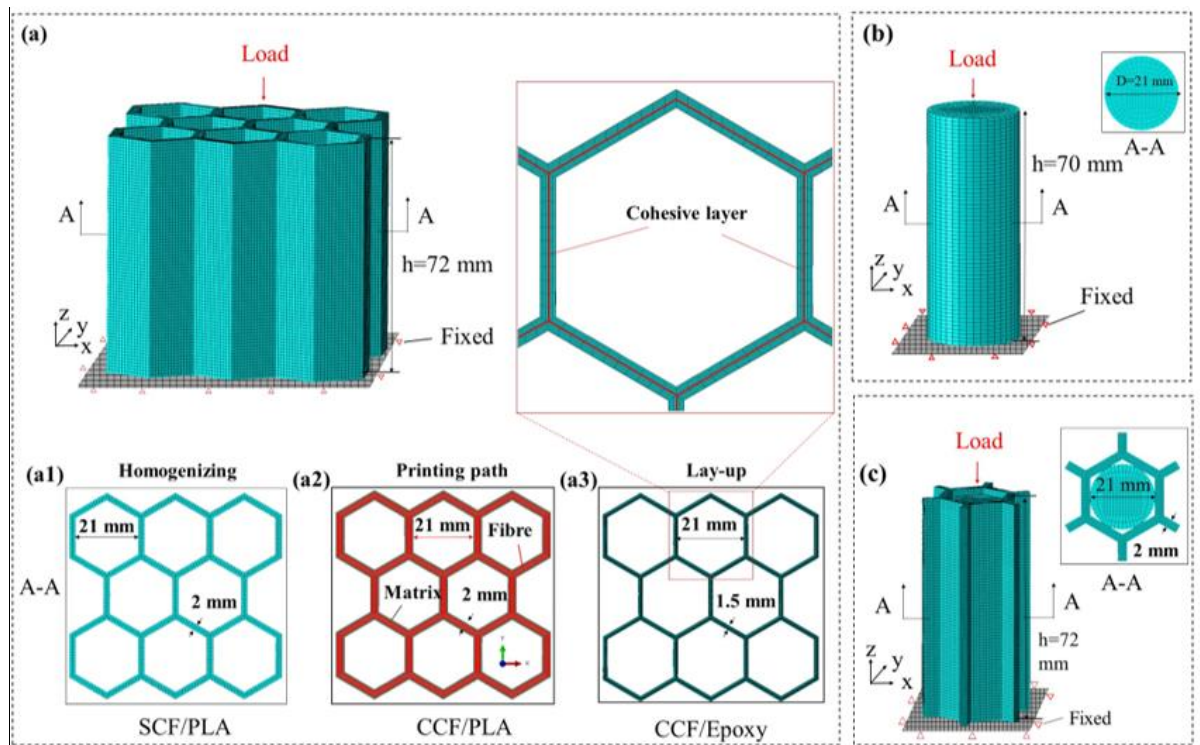


Figure 4. (a) Full-scale FE model of honeycomb of different manufactures (a1, 3D printing thermoplastic short carbon fibre (SCF/PLA) composites, a2, 3D printing continuous carbon fibre (CCF/PLA) composites, a3, vacuum bag moulding method (CCF/Epoxy) composites), (b) the geometry and mesh generation of the FE model for the cylindrical battery.

The height is 72 mm, the inner diameter is 21 mm, and the honeycomb thickness is 2 mm. The top view of

the model is also exhibited. The honeycomb was meshed using 8-noded solid elements with reduced integration (C3D8R) to model the out-of-plane mechanical behaviour. Based on mesh sensitivity studies (Appendix A), a mesh size of 1 mm × 1 mm was adopted, as shown in Figure 4(a1). The interaction properties were applied to the face-to-face contact to feature the tangential behaviour with a friction penalty of 0.2 ^[40] and hard normal contact to prevent self-penetration of elements. The material properties are listed in Table 3. Among them, the stress triaxiality of -0.33 was calculated to represent the compressive shear dominated stress state during honeycomb wall buckling under quasi-static compression.

Table 3. Properties of short carbon fibre composite honeycomb ^[40].

Parameter		Value	Units
Density		1.35	g/cm ³
Young's Modulus		2294	MPa
Poisson's Ratio		0.3	/
Ductile damage (Damage evolution)	Fracture strain	1.45	%
	Stress triaxiality	-0.33	/
	Strain rate	0.000656	s ⁻¹
	Failure displacement	0.14143	mm
Shear damage (Damage evolution)	Fracture strain	1.35	%
	Shear stress ratio	1.6	/
	Strain rate	0.00416	s ⁻¹
	Failure displacement	1.1343	mm

3.1.2 Modelling of 3D printed continuous carbon fibre reinforced composite honeycomb structure

The finite element model of 3D printed continuous carbon fibre reinforced composite honeycomb structure is established using the commercial finite element software ABAQUS/Explicit, to simulate the compressive behaviour of the honeycomb structure. The established FE models of 1-cell and 8-cell honeycombs are shown in Figures 4(a) and (a2), and their finite element models with structural geometric parameters are established by combining the printing path, material properties of matrix and continuous carbon fibre, as shown in Table 4. When establishing the FE model, PLA is defined as an isotropic material, and its elastic and plastic parameters are obtained from the previous experimental study ^[41].

To improve computational efficiency, a simplified numerical model was adopted, in which the curvature of real fibres, intermittent deposition, and interlayer defects were neglected. Also, the printed continuous fibre composite was treated using homogeneous anisotropic material properties ^[42], where their stiffness and strength characteristics along three main directions were incorporated through homogenised orthotropic properties.

Table 4. Basic mechanical parameters of continuous carbon fibre bindle and PLA ^[41].

Parameter	Tensile strength/MPa	Tensile modulus/GPa	Density/g·cm ⁻³
Fibre bindle	4100	240	1.78
PLA filament	62.63	3.20	1.24

Consequently, this modeling framework is suitable for predicting the overall structural response and crushing trends, but is not intended for a detailed description of micro-scale failure mechanisms. The matrix and fibre bundle parts are also meshed by eight-noded hexahedral linear element with reduced integration (C3D8R). The same contact properties as described in the previous section are used for both tangential and normal interaction. Face-to-face contact is also applied between the fibre bundle and matrix parts. A displacement control boundary condition is applied on the top rigid surface. All degrees of freedom (Translational and rotational ones) on the bottom rigid surface and rotational degrees of freedom on the top rigid surface are constrained (Figure 4a).

3.1.3 Modelling of vacuum bag mould continuous carbon fibre reinforced composite honeycomb structure

The simulations were conducted by using the explicit solver in Abaqus/Explicit. The mesh, geometry and boundary conditions of the FE model are shown in Figures 4(a) and 4(a3). Composite samples have a layup of [45/-45/0/-45/45]_s and dimensions of the inner diameter of 21 mm, the height of 72 mm, and the cell thickness of 1.5 mm. A user-defined material model (VUMAT) for fabric reinforced composite materials, namely “ABQ_PLY_FABRIC”, was adopted to define the intralaminar material. The continuum shells were individually meshed using 8-noded quadrilateral stress/displacement continuum shell elements with reduced integration (SC8R)

^[43]. Each shell represents one fibre ply with a thickness of 0.15 mm. A regular mesh size of 1 mm × 1 mm was employed after a mesh sensitivity study (Appendix A). There are 66528 elements for 1-cell and 144000 elements for 8-cell structures used in the numerical analysis. A 45° chamfer triggering was modelled in order to ensure a stable and progressive collapse of the structure. The model was positioned between two steel plates that were modelled as discrete rigid shell plates. Based on the traction-separation law, the bonding behaviour was defined. As shown in Figure 4(a3), zero-thickness 8-node cohesive elements (COH3D8) were employed to model the inter-laminar interfaces of the thermosetting composite laminates (the red line). The bottom plate was fixed, whilst constant displacement loading was assigned to the top plate.

The elastic and strength properties adopted in the numerical simulations were provided by the material manufacturers and the remaining material properties were taken from Ref. ^[44]. The material properties used in this study are shown in Table 5.

Table 5. Mechanical properties of honeycomb composite ^[44].

Properties	Values	Units
Density	$\rho=1.7$	g/cm^3
Stiffness properties	$E_{11}=130$ $E_{22}=E_{33}=9$	GPa
Poisson's ratio	$\nu_{12+}=0.35$ $\nu_{12-}=0.3$	/
Material strengths	$X_{1+}=1954.1$ $X_{1-}=2000$ $Y_{2+}=90$ $Y_{2-}=724$ $S=119$	MPa
Critical fracture energy	$G_f^{1+}=280$ $G_f^{1-}=200$ $G_f^{2+}=19$ $G_f^{2-}=31.5$	kJ/m^2

During the fracture process, open-section composite structures may exhibit multiple failure modes, including intralaminar shear, axial cracking, and transverse delamination. These modes result in spreading or petal-like

damage patterns. To fully represent these failure modes, detailed numerical analysis methods capable of capturing them are required. Single-shell modeling approaches cannot capture failure modes ^[45]. The interlayer cohesive properties are shown in Table 6 ^[46].

Table 6. Inter-laminar properties defined for the composite cohesive element layer shell interfaces ^[46].

Properties	Values	Units
Cohesive strengths	$t_n=100, t_s=t_t=130$	MPa
Initial cohesive stiffness	$K_{nn}=5.4E^6$ $K_{ss}=K_{tt}=4.2E^6$	N/mm ³
Fracture energy	$G_{Ic}=0.554$ $G_{IIc}=G_{IIIc}=2.8116$	N/mm ³
Density	$\rho=1.7$	g/cm ³
Benzeggagh–Kenane exponent	$\eta=2$	/

3.1.4 Modelling of battery

A FE model, representing almost all the details of the battery cell, was established in Abaqus/explicit. Here, cylindrical Lithium-ion 21700 cells (GAIA, HP 602030 NCA-45 A/162 Wh) were chosen to be simulated. The main dimensions and components of the battery are shown in Figure 4(b). The battery shell is made of stainless steel ^[47]. The outer diameter and the axial length of the battery are 21 mm and 70 mm, respectively. The battery was normalised in this model, so that the model parameters correspond to the actual size of the battery studied. All mechanical properties of the battery are shown in Table 7 ^[48].

Table 7. Properties of cylindrical battery ^[48].

Properties	Values	Units
Young's Modulus	8467.56	MPa
Poisson's Ratio	0.3	/
Density	3.5	g/cm ³
Fracture Strain	0.35	/
Stress Triaxiality	-0.33	/
Strain Rate	0.0002	s ⁻¹
Displacement at Failure	10	mm

3.2 Modelling of honeycomb structure with battery

To facilitate the study of the effects of honeycomb structures on battery protection, predictions of compression protection using honeycomb structures of different heights are particularly important. The FE model needs to include both the honeycomb unit cell structure and the battery. Based on the models of the composite honeycomb structure and the battery, we have calculated the mechanical matching between the honeycomb structures with heights of 72, 74, 76, 78, and 80 mm and the battery, to predict and analyze the mechanical matching design between the protective structure and the battery.

4. The test results and discussion

In the initial part of this investigation, a quasi-static compression test with a displacement loading rate of 1 mm/min was carried out to study the mechanical properties of the cylindrical battery. Then, the strength and energy-absorbing characteristics of 1-cell honeycomb samples, based on 3D printing and vacuum bag moulding, were investigated through a series of quasi-static compression tests at a crosshead displacement rate of 5 mm/min. The results of these tests are presented and discussed as follows.

4.1 Results of quasi-static compression test of cylindrical battery

Based on the load-, voltage- and temperature-displacement curves in Figure 5, the battery response can be divided into three stages (Figure 5a). At Stage I (Mechanical deformation), the battery undergoes elastic and plastic deformations until structural buckling occurs, whilst voltage and temperature remain nearly constant. The increase on load contributes to the accumulation of progressive strain energy. At Stage II (Separator deformation), after buckling, the separator with a high failure strain begins to carry the load. The applied load gradually decreases as the separator deforms under continuous platen displacement, whilst the voltage remains around 3.4 V. This stage highlights stress redistribution and progressive damage within the cell. At Stage III (Short circuit and thermal response), once the separator reaches its mechanical limit, a cathode-anode short circuit occurs, leading to a rapid

voltage drop and temperature rise to 129 °C. The accelerated changes are attributed to localised shorting and Joule heating. Further crush results in the ultimate mechanical failure and complete voltage collapse. This stage illustrates the coupling among the structural degradation, electrochemical response and thermal effect.

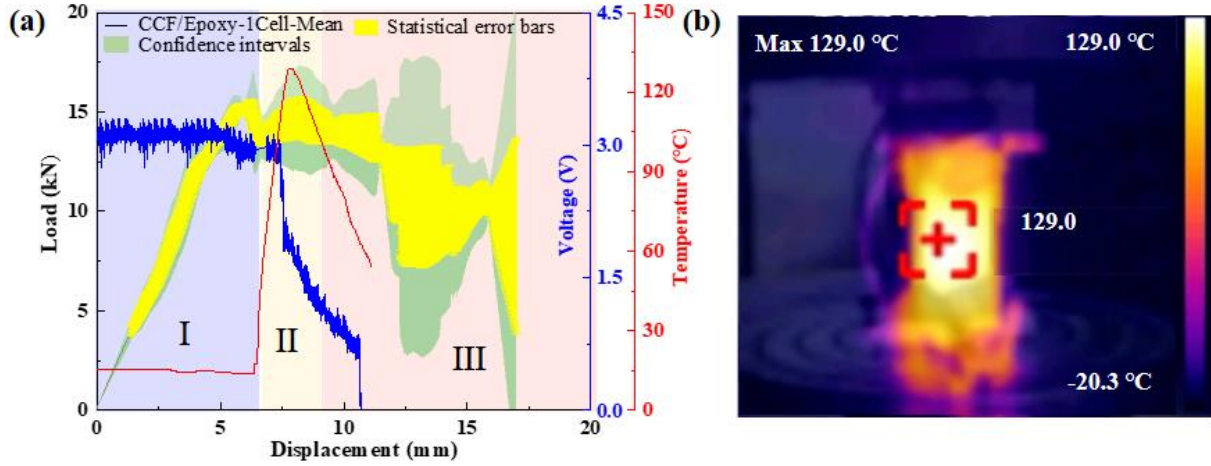


Figure 5. (a) The curves of load, voltage, and battery temperature versus time under quasi-static compression, and (b) the temperature results of the failure center.

4.2 Test results of honeycomb structures made of fibre composites with different manufacturing processes

The compressive response of the three honeycomb structures is illustrated in Figure 6, which depicts both the load-displacement curves and the corresponding progressive failure modes. The densification points are also marked on the curves. To ensure statistical robustness, the curves were analysed to derive the mean, variance, and confidence intervals, as well as the boundary intervals for the total energy absorption. For SCF/PLA honeycombs, both 1-cell (Figure 6a) and 8-cell (Figure 6b) configurations exhibit a rapid increase in load to peak values (8.16 kN and 39.81 kN), followed by a gradual load decrease and a relatively stable crushing plateau. This behaviour is governed by progressive cell-wall buckling and matrix-dominated failure. Damage initiates at stress concentration regions and propagates through matrix cracking and interlayer debonding, whilst the load was subsequently sustained by wall folding and frictional contact between fractured segments. The dominant failure modes are

therefore matrix cracking, wall buckling, and delamination (Figures 7a1, a2), which are the damage characteristics of composite cellular solids under axial loading [49]. The SEA values are 12.88 J/g for the 1-cell and 10.32 J/g for the 8-cell one.

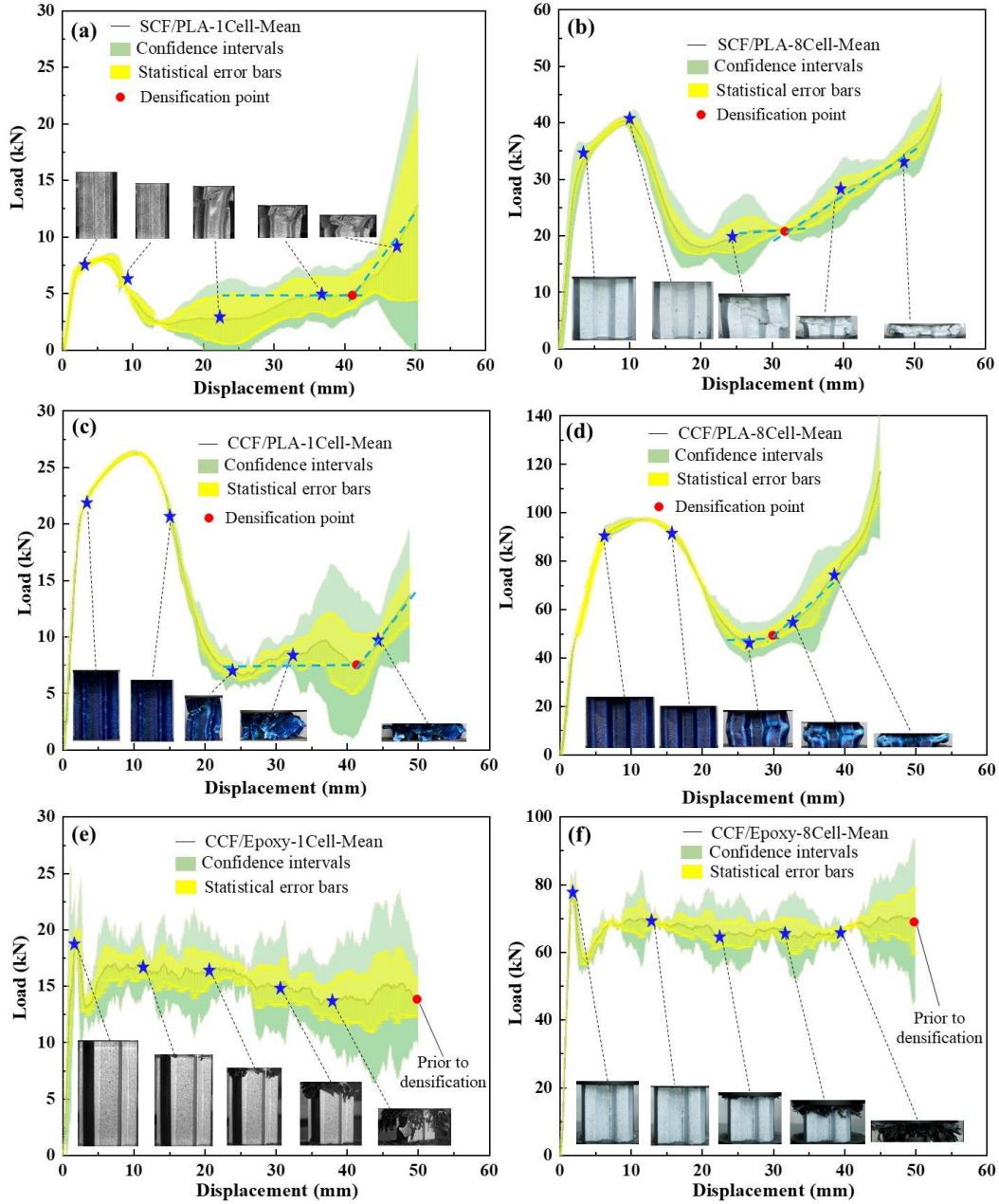


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static crushing tests on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

For the CCF/PLA honeycombs, the peak loads were measured as 25.29 kN for 1-cell (Figure 6c) and 97.68

kN for 8-cell samples (Figure 6d). A more pronounced post-peak load drop is observed. This sharp reduction is mechanistically associated with interface-controlled damage in additively manufactured continuous fibre composites, where limited interlayer fusion and in-situ impregnation promote early delamination and fibre-matrix debonding. After the initial drop of load, a secondary plateau is maintained through progressive fibre fracture, wall buckling, and interfacial sliding. The dominant failure mechanisms include fibre fracture, matrix cracking, and delamination (Figures 7b1, b2), consistent with reported post-peak instability features in continuous fibre AM composites. [50] The 1-cell and 8-cell honeycombs exhibit SEA values of 18.48 J/g and 20.07 J/g, respectively.

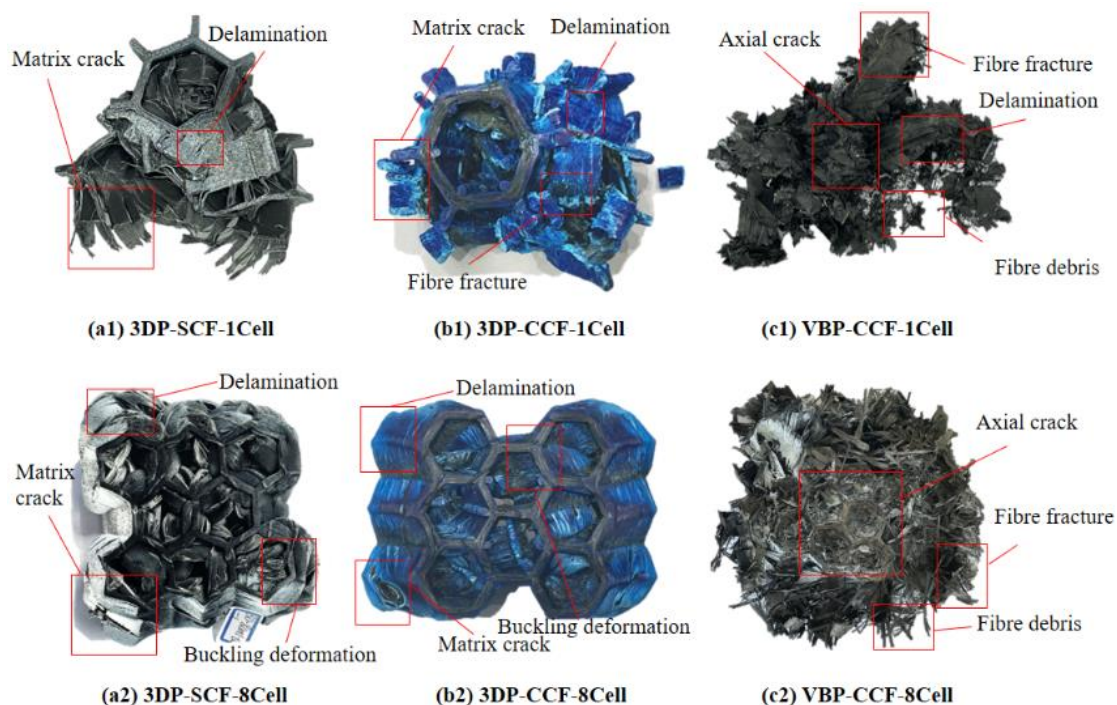


Figure 7. Failure modes of 1-cell (a1 for SCF/PLA, b1 for CCF/PLA, c1 for CCF/Eopxy) and 8-cell (a2 for SCF/PLA, b2 for CCF/PLA, c2 for CCF/Eopxy) honeycomb structures made of different materials after quasi-static compression.

Similar staged crushing behaviour is also observed in vacuum-bag-molded CCF/Epoxy honeycombs, but with a more stable post-peak response (Figures 6e, f). The 1-cell and 8-cell structures reach peak loads of 20.33 kN and 80.18 kN, respectively, followed by sustained plateau loads prior to densification. Failure initiates with matrix cracking and delamination, followed by wall sliding and progressive lamina fragmentation (Figures 7c1, c2).

Interfacial sliding and friction between fractured walls provide a sustained energy dissipation mechanism via friction, which delays collapse locally and directly contributes to the superior SEA (43.27 J/g for the 1-cell and 38.25 J/g for the 8-cell honeycomb) ^[51]. This contrasts with abrupt failure modes (e.g., sudden matrix/fibre fracture) of the previous two materials (SCF/PLA and CCF/PLA).

Table 8. Energy absorption characteristics of honeycomb structures made of 3D-printed short carbon fibre, 3D-printed continuous carbon fibre and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composites, respectively.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
SCF/PLA-1Cell-a	20.10	8.02		11.58		13.62	
SCF/PLA-1Cell-b	20.40	7.97		11.51		10.51	
SCF/PLA-1Cell-c	20.20	8.08	8.02	11.66	11.58	13.01	12.88
SCF/PLA-1Cell-d	20.15	8.10		11.70		13.58	
SCF/PLA-1Cell-e	20.22	7.98		11.50		13.67	
Variance	/	0.032		0.069		1.45	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	
CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

All data on the peak load, compressive strength, energy absorption and specific energy absorption of the

honeycomb structures are listed in Table 8. For 8-cell honeycomb structures made of different materials, it is observed that the 3D-printed CCF/PLA composites provide the highest compressive strength (18.61 MPa), surpassing those of CCF/Epoxy (15.95 MPa) and SCF/PLA (7.58 MPa).

Due to severe fragmentation after crush, the CCF/Epoxy specimens could not be examined by CT scanning. Therefore, post-crush CT analyses were performed on the 3D-printed SCF/PLA and CCF/PLA honeycomb structures. Figure 8(a) presents the overall 3D CT reconstruction of the crushed SCF/PLA honeycomb, which is consistent with the macroscopic observations and reveals interlayer delamination and matrix cracking. Figure 8(a1) shows sectional views from different orientations, which more distinctly illustrate the internal damage features, including interfacial separation, local buckling and fracture.

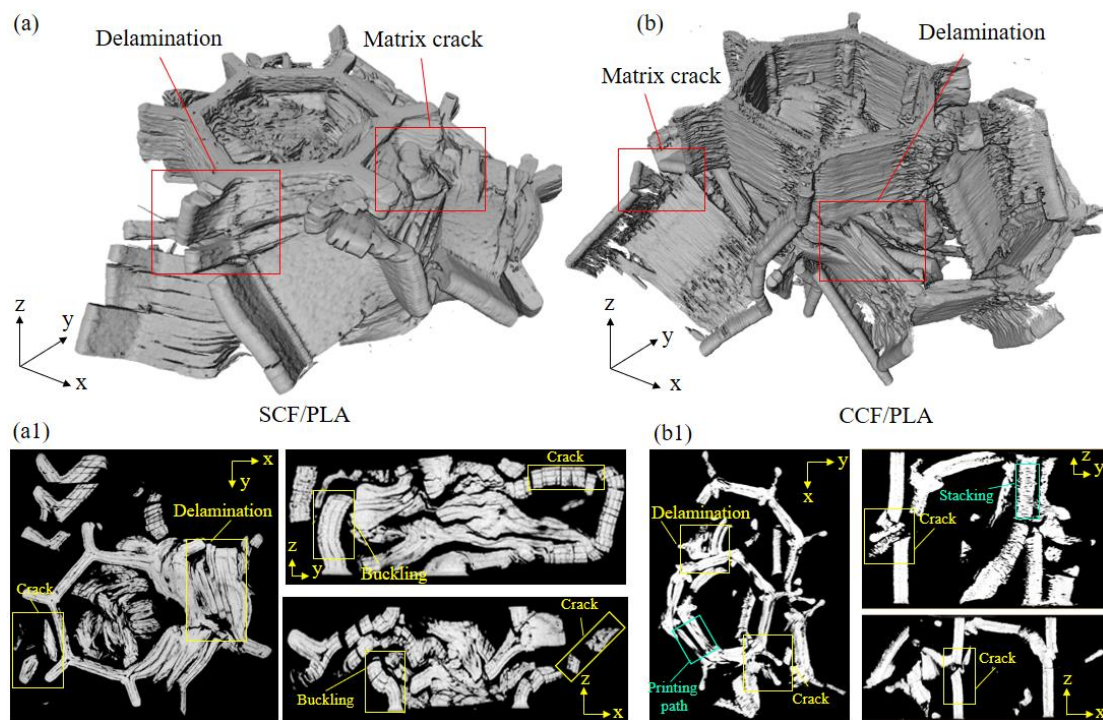


Figure 8 CT results of SCF/PLA (a, a1) and CCF/PLA (b, b1) after quasi-static compression.

For the CCF/PLA composite honeycomb, Figure 8(b) provides the 3D CT reconstruction after compression crush, exhibiting failure characteristics as observed during the physical test. The sectional views in Figure 8(b1) further reveal the layer-by-layer deposition features inherent to the 3D printing process, as well as the fibre

deposition paths. In addition, internal damage modes, such as delamination and fracture, can be clearly identified within the structure. These CT results provide direct internal damage evidence, with supporting the observed macroscopic failure mechanisms.

As shown in Figure 9, the compressive performances of the 3D-printed SCF/PLA and CCF/PLA honeycomb structures, as well as the VBM-fabricated CCF/Epoxy honeycomb structure developed in this study, are compared with various competitive cellular architectures reported in the literature. These include conventional aluminum honeycombs ^[52], aluminum pomelo-inspired hierarchical hexagonal honeycombs (Al Pomelo-HHH) ^[53], aluminum hierarchical hexagonal honeycombs (Al HHH) ^[54]. Other advanced cellular structures include polymer tube-reinforced honeycombs ^[55], CFRP hexagonal honeycombs, and 3D CFRP honeycombs ^[56]. The results demonstrate that the CCF/Epoxy honeycomb structure developed in this study exhibits a significantly higher specific energy absorption (SEA) compared to most of the comparable cellular designs, indicating its superior energy absorption capability. However, its density is higher than CFRP hexagon honeycomb.

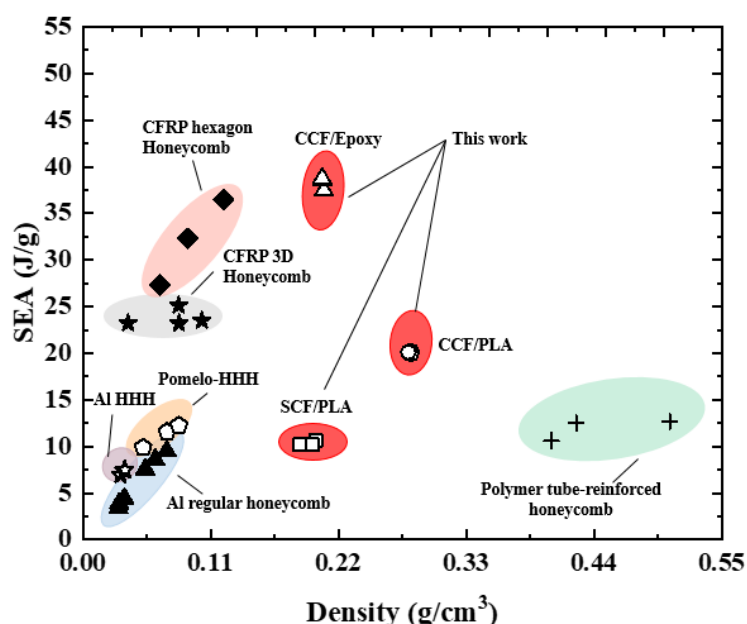


Figure 9. Comparison between the SEA of this work with other cellular structures: regular honeycomb ^[53] aluminium pomelo-inspired hierarchical hexagonal honeycomb (Al Pomelo-HHH) ^[54], aluminium hierarchical hexagonal honeycomb (Al HHH) ^[53] and other competing structures, including polymer tube-reinforced honeycombs ^[54], CFRP Hexagon and CFRP 3D Honeycomb ^[56].

In this study, the honeycomb structures made of three types of materials with embedded battery were tested under compressive loading. The experimental results are presented in Figure 10. All load-displacement curves were processed to obtain the mean values, standard deviations and confidence intervals. For the 3D honeycomb structures with embedded battery (Figures 10(a-c)), the structure failed prior to the battery, followed by combined load-bearing and subsequent failure together with the battery. The peak load of the SCF/PLA structure with embedded battery is approximately 18.5 kN (Figures 10(a, a1)). Subsequently, the battery also began to fail, with the temperature gradually rising to 70.7 °C. Similarly, for the CCF/PLA honeycomb structure with embedded battery (Figures 10(b, b1)), the overall structure underwent two elastic stages (0-2 mm, and 4.8-10 mm) before eventually entering the plastic stage, with a peak load of 33.2 kN. Ultimately, the structure and the battery experienced instability and failure together, with the temperature increasing to 59.5 °C. For the CCF/Epoxy structure with embedded battery, with the highest peak load of 35.2 kN (the compressive strength of 45.88 MPa). The structure exhibited a petal-like failure mode, leading to ultimately failure, with the temperature reaching 41 °C.

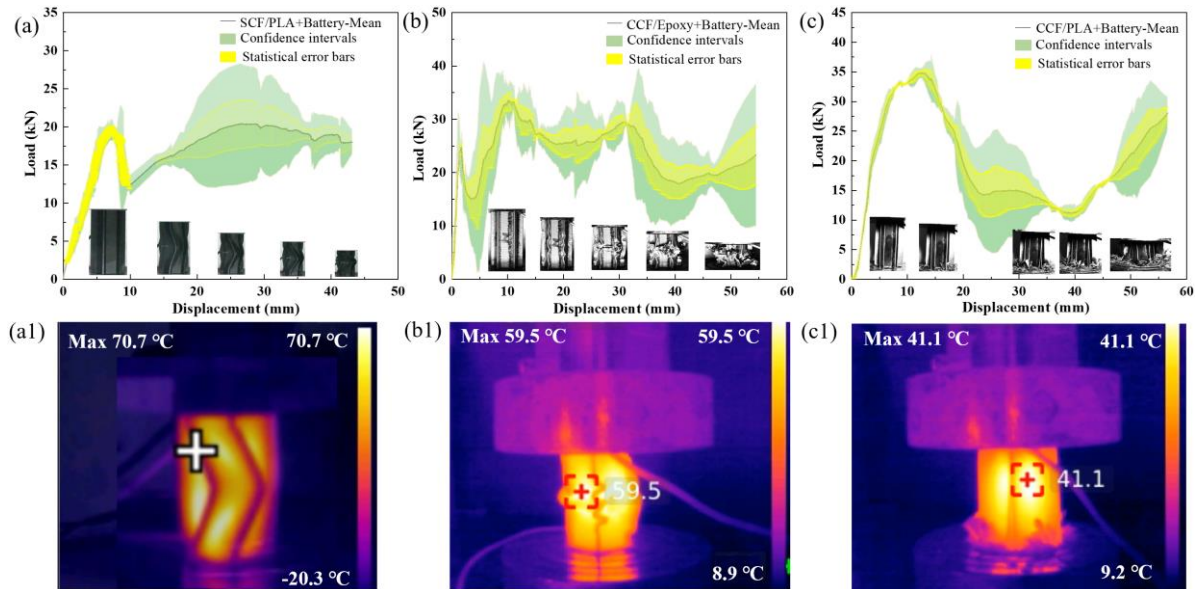


Figure 10. Quasi-static compression load–displacement responses, corresponding failure modes, and temperature evolution of 21700 cylindrical batteries embedded in honeycomb structures: (a, a1), SCF/PLA + battery, (b, b1) CCF/PLA + battery, (c, c1) CCF/epoxy + battery.

Comparative results indicate that CCF/PLA with battery achieves the highest compressive strength (50.44

MPa), while CCF/epoxy with battery provides the maximum specific energy absorption (11.4 J/g). Owing to the high material utilisation, design flexibility, and capability for manufacturing complex geometries, additive manufacturing presents significant advantages for structural applications. The findings offer useful references for material and process selection in engineering battery-integrated structures.

The 21700 lithium-ion cell used in this study has a nominal capacity of 5000 mAh, a rated voltage of 3.7 V, and a mass of 68 g, corresponding to an energy of approximately 18.6 Wh. Based on this value, the gravimetric energy density of the honeycomb-embedded battery system can be estimated for different material configurations. Specifically, the energy density is 128.28 Wh/kg for the SCF/PLA configuration, 101.64 Wh/kg for the CCF/PLA one, and 129.53 Wh/kg for the CCF/epoxy one. The gravimetric energy density reported here is a simplified structural-level estimate based only based on the battery cell and the honeycomb protective structure, without including other pack components such as BMS, interconnects, thermal management devices, or supporting frame. It is therefore used only for conceptual comparison. In addition, the SEA values are derived under axial crushing conditions only and are intended for relative comparison within this specific loading scenario, rather than representing full battery pack performance under general service conditions.

5. Verification of numerical models

5.1 The honeycomb structure

As shown in Figure 11, the finite element predictions are compared with the experimental results to evaluate the predictive capability of the proposed damage model. Overall, the simulations reproduce the load-displacement responses and the main failure characteristics observed in the experiments. For the SCF/PLA honeycomb structures (Figures 11a and b), the numerical results show good agreement with the experimental curves, with the error in peak load within 6 %. The model also captures the characteristic deformation stages, including the elastic response

(initial stiffness), peak load, progressive collapse and final densification. Also, micro-buckling is observed in both the experimental and simulation results, as shown in Figure 11a. These phenomena indicate that the progressive buckling and damage accumulation of the cell walls are reasonably simulated.

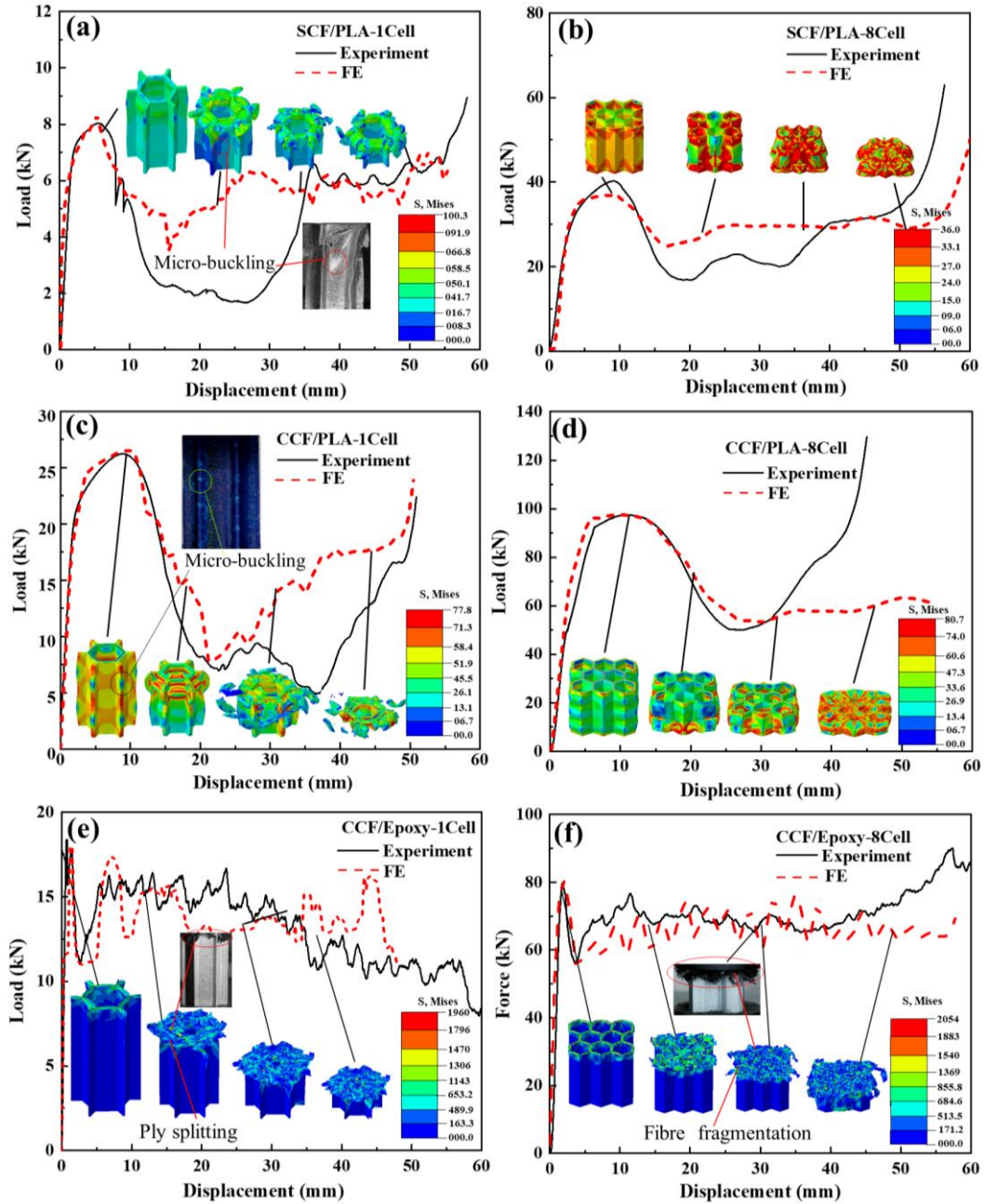


Figure 11. The load-displacement curves and predicted failure modes following crush tests on 1-cell (a) and 8-cell (b) honeycomb of 3D-printed SCF/PLA composite, 1-cell (c) and 8-cell (d) honeycomb of 3D-printed CCF/PLA composite and 1-cell (e) and 8-cell (f) honeycomb of vacuum bag moulding CCF/Epoxy composite. (the solid lines correspond to test data and the dashed lines to the FE predictions).

For the CCF/PLA honeycomb structures (Figures 11c and d), the predicted responses also agree well with the

experimental results. The discrepancies in peak load are between 5 and 9 %, whilst the differences on plateau load remain below 10 %. The micro-buckling phenomenon of the structure is captured in Figure 11(c), as well.

However, in the present model the honeycomb walls are represented using homogenised material properties. Fibre-matrix micromechanical interactions, discrete fibre fracture and layer-wise additive manufacturing defects are neglected. Consequently, the simulations exhibit fewer fibre fragments and more continuous damage patterns than those observed experimentally. Nevertheless, the model accurately reproduces the global structural response of the CCF/PLA honeycombs, including the initial stiffness, peak load, post-peak softening and overall crushing behaviour (Figures 11c and d). This indicates that the model is suitable for capturing the macroscopic deformation and progressive collapse behaviour of the integrated structure [57, 58].

For the CCF/epoxy honeycomb structures (Figures 11e and f), the simulations also reproduce the key deformation stages, including the elastic response, sudden load drop, and subsequent plateau region. The discrepancies between the predicted and experimental mechanical responses remain approximately 10-11 %. Figures 11(e) and (f) show the ply splitting and fibre fragmentation phenomenon of the structure, which are reasonably captured using the Abq_Ply_Fabric model. Overall, the FE models provide reliable predictions of the global mechanical response and dominant failure mechanisms of the battery protective system under crush compression.

5.2 The FE result of battery and battery with honeycomb structure

Figure 12 shows strong agreement between the experimental and simulated load-displacement curves for both the battery alone and the battery coupled with an SCF/PLA honeycomb structure. In Figure 12(a), the battery undergoes a linear elastic deformation, followed by a sharp load drop and steady plateau decline until the ultimate failure. Due to the simplifications inherent in the simulation model developed, the current numerical outputs

primarily indicate the agreement with the corresponding experimental results, in terms of overall profiles of load-displacement curves and damage evolutions of the structures. In numerical terms, the simulated peak load deviates from the experimental value by 20%, and the simulated energy absorption deviates by 5%. Damage patterns, including buckling and collapse, are consistent in both. Figure 12(b) shows that for the battery-structure system the simulation captures the buckling and failure of the honeycomb reasonably well, with almost identical initial stiffness and peak load. These results confirm the reliability of the simulation in predicting the mechanical response and failure behaviour of the battery and its integrated structure.

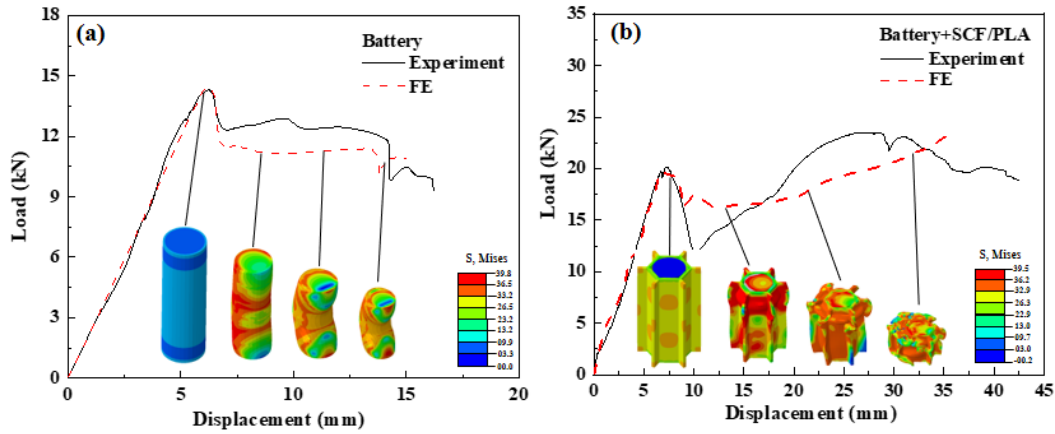


Figure 12. The load-displacement curves and the predicted failure modes following crush tests on a 21700 battery (a) and SCF/PLA composites (b) (the solid lines correspond to test data and the dashed lines to the FE predictions).

5.3 Result of the honeycomb-battery integrated structures with different heights

Finite element simulations were conducted to evaluate the mechanical interaction between the honeycomb structure and the cylindrical battery at different structural heights (72-80 mm), as shown in Figure 13. For the 3D-printed SCF/PLA honeycombs (Figure 13a), when the structural height of 72 mm is close to the battery height of 70 mm, the structure and battery deform cooperatively, resulting in the highest peak load of 23.08 kN. As the structural height increases, the protective structure collapses earlier than the battery, leading to a gradual reduction in peak load and energy absorption. Figure 13(c) further shows that the optimal structural matching height lies between 72-74 mm. in terms of peak load. Although the peak load at 74 mm is similar to that at 72 mm, the energy

absorption at 72 mm (Figure 13d) is about 3.2% higher, indicating a more effective structural matching condition.

A similar trend is observed for the CCF/PLA honeycombs (Figure 13b). At larger structural heights (e.g., 78-80 mm), premature structural failure reduces the load-sharing capability between the honeycomb and the battery, resulting in decreased peak load and energy absorption. Overall, the results indicate that a structural height of 72 mm, slightly higher than the battery height, provides the optimal mechanical compatibility and maximizes the load-bearing capacity and energy absorption of the integrated system.

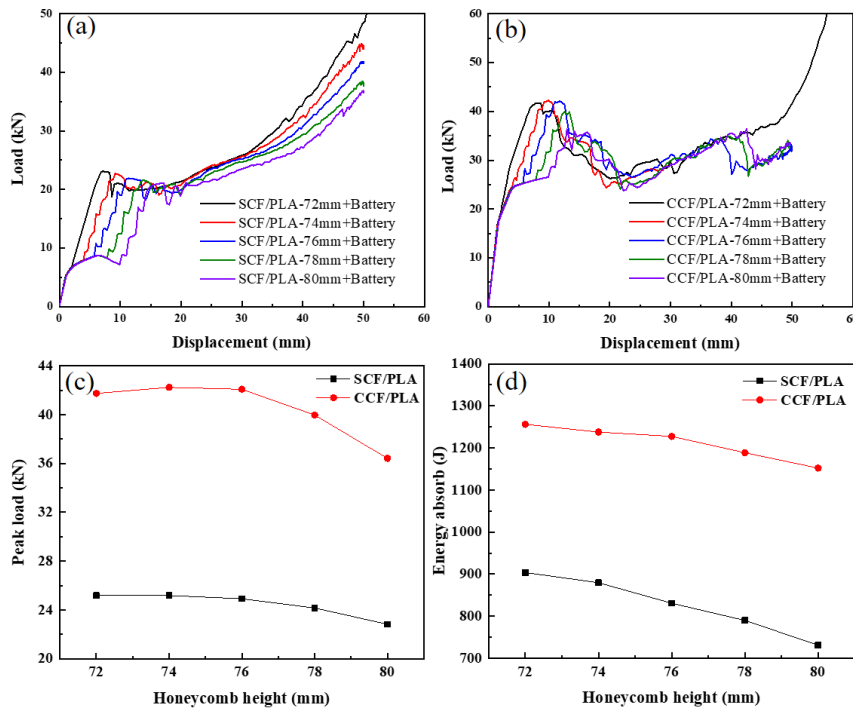


Figure 13. Mechanical responses of integrated honeycomb-battery systems with varying honeycomb heights: (a-b) load-displacement curves of 3D-printed SCF/PLA and CCF/PLA honeycombs, and (c-d) corresponding peak load and energy absorption.

To clarify the mechanical compatibility between the protective structure and the battery, a quantitative analysis was performed. As shown in Figure 14(a), the protective structure is designed to be slightly taller than the battery, allowing it to undergo elastic deformation and provide buffering before battery failure. The load-bearing enhancement of the integrated system therefore depends on the height difference between the two components.

As illustrated in Figure 14(b), the energy absorption behaviour during compression was evaluated using an analytical model. Based on experimental and simulation results, a semi-empirical relationship was established to

quantify the energy absorption of both the structure and the battery. With the fibre volume fraction and reinforcement distribution being kept constant across all specimens, the reinforcement phase density is not explicitly included as a variable. It should be noted that the present study does not aim to perform a global optimization for the battery-structure system. Instead, the novelty of this study is to identify the mechanical compatibility between the honeycomb structure and the cylindrical battery within the design range investigated. The empirical relations were calibrated using the results from four height configurations, whilst the remaining configurations were used as validation cases to assess the predictive capability of the proposed expressions. The resulting expressions are given as follows:

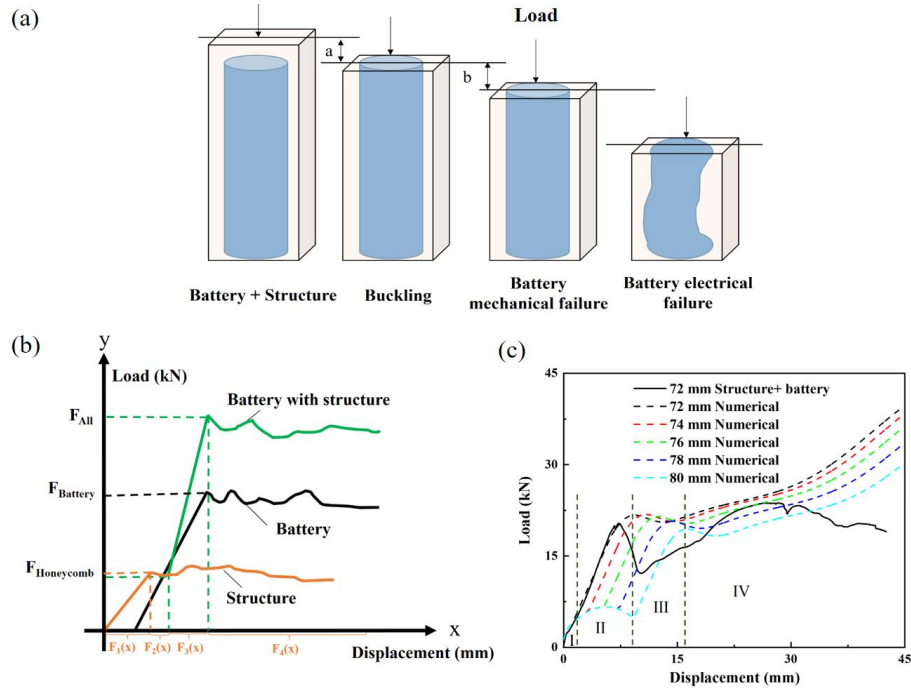


Figure 14. (a) Schematic diagram of structure and battery loading process, (b) Matching design with mechanics, (c) Theoretical calculation results at different heights.

$$E_{all} = \int_0^x F(x)dx, (F(x)=F_{all}) \quad (4)$$

$$F_1(x) = a * x + b \quad (0 \leq x \leq 1) \quad a=4.63 \quad b=0.013 \quad (5)$$

$$F_2(x) = a * x^2 + bx + c \quad (1 \leq x \leq 10 - db) \quad a=-0.1405, b=1.7577, c=3.1794 \quad (6)$$

$$F_3(x) = a * x + b \quad (10 - db \leq x \leq (14 - db) - (dely - dy) / 2.8) \quad a=2.8 \quad b=-21.05 \quad (7)$$

$$F_4(x) = a_1 \cdot \exp(-((x - b_1) / c_1)^2) + a_2 \cdot \exp(-((x - b_2) / c_2)^2) + a_3 \cdot \exp(-((x - b_3) / c_3)^2)$$

$$(x \geq (14 - db) - (delty - dy) / 2.8) \quad a_1 = 62.089, \quad b_1 = 49.1409, \quad c_1 = 31.1435, \quad a_2 = -30.0051, \quad b_2 = 42.3597, \quad c_2 = 18.4386, \\ a_3 = 4.9067, \quad b_3 = 15.9977, \quad c_3 = 3.5338. \quad (8)$$

The y-direction translation dy is given by:

$$dy = -0.568 \cdot db^2 + 0.741 \cdot db + 0.0308 \quad (9)$$

The auxiliary x-direction translation $delty$ is given by:

$$delty = -0.1189 \cdot (8 - db)^2 + 1.4962 \cdot (8 - db) + 3.8481 \quad (10)$$

where db represents the height difference between the honeycomb and the battery.

As shown in the Figure14, $F_1(x)$ represents the expression for the elastic stage of the honeycomb structure, while $F_2(x)$ corresponds to its plastic stage. $F_3(x)$ denotes the expression for the combined elastic response of the structure and the battery, and $F_4(x)$ describes their combined plastic response.

6. Conclusions

In this study, honeycomb structure-battery packs have been fabricated using carbon fibre composite materials through 3D printing and vacuum bag moulding processes. In terms of specific energy absorption (SEA), the vacuum-bag-moulded CCF/epoxy honeycomb has exhibited the highest SEA (38.25 J/g), the 3D-printed CCF/PLA honeycomb has showed the highest compressive strength (18.61 MPa). The mechanical interaction between the honeycomb structure and the battery has been successfully analysed through experiments and finite element simulations. The results indicate that the honeycomb structure can provide initial protection through sacrificial deformation before the battery reaches its failure stage, followed by a coupled load-bearing response of the integrated system.

Using the validated FE model, parametric FE analyses have been conducted to evaluate the influence of structural height on the integrated performance. An optimal matching height is identified for both SCF/PLA and CCF/PLA honeycombs, with a balance being obtained between peak load and energy absorption. Overall, the

results highlight the importance of deformation compatibility and load-transfer mechanisms in governing the protective performance of honeycomb battery-pack structures. It provides a mechanics-based guidance for the design of integrated lightweight battery systems.

Acknowledgements

The strong technical support from the state-of-the art equipment and experimental research, performed in the Core Facilities and Experiment Center of Xi'an Jiaotong University is very much appreciated.

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Appendix.

Mechanical characterisation tests are carried out on SCF/PLA composite specimens manufactured under identical printing conditions. The composites exhibited a tensile strength of 32.40 MPa and a tensile modulus of 2.94 GPa. Under compressive loading, the measured strength and modulus are 47.63 MPa and 1.51 GPa, respectively. The corresponding flexural strength and flexural modulus are found to be 84.54 MPa and 6.32 GPa, whilst the shear strength and shear modulus reached 20.74 MPa and 0.30 GPa, respectively (Figure A1).

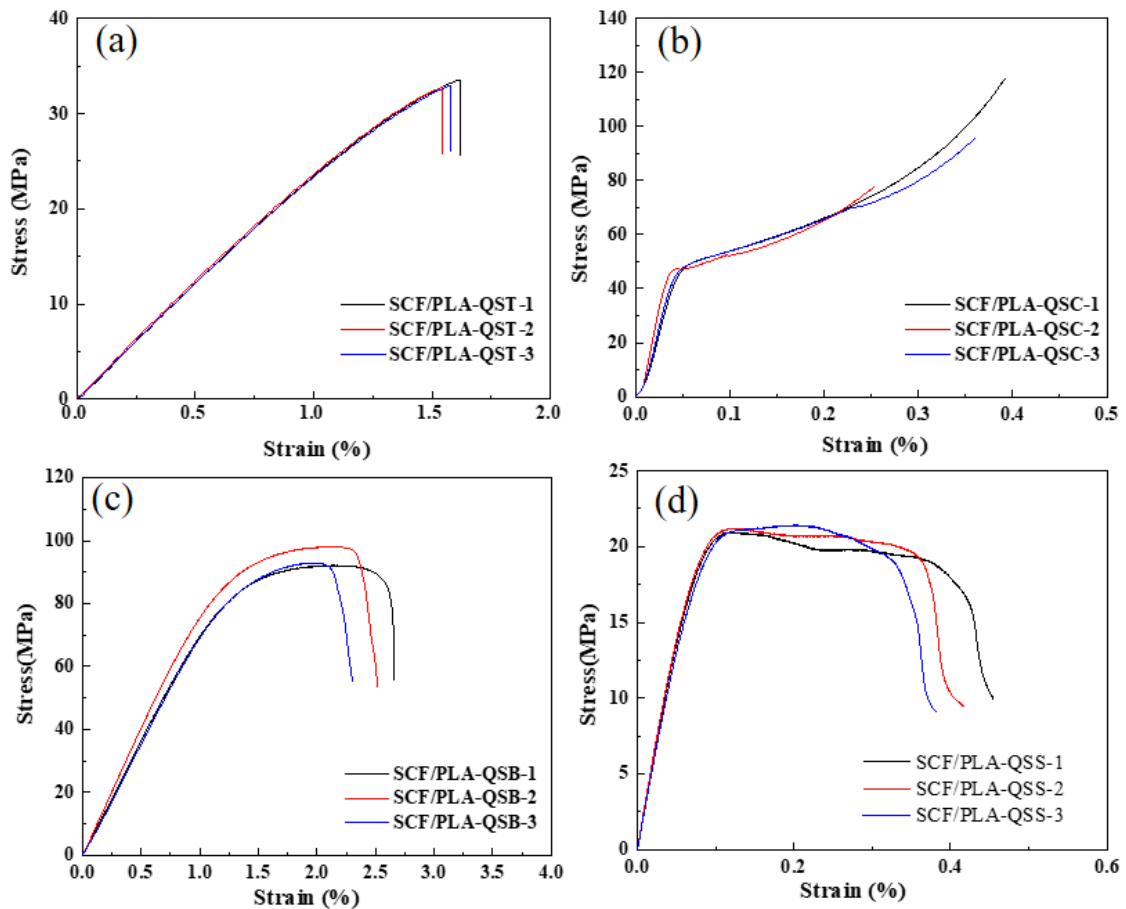


Figure A1. Mechanical test results of SCF/PLA in (a) tension, (b) compression, (c) bending and (d) shearing.

According to experimental measurements conducted on specimens fabricated in the identical printing conditions, the tensile strength and modulus of the CCF/PLA composites are determined as 333.46 MPa and 21.97 GPa, respectively. The measured flexural strength and modulus are 273.37 MPa and 16 GPa, respectively, whilst the shear strength is 32.47 MPa (Figure A2).

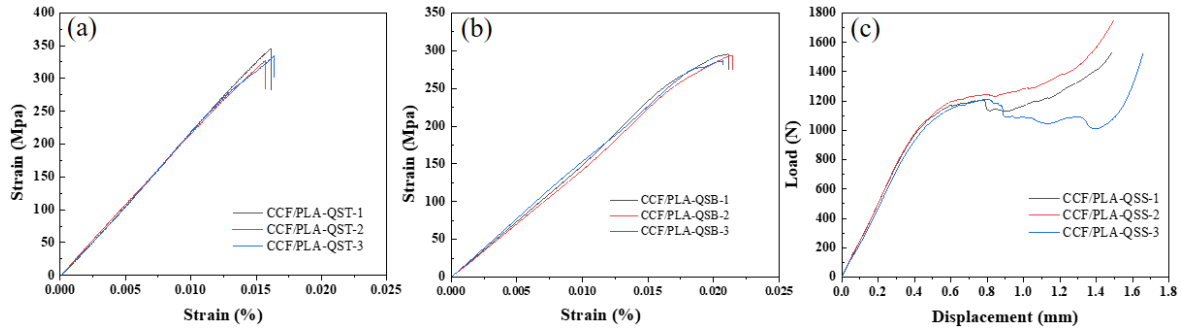


Figure A2. Mechanical test results of CCF/PLA in (a) tension, (b) bending and (c) shearing.

In order to prove the mesh independence of the proposed damage model, three mesh sizes (0.5, 1 and 1.5 mm) were used to investigate the mesh sensitivity in the honeycomb structure. As illustrated in Figure A3(b) (Appendix), the simulations with mesh sizes of 0.5 mm and 1 mm show good agreement with the experimental peak load, indicating that the numerical predictions are relatively insensitive to further mesh refinement. In contrast, the model with a mesh size of 1.5 mm significantly underestimates the peak load, as the coarse mesh cannot adequately capture the local deformation and damage evolution of the honeycomb cell walls.

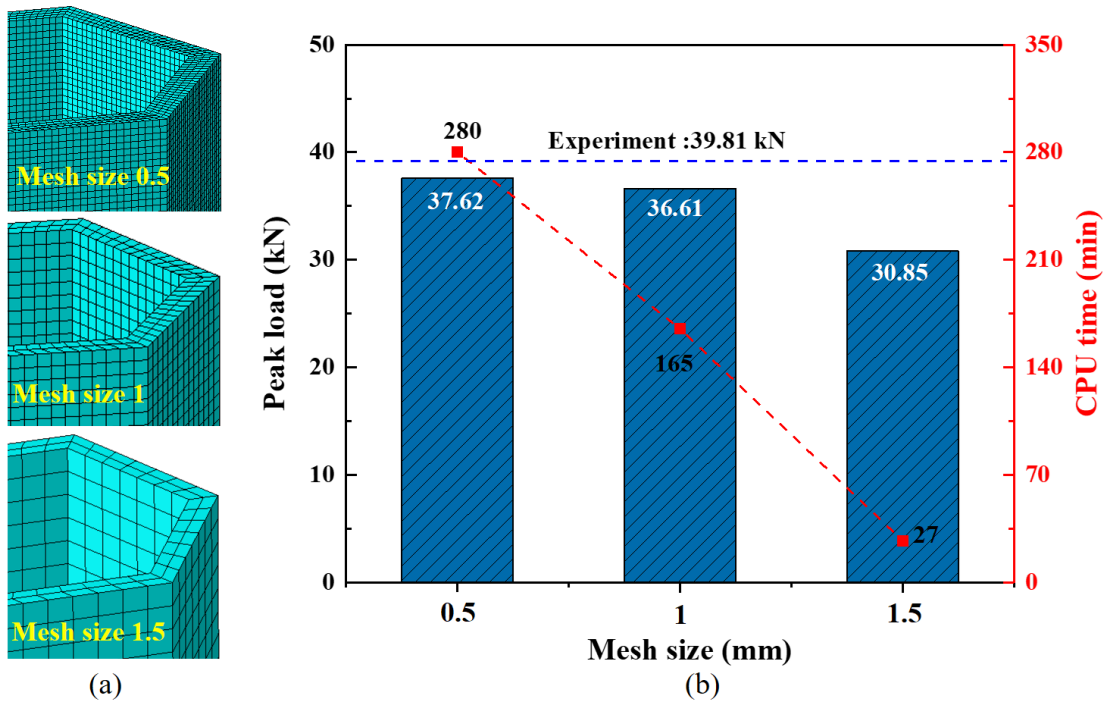


Figure A3. Mesh geometries in mesh sensitivity study: (a) specimens for mesh geometries and (b) corresponding peak load histories obtained from quasi-static compress test.

In addition, the computational efficiency was evaluated. The CPU time for mesh sizes of 0.5, 1 and 1.5 mm was approximately 280, 165 and 27 min, respectively. Considering both prediction accuracy and computational

1 648 efficiency, a mesh size of 1 mm was selected for the subsequent simulations. This sensitivity analysis demonstrates
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3 649 that the selected mesh size provides a reasonable balance between the computational cost and accuracy.
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Manuscript Number: ENGSTRUCT-D-25-09552

Title: The energy-absorbing behaviour of fibre reinforced honeycombs for battery pack structures

Dear Editor and Reviewers,

Thank you for your rigorous comments and suggestions, which are vital to improve the manuscript. The manuscript has been revised accordingly with all changes marked in **red** in the revised manuscript. (Reviewer's comments in **black**, authors' response in **blue**).

Response to reviewers#1' comments

Reviewer #1:

Reviewer #1: This manuscript presents a comprehensive experimental and numerical investigation into the energy-absorbing characteristics of carbon fiber-reinforced honeycomb structures for battery pack applications. The study is timely, given the urgent need for lightweight and safe energy storage solutions in electric transportation. The work is methodologically sound, combining material fabrication (3D printing and VBM), quasi-static testing, and detailed finite element analysis. The comparison between different manufacturing processes and material systems is a particular strength. However, the manuscript requires significant revisions in terms of clarity, depth of discussion, and contextualization of its findings before it can be considered for publication.

Comment 1

1. the description of the continuous carbon fiber (CCF) 3D printing process is unclear and potentially misleading. The text and Figure 1(a) suggest a process where a dry carbon fiber bundle and a PLA filament are mixed in a heated guide tube. This is atypical for most commercial continuous fiber 3D printing, which typically uses a pre-impregnated fiber filament.

Authors' Response to comment 1

Thank you for your valuable comments. In this study, the continuous carbon fibre (CCF/PLA) 3D printing process was adopted, instead of the pre-impregnated fibre filament method. To avoid confusion, the manuscript has been revised as follows:

Page 6, line 88-90:

“...Both materials are subsequently delivered into the heating zone, where the molten PLA impregnates the carbon fibre bundle in-situ. The resulting impregnated composite filament is then extruded through the nozzle and deposited to form the printed structure...”

Comment 2

2. The rationale behind the chosen layup sequence for the VBM CCF/Epoxy samples is not provided. Why was this specific stacking sequence selected? A brief justification based on crashworthiness or energy absorption principles would strengthen the methodological rigor.

Authors' Response to comment 2

Thank you for your valuable comments and suggestions. The stacking sequence of $[45/-45/0/-45/45]_s$ was

selected for the VBM CCF/Epoxy specimens to balance the axial stiffness and energy absorption capability under crushing loads. This design is supported by a relevant reference ^[39], which demonstrates that such quasi-balanced lay-ups are commonly adopted in crashworthiness design to distribute damage more uniformly and to increase specific energy absorption (SEA). Accordingly, the selected sequence provides a compromise between load-bearing stiffness and energy dissipation. In accordance with the reviewer's suggestions, this justification, along with the corresponding citation, has been incorporated into the revised manuscript.

Page 7 line 104-106:

“...The symmetric quasi-angle lay-up [45/−45/0/−45/45]_s was adopted to balance axial stiffness and shear-dominated progressive failure, since $\pm 45^\circ$ plies induce progressive damage and enhance energy absorption capacity under crushing loads ^[39]....”

[39] Alshahrani H, Ahmed A. Enhancing Impact Energy Absorption, Flexural and Crash Performance Properties of Automotive Composite Laminates by Adjusting the Stacking Sequences Layup. *Polymers* (Basel). 2021 Oct 3;13(19):3404.

Comment 3

3. The thicknesses of the manufactured samples vary significantly (SCF/PLA: ~2mm, CCF/PLA: ~3.3mm, CCF/Epoxy: ~1.5mm). This makes a direct comparison of "absolute" performance (like peak load) less meaningful. The authors should normalize the compressive strength and stiffness by the thickness, or at least discuss the performance differences in the context of their areal density or specific thickness. This would lead to a fairer comparison.

Authors' Response to comment 3

Thank you for pointing out this. The comparisons of peak load and energy absorption were limited to the same material type, whereas comparisons between different materials were made using compressive strength and specific energy absorption to eliminate the influence of sample dimensions and mass.

Corresponding discussions have been incorporated into the revised manuscript, with a focus on elucidating the effect of material type on the compressive performance of honeycomb structures. This improvement enables a more rigorous and equitable comparison among different materials.

Page 11-12, line167-169:

“...Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated up to the onset of densification to ensure consistent comparisons....”

Page 21-22, line 315-318:

“...All data on the peak load, compressive strength, energy absorption and specific energy absorption of the honeycomb structures are listed in Table 8. For 8-cell honeycomb structures made of different materials, it is observed that the 3D-printed CCF/PLA composites provide the highest compressive strength (18.61 MPa), surpassing those of CCF/Epoxy (15.95 MPa) and SCF/PLA (7.58 MPa)....”

The data regarding compressive strength has also been included in Table 8 below.

Table 8. Energy absorption characteristics of honeycomb structures made of 3D-printed short carbon fibre, 3D-printed continuous carbon fibre and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composites, respectively.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
SCF/PLA-1Cell-a	20.10	8.02		11.73		13.62	
SCF/PLA-1Cell-b	20.40	7.97		12.27		10.51	
SCF/PLA-1Cell-c	20.20	8.08	8.16	11.66	11.80	13.01	12.88
SCF/PLA-1Cell-d	20.15	8.10		11.70		13.58	
SCF/PLA-1Cell-e	20.22	7.98		11.50		13.67	
Variance	/	0.032		0.069		1.45	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	
CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

Comment 4

4. The discussion of failure modes in Section 4, while supported by images, is largely descriptive. A deeper mechanistic explanation is lacking.

Authors' Response to comment 4

Thank you for your valuable suggestions. In response, the discussion of failure modes in Section 4 has been expanded to provide a more mechanistic interpretation, beyond the descriptive analysis supported by the images. The revised content is as follows:

Page 17-18, line 259-269:

“Based on the load-, voltage- and temperature-displacement curves presented in Figure 5, the battery response can be divided into three stages (Figure 5a). At Stage I (Mechanical deformation), the battery undergoes elastic and plastic deformations until structural buckling occurs, whilst the voltage and temperature remain almost constant. The increase in load reflects progressive strain energy accumulation. At Stage II (Separator deformation), after buckling, the separator, which has a high failure strain, begins to carry the load. The applied load gradually decreases as the separator is deformed under continuous platen displacement, whilst the voltage remains at approximately 3.4 V. This stage highlights stress redistribution and progressive damage within the cell. At Stage III (Short circuit and thermal response), once the mechanical limit of the separator is reached, a cathode-anode short circuit occurs, leading to a rapid voltage drop and a temperature rise to 129 °C. These accelerated changes are primarily attributed to localised electrical shorting and Joule heating. Further penetration results in ultimate mechanical failure and complete voltage collapse. This stage illustrates the coupling among the structural degradation, electrochemical response and thermal effects.”

Page 18-21, line 275-332:

“The compressive response of the three honeycomb structures is illustrated in Figure 6, which exhibits both the load-displacement curves and the corresponding progressive failure modes. The densification points are also marked on the curves. To ensure statistical robustness, the curves are analysed to derive the mean, variance and confidence intervals, as well as the boundary intervals for energy absorption. For SCF/PLA honeycombs, both 1-

cell (Figure 6a) and 8-cell (Figure 6b) configurations exhibit a rapid increase in load to the corresponding peak values (8.16 kN and 39.81 kN), followed by a gradual load decrease and a relatively stable crushing plateau. This behaviour is governed by progressive cell-wall buckling and matrix-dominated failure. Damage initiates at stress concentration regions and propagates through matrix cracking and interlayer debonding, whilst the load is subsequently sustained by wall folding and frictional contact between fractured segments. The dominant failure modes are therefore matrix cracking, wall buckling and delamination (Figures 7a1, a2), which are the damage characteristics of composite cellular solids under axial loading ^[49]. The SEA values are 12.88 J/g for the 1-cell and 10.32 J/g for the 8-cell one.

For the CCF/PLA honeycombs, the peak loads were measured as 25.29 kN for 1-cell (Figure 6c) and 97.68 kN for 8-cell samples (Figure 6d). A more pronounced post-peak load drop is observed. This sharp reduction is mechanistically associated with interface-controlled damage in additively manufactured continuous fibre composites, where limited interlayer fusion and in-situ impregnation promote early delamination and fibre-matrix debonding. After the initial drop of load, a secondary plateau is maintained through progressive fibre fracture, wall buckling and interfacial sliding. The dominant failure mechanisms also include fibre fracture, matrix cracking and delamination (Figures 7b1, b2), consistent with reported post-peak instability features in continuous fibre AM composites ^[50]. The 1-cell and 8-cell honeycombs exhibit SEA values of 18.48 J/g and 20.07 J/g, respectively.

Similar staged crushing behaviour is also observed in vacuum-bag-molded CCF/Epoxy honeycombs, but with a more stable post-peak response (Figures 6e, f). The 1-cell and 8-cell structures reach peak loads of 20.33 kN and 80.18 kN, respectively, followed by sustained plateau loads prior to densification. Failure initiates with matrix cracking and delamination, followed by wall sliding and progressive lamina fragmentation (Figures 7c1, c2). Interfacial sliding and friction between fractured walls provide a sustained energy dissipation mechanism via friction, which delays collapse localization and directly contributes to the superior SEA (43.27 J/g for the 1-cell

and 38.25 J/g for the 8-cell honeycomb)^[51]. This contrasts with abrupt failure modes (e.g., sudden matrix/fibre fracture) of the previous two materials.

All data on the peak load, compressive strength, energy absorption and specific energy absorption of the honeycomb structures are listed in Table 8. For 8-cell honeycomb made of different materials, it is observed that the 3D-printed CCF/PLA composites provide the highest compressive strength (18.61 MPa), surpassing those of CCF/Epoxy (15.95 MPa) and SCF/PLA (7.58 MPa).

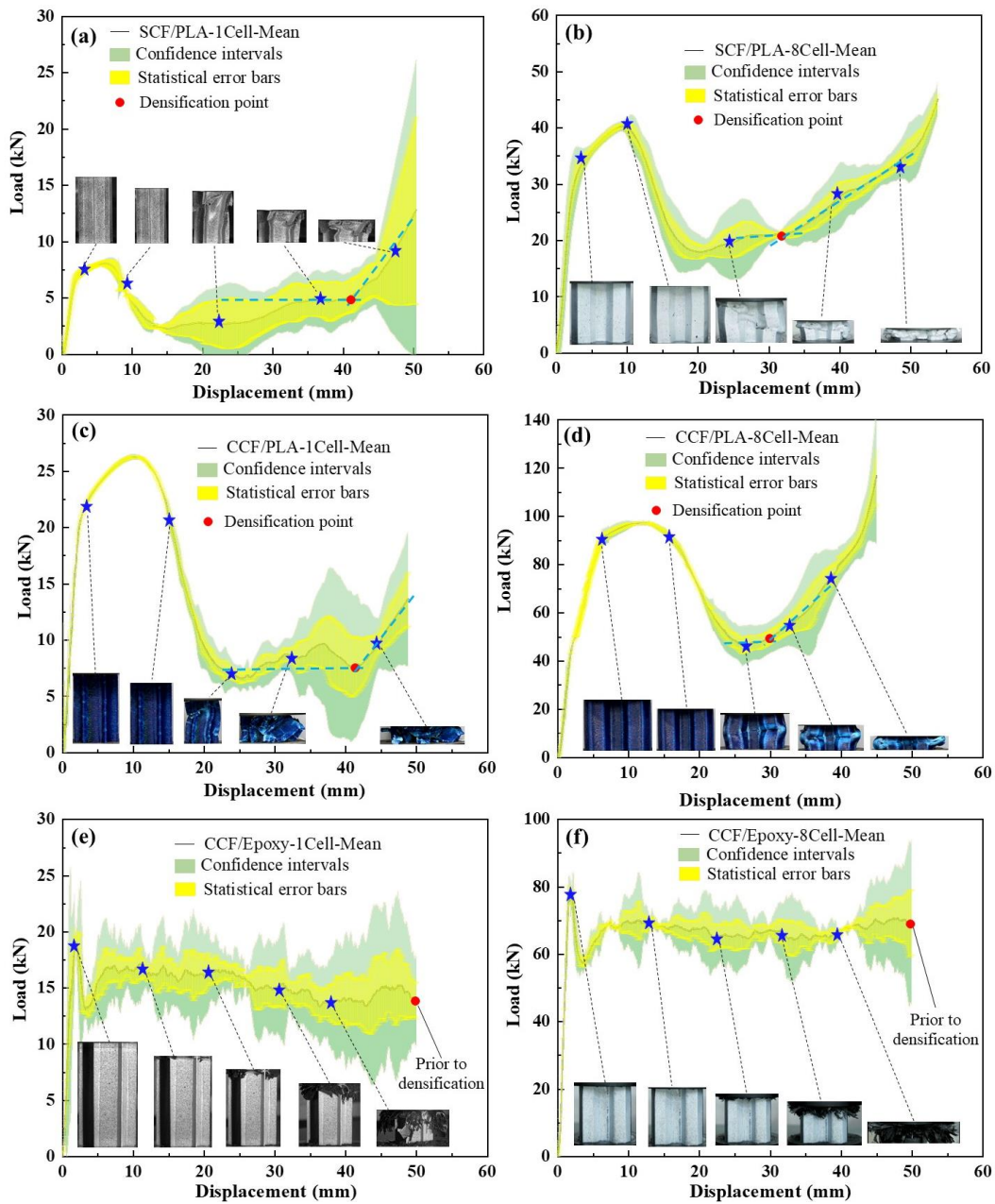


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static crushing tests

on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

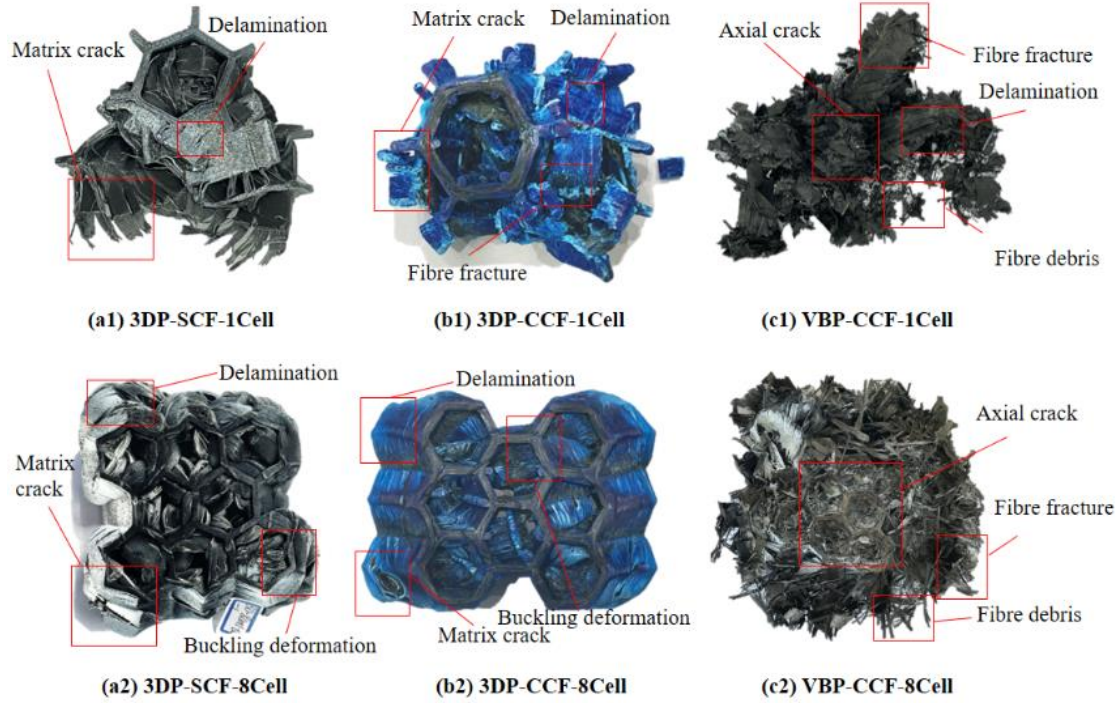


Figure 7. Failure modes of 1-cell (a1 for SCF/PLA, b1 for CCF/PLA, c1 for CCF/Eopxy) and 8-cell (a2 for SCF/PLA, b2 for CCF/PLA, c2 for CCF/Eopxy) honeycomb structures made of different materials after quasi-static compression.

Due to severe fragmentation after crush, the CCF/Epoxy specimens could not be examined by CT. Therefore, post-crush CT analyses were performed on the 3D-printed SCF/PLA and CCF/PLA honeycomb structures. Figure 8(a) presents the overall 3D CT reconstruction of the crushed SCF/PLA honeycomb, which is consistent with the macroscopic observations and reveals interlayer delamination and matrix cracking. Figure 8(a1) shows sectional views from different orientations, which more distinctly illustrate the internal damage features, including interfacial separation, local buckling and fracture. For the CCF/PLA composite honeycomb, Figure 8(b) provides the 3D CT reconstruction after compression crush, exhibiting failure characteristics as observed during the physical test. The sectional views in Figure 8(b1) further reveal the layer-by-layer deposition features inherent to the 3D printing process, as well as the fibre deposition paths. In addition, internal damage modes, such as delamination

and fracture, can be clearly identified within the structure. These CT results provide direct internal damage evidence, with supporting the observed macroscopic failure mechanisms.”

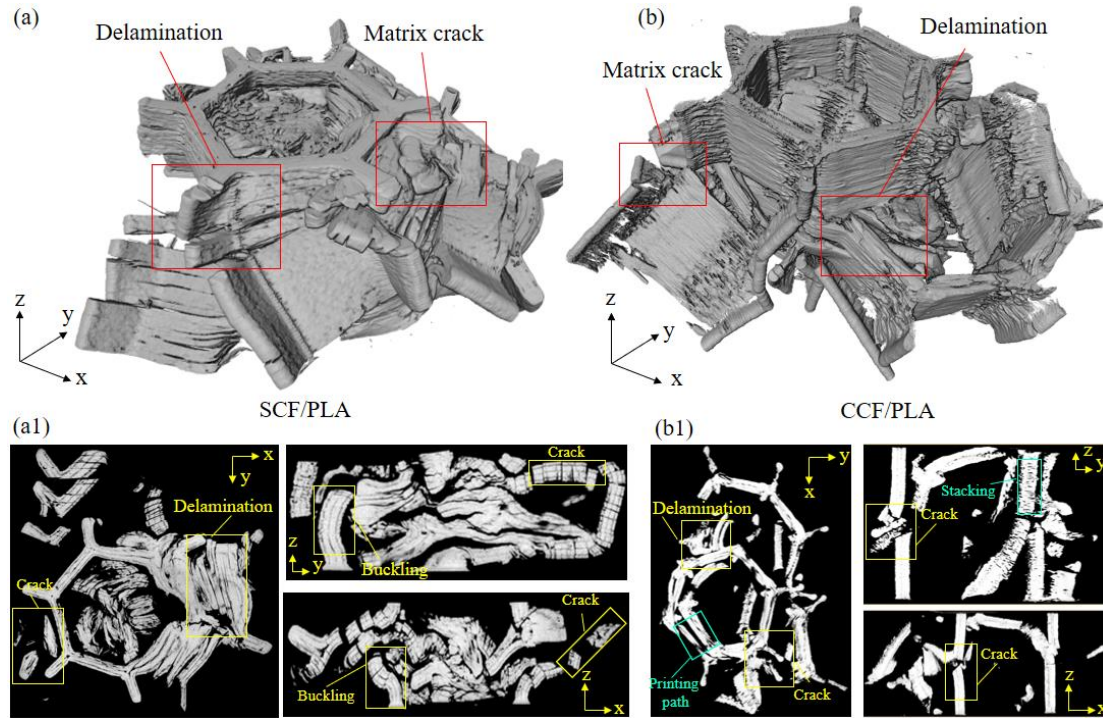


Figure 8 CT results of SCF/PLA (a, a1) and CCF/PLA (b, b1) after quasi-static compression.

Comment 5

5. Why does CCF/PLA exhibit a more severe post-peak load drop than CCF/Epoxy? Is this attributed to the weaker inter-layer bond in the 3D-printed parts?

Authors' Response to comment 5

Thank you for this insightful question. The more pronounced post-peak load drop observed in the CCF/PLA samples compared to the CCF/Epoxy ones can be indeed attributed primarily to the weaker inter-layer bonding inherent in FDM 3D-printed parts. In the CCF/PLA printing via fused deposition, the interface between successive layers and fibre-matrix bonding is limited by the in-situ impregnation of molten PLA, which typically results in a lower interlaminar shear strength and reduced ability to sustain load after initial failure.

In contrast, CCF/Epoxy composites are manufactured using traditional lamination processes, in which the epoxy resin is used to ensure uniform impregnation and strong adhesion between layers, providing better post-peak load retention and more gradual failure. Consequently, the post-peak softening is more severe in the 3D-printed CCF/PLA due to weaker inter-layer bond.

This observation is consistent with previous studies on 3D-printed continuous fibre composites, where layer

interfaces were identified as the primary factor controlling post-peak load behaviour ^{[1], [2]}.

[1] Fan C, Shan Z, Zou G. et al. Interfacial Bonding Mechanism and Mechanical Performance of Continuous Fiber Reinforced Composites in Additive Manufacturing. *Chin. J. Mech. Eng.* 34, 21 2021. <https://doi.org/10.1186/s10033-021-00538-7>

[2] Wu Y, Wang K, Neto V. et al. Interfacial behaviors of continuous carbon fiber reinforced polymers manufactured by fused filament fabrication: A review and prospect. *Int J Mater Form* 15, 18 2022. <https://doi.org/10.1007/s12289-022-01667-7>

Comment 6

6. How does the failure mode (e.g., "wall sliding" in CCF/Epoxy) correlate with the superior SEA? Link the observed failure mechanisms directly to the energy dissipation metrics.

Authors' Response to comment 6

Thank you for the insightful comments. We fully agree that it is necessary to directly link the observed failure mechanisms to energy dissipation metrics to have a more in-depth mechanistic interpretation. Therefore, the following discussion has been added in the revised manuscript.

Page 20-21 line 304-311:

“Similar staged crushing behaviour is also observed in vacuum-bag-molded CCF/Epoxy honeycombs, but with a more stable post-peak response (Figures 6e, f). The 1-cell and 8-cell structures reach peak loads of 20.33 kN and 80.18 kN, respectively, followed by sustained plateau loads prior to densification. Failure initiates with matrix cracking and delamination, followed by wall sliding and progressive lamina fragmentation (Figures 7c1, c2). Interfacial sliding and friction between fractured walls provide a sustained energy dissipation mechanism via friction, which delays collapse locally and directly contributes to the superior SEA (43.27 J/g for the 1-cell and 38.25 J/g for the 8-cell honeycomb) ^[51]. This contrasts with abrupt failure modes (e.g., sudden matrix/fibre fracture) of the previous two materials (SCF/PLA and CCF/PLA).”

Comment 7

7. The manuscript briefly mentions that "fewer fiber fragments are observed in the simulation" but does not fully acknowledge the limitations of the modelling approach. The authors should explicitly discuss the limitations of their models.

Authors' Response to comment 7

Thank you for your valuable suggestions. The limitations related to the modelling are added in the revised

manuscript.

Page 27, line 396-402:

“...However, in the present model, the honeycomb walls are represented using homogenised material properties. Fibre-matrix micromechanical interactions, discrete fibre fracture and layer-wise additive manufacturing defects are not considered in the current modelling. Consequently, less fibre fragments are produced in the simulations in comparison to those observed experimentally. Nevertheless, the model accurately reproduces the global structural response of the CCF/PLA honeycombs, including the initial stiffness, peak load, post-peak softening and overall crushing behaviour (Figures 11c and d). This indicates that the model is suitable for capturing the macroscopic deformation and progressive collapse behaviour of the structure [57, 58]”.

Reviewer #2: The manuscript investigates the energy-absorbing behaviour of fibre-reinforced honeycomb structures for lightweight battery pack applications. These structures are mechanically tested under quasi-static axial compression, both as standalone honeycombs and in combination with a 21700 cylindrical Li-ion battery. The authors also construct finite element (FE) models in Abaqus/Explicit to simulate deformation and failure of each honeycomb configuration, including a detailed composite model for the CCF/Epoxy system. However, The paper does not sufficiently articulate what is fundamentally new relative to extensive existing literature. The work reads primarily as a technical report, not a research paper with novel findings.

Comment 1

-- Section 2.2.1. The authors assume a fixed densification displacement of 40 mm (55% of height) for all samples, regardless of material, thickness, stiffness, or geometry (p. 10). This is not scientifically valid, since densification depends strongly on material modulus and wall buckling behavior. This choice directly affects calculated EA and SEA values, making comparisons unreliable.

Authors' Response to comment 1

Thank you for the important comments. It is agreed that the densification displacement of cellular structures may vary depending on material properties and deformation behaviour.

Following the reviewer's suggestions, the energy absorption values have been recalculated based on the actual deformation response of each specimen. The densification points have been clearly indicated in the load-displacement curves of the revised figures. Based on this updated calculation method, the energy absorption (EA) and specific energy absorption (SEA) values have been recalculated and updated in the corresponding tables.

The relevant descriptions in Section 2.2 have been revised accordingly in the manuscript.

Page 11-12, line 167-169:

“Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated based on the load-displacement curve up to the onset of individual densification to ensure consistent comparison...”

Page 18, line 275-278:

“The compressive response of the three honeycomb structures is illustrated in Figure 6, which depicts both the load-displacement curves and the corresponding progressive failure modes. The densification points are also marked on the curves. To ensure statistical robustness, the curves were analysed to derive the mean, variance, and confidence intervals, as well as the boundary intervals for the total energy absorption....”

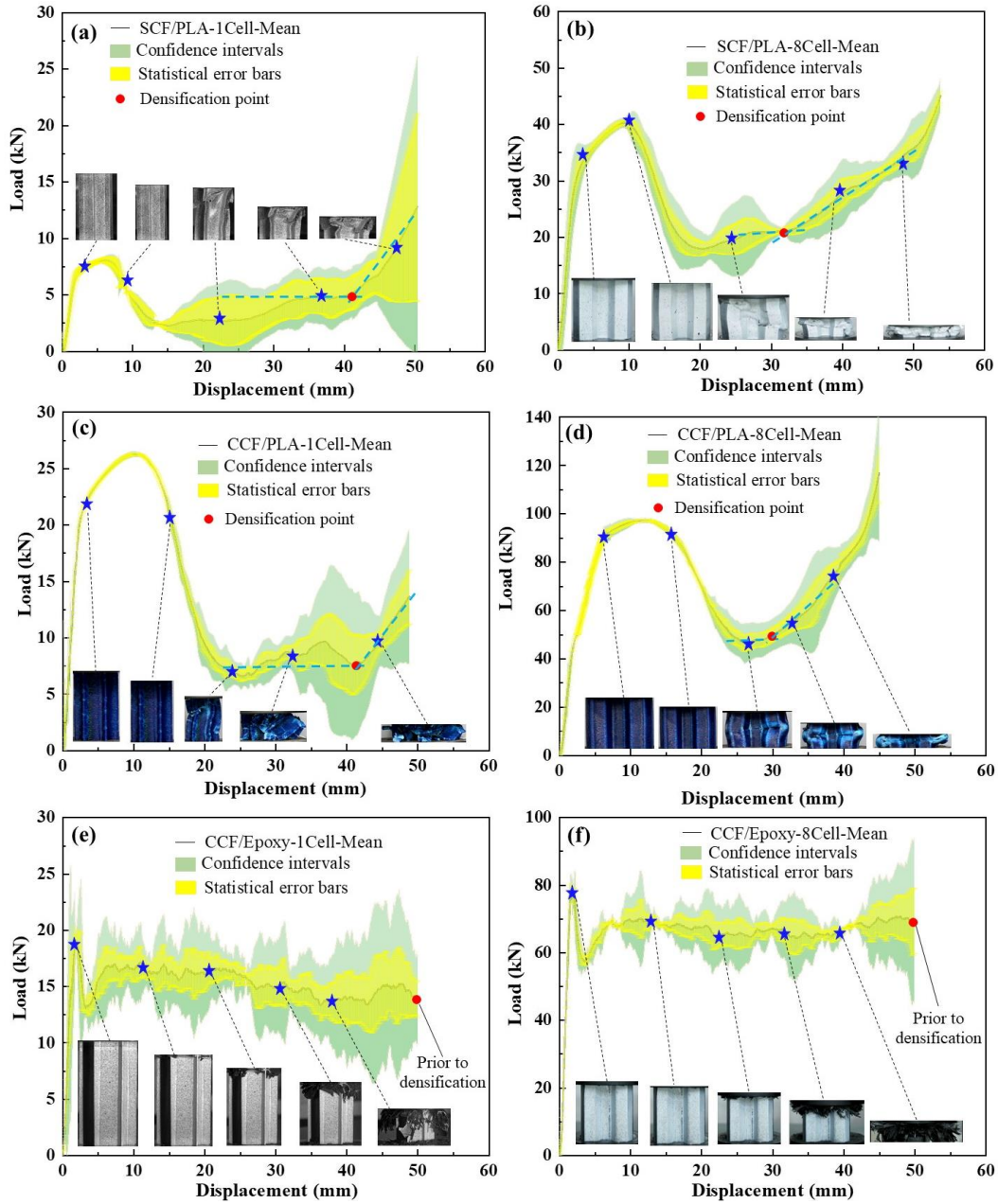


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static crushing tests on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

Comment 2

-- Section 2.2.2. Only 3 replicates per condition are tested (Table 2). No statistical error bars, confidence intervals, or ANOVA are performed. SEA values vary considerably (e.g., SCF/PLA-1Cell: 13.50, 8.02, 11.66 J/g). Without proper statistical treatment, many comparisons—particularly between materials—are not meaningful.

Authors' Response to comment 2

Thank you for your comments. It is fully agreed that statistical indicators such as confidence intervals are necessary. Based on the reviewers' valuable suggestions, revisions have been made to the experimental results and statistical methods, as described below.

Two additional sets of data are supplemented for the SCF/PLA-1cell composite honeycomb structure, from which five experimental results are obtained and analysed. All relevant curve data have been processed, with the mean values, variances and confidence intervals being derived. These results are now presented in the revised Figure 6 presented the response to Comment 1.

Table 8 is re-analysed to obtain the mean values and variances for peak load, compressive strength, and specific energy absorption (SEA), thereby illustrating the consistency of the experimental data.

Table 8. Energy absorption characteristics of honeycomb structures made of 3D-printed short carbon fibre, 3D-printed continuous carbon fibre and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composites, respectively.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
SCF/PLA-1Cell-a	20.10	8.02		11.58		13.62	
SCF/PLA-1Cell-b	20.40	7.97		11.51		10.51	
SCF/PLA-1Cell-c	20.20	8.08	8.02	11.66	11.58	13.01	12.88
SCF/PLA-1Cell-d	20.15	8.10		11.70		13.58	
SCF/PLA-1Cell-e	20.22	7.98		11.50		13.67	
Variance	/	0.032		0.069		1.45	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	

CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

Comment 3

-- Section 2.2.3. For the 3D-printed CCF/PLA structures: The fibre volume fraction is never reported.

The degree of fibre alignment, waviness, or resin impregnation quality is not quantified. No microstructural imaging is provided (SEM or optical cross-sections). These omissions make it impossible to understand the mechanical performance or validate FE assumptions.

Authors' Response to comment 3

Thank you for your comments. The printing parameters and description of the fibre volume fraction have been incorporated into the revised manuscript.

Page 6 line 93-95:

“...The printing spacing was set to 1 mm, a printing speed to 150 mm/min, a layer height to 0.3 mm, and a nozzle temperature of 220 °C. The fibre volume fraction of the structure in this study was determined to be 17 %....”

The fibre alignment in the printed structure is primarily governed by the deposition path during the melt extrusion manufacturing process.

Page 6-7, line 95-99:

“The printing path of the 3D-printed CCF/PLA composite is presented in Figure 1(a). The as-printed material exhibits a good impregnation quality, with uniform resin distribution around the fibres and no significant fibre exposure. Due to weakened fibre fixation after resin melting, pull-back phenomena are observed (approximately 20% angular deviation at corners and 45% at path reversals), which are compensated by reserved path design.”

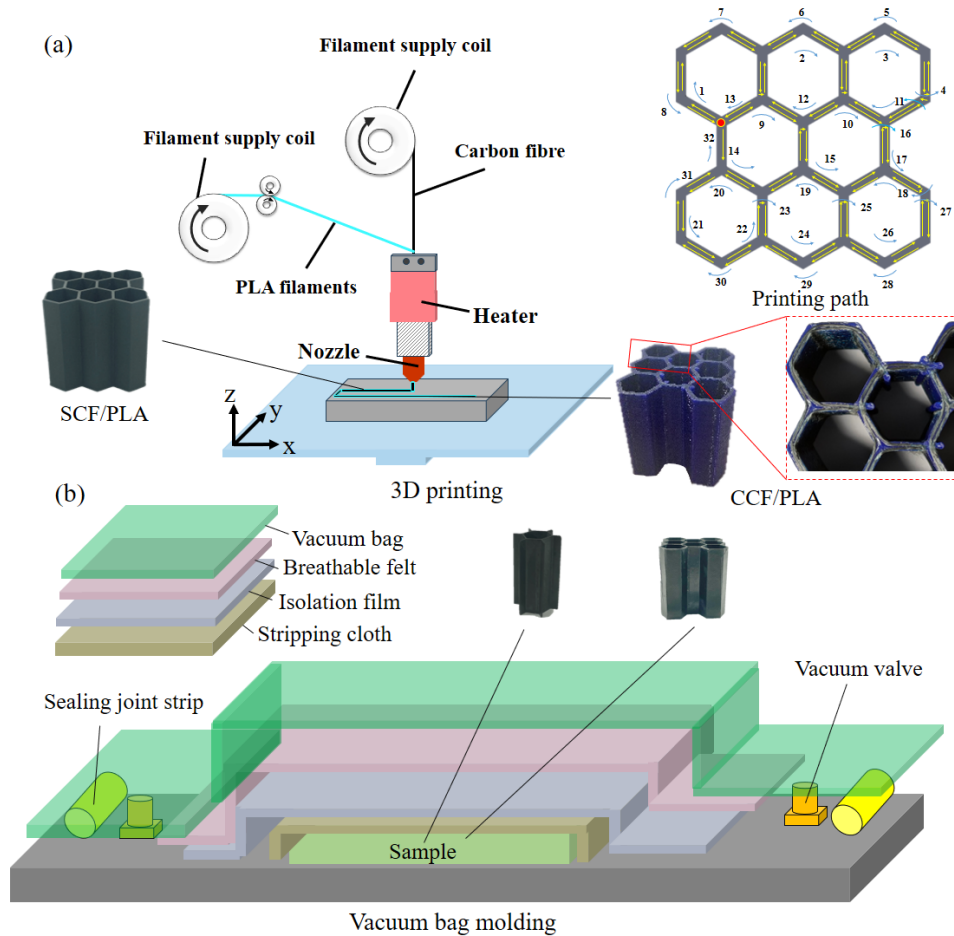


Figure 1. (a) Schematic of the 3D printing process of short/continuous carbon fibre thermoplastic composite, (b) preparation process of continuous carbon fibre thermosetting composite honeycomb structure by vacuum bag moulding method.

Additionally, X-ray computed tomography (CT) was performed on a single cell to determine the porosity and the degree of impregnation. These analyses are also explained in the revised manuscript:

Page 7, line 96-97:

“...the as-printed material exhibits good impregnation quality, with uniform resin distribution around the fibres and no significant fibre exposure...”

Page 9 line 133-134:

“...The measured void contents are 2.7% for SCF/PLA (Figure 2a), 3.46% for CCF/PLA (Figure 2b)...”

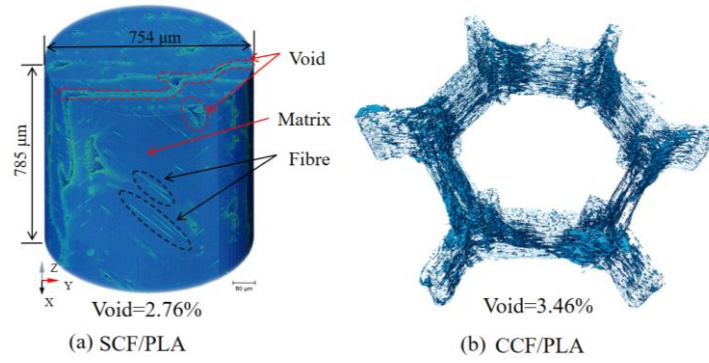


Figure 2. Comparison of void content (%) in honeycomb composite structures manufactured by different processes: (a) 3D-printed SCF/PLA, (b) 3D-printed CCF/PLA, as determined by X-ray computed tomography (CT).

Comment 4

-- Section 2.2.4. The FE models include numerous questionable assumptions: "Equivalent anisotropic material" for printed continuous fibres (p. 12) ignores real fibre curvature, intermittent deposition, and interlayer defects. Cohesive elements in the CCF/Epoxy model are included only between layers, not along honeycomb vertical joints where delamination is visibly prominent in Figure 7(a-c). Battery model is adopted from literature with no calibration against the authors' own cell, yet they claim good agreement. The overall modelling framework is not sufficiently validated, especially for complex failure modes like fibre fragmentation, ply splitting, and microbuckling.

Authors' Response to comment 4

Thank you for the detailed and constructive comments. It is agreed that the numerical framework involves several simplification assumptions. The rationale, scope, and limitations have been clarified in the revised manuscript.

For the CCF/Epoxy model, cohesive elements are also introduced between the vertical joints of the honeycomb structure (Figure 4a), which was not indicated before.

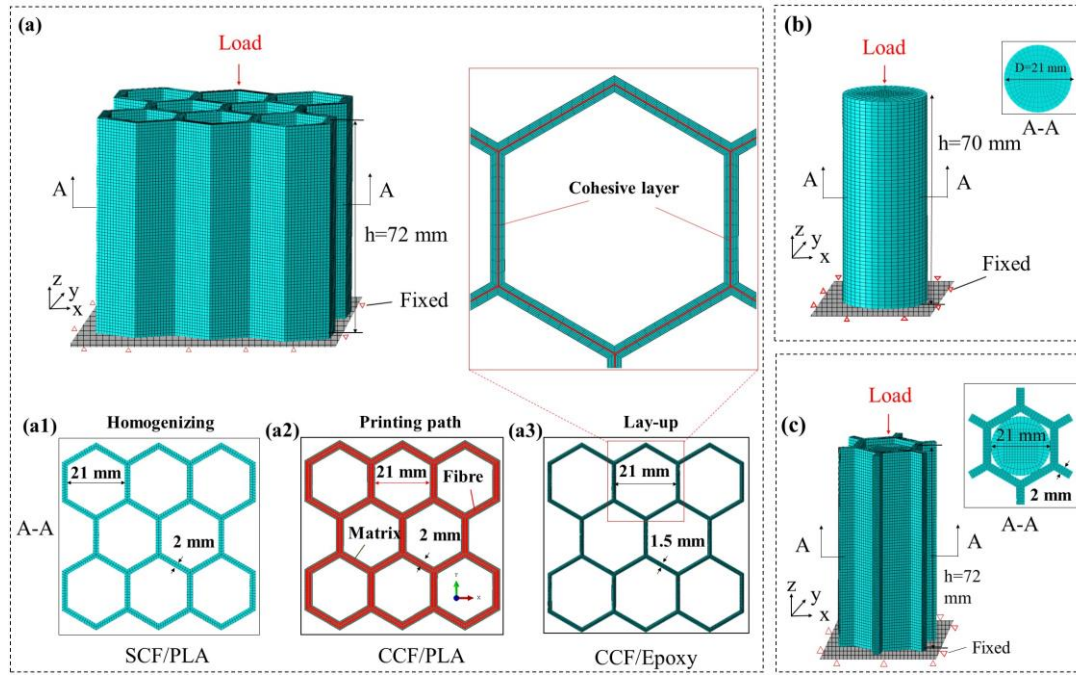


Figure 4. (a) Full-scale FE model of honeycomb of different manufactures (a1, 3D printing thermoplastic short carbon fibre (SCF/PLA) composites, a2, 3D printing continuous carbon fibre (CCF/PLA) composites, a3, vacuum bag moulding method (CCF/Epoxy) composites), (b) the geometry and mesh generation of the FE model for the cylindrical battery.

The battery model was adopted from validated literature sources and used as a baseline electro-mechanical representation. Although the simulated deformation trends and load-displacement responses show good agreement with experimental observations, the manuscript has been revised to avoid overstatement.

Page 28, line 414-416:

“...Due to the simplifications inherent in the simulation model developed, the current numerical outputs primarily indicate the agreement with the corresponding experimental results, in terms of overall profiles of load-displacement curves and damage evolutions of the structures....”

Within this modeling framework, the failure modes, including fibre breakage, delamination and micro-buckling, were also validated by the corresponding experimental results. The comparison between experimental and simulated results is presented in Figure 11. A paragraph has been added to the revised manuscript to discuss these findings.

Page 14, line 199-205:

“...To improve computational efficiency, a simplified numerical model was adopted, in which the curvature of real fibres, intermittent deposition and interlayer defects were neglected. Also, the printed continuous fibre

composite was treated as a homogeneous anisotropic material ^[42], where their stiffness and strength characteristics along three main directions were incorporated through homogenised orthotropic properties. Consequently, this modeling framework is suitable for predicting the overall structural response and crushing trends, but is not intended for a detailed description of micro-scale failure mechanisms....”

Page 26, line 385-386:

“...Also, micro-buckling is observed in both the experimental and simulation results, as shown in Figure 11a....”

Page 27, line 395:

“...The micro-buckling phenomenon of the structure is captured in Figure 11(c), as well....”

Page 27, line 406:

“...Figures 11(e) and (f) show the ply splitting and fibre fragmentation phenomenon of the structure,....”

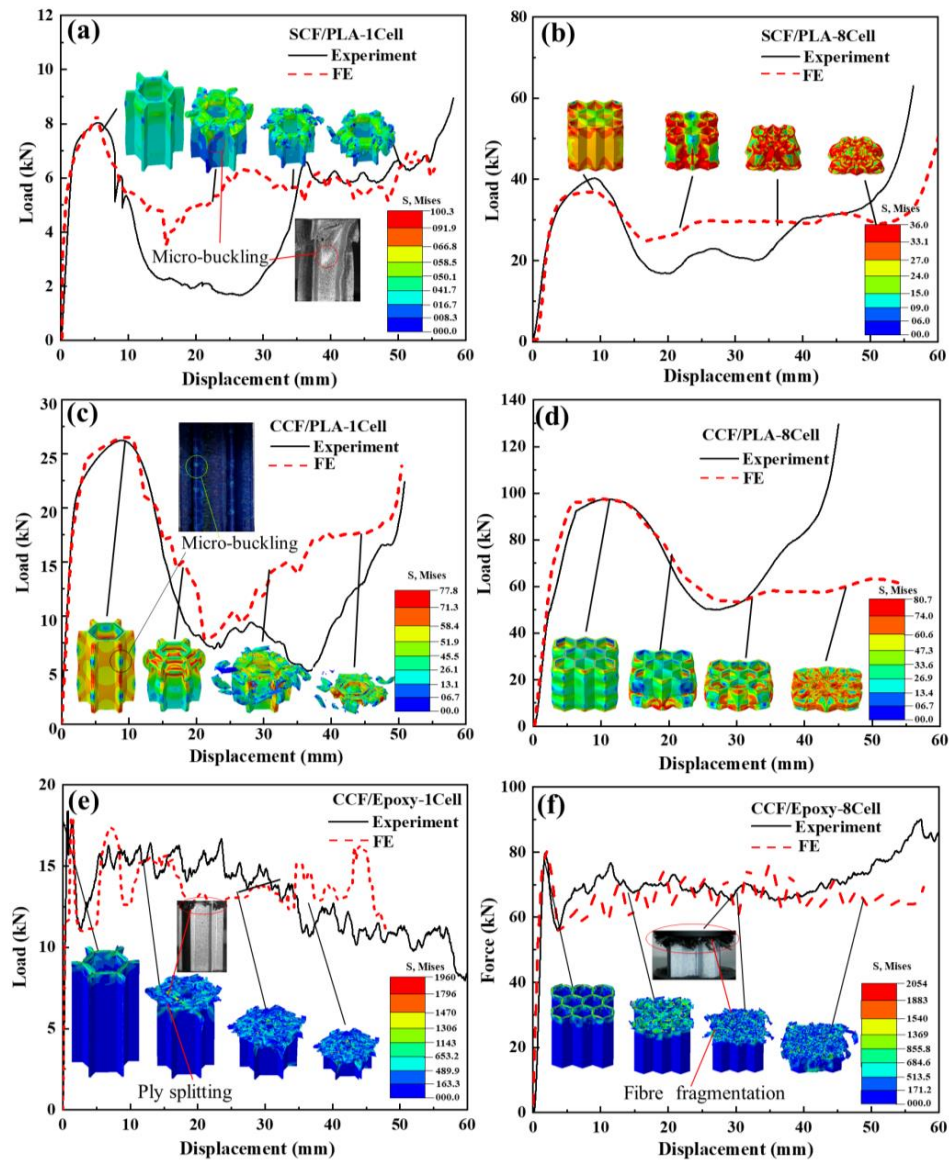


Figure 11. The load-displacement curves and predicted failure modes following crush tests on 1-cell (a) and 8-cell

(b) honeycomb of 3D-printed SCF/PLA composite, 1-cell (c) and 8-cell (d) honeycomb of 3D-printed CCF/PLA composite and 1-cell (e) and 8-cell (f) honeycomb of vacuum bag moulding CCF/Epoxy composite. (the solid lines correspond to test data and the dashed lines to the FE predictions).

Comment 5

-- Section 2.3.1. Quasi-static crushing of Li-ion cells is a hazardous process requiring strict control. However: The battery's state-of-charge is vaguely described ("less than 25%"); no exact SOC is reported. No mention of thermal runaway mitigation equipment (fume hood, vented chamber, fire suppressants). Temperature data (Fig. 5) shows unrealistic gradients (e.g., 129°C maxima with no thermal runaway description). Without rigorous battery safety methodology, results are unreliable.

Authors' Response to comment 5

Thank you for pointing out this. The actual service conditions of the battery pack, such as cell arrangement and thermal management systems, have not been strictly considered in this study, given that the primary focus is on the mechanical performance analysis of the battery integrated with protective structure. These limitations are now clearly stated in the revised manuscript.

In addition, it is noted that fire blankets for thermal runaway protection were available in the laboratory during the experiments, which has also been added to the revised manuscript. The state of charge (SOC) of all the batteries used in this test has been clearly stated in the revised manuscript as well. To avoid misunderstanding, we have now explicitly stated that the temperature measurements reflect localized thermal response under controlled low-SOC crushing conditions, rather than uncontrolled thermal runaway behaviour.

Page 9-10 line 147-151:

“...All quasi-static crushing tests of the lithium-ion cells were conducted in a well-ventilated and open testing environment equipped with remote loading control and a fire blanket for emergency protection. The state of charge (SOC) of all tested cells was maintained at $(20 \pm 5) \%$. The recorded temperature values represent localised surface heating during deformation rather than a full thermal runaway event...”

Page 17-18, line 265-267:

“...At Stage III (Short circuit and thermal response), once the separator reaches its mechanical limit, a cathode-anode short circuit occurs, leading to a rapid voltage drop and temperature rise to 129 °C...”

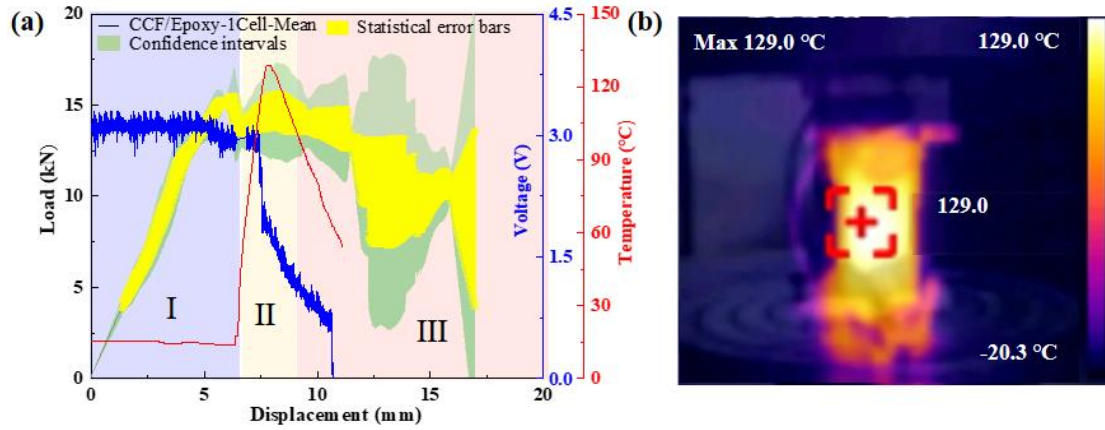


Figure 5. (a) The curves of load, voltage, and battery temperature versus time under quasi-static compression, and (b) the temperature results of the failure center.

Comment 6

-- The authors only test SCF/PLA + battery, but not CCF/PLA or CCF/Epoxy with battery. The partial evaluation prevents any design comparison. Given the paper's stated goal ("guidance for lightweight battery pack design"), this omission is critical.

Authors' Response to comment 6

Thank you for the comments. Testing only one material configuration is indeed insufficient to meet the stated objectives of this paper.

Following the reviewer's suggestions, the experimental scope has been significantly expanded. Specifically, battery pack structures integrated with all three materials are now fabricated and tested.

Page 23-24, line 347-358:

“...In this study, the honeycomb structures made of three types of materials with embedded battery were tested under compressive loading. The experimental results are presented in Figure 10. All load-displacement curves were processed to obtain the mean values, standard deviations and confidence intervals. For the 3D honeycomb structures with embedded battery (Figures 10(a-c)), the structure failed prior to the battery, followed by combined load-bearing and subsequent failure together with the battery. The peak load of the SCF/PLA structure with embedded battery is approximately 18.5 kN (Figures 10(a, a1)). Subsequently, the battery also began to fail, with the temperature gradually rising to 70.7 °C. Similarly, for the CCF/PLA honeycomb structure with embedded battery (Figures 10(b, b1)), the overall structure underwent two elastic stages (0-2 mm, and 4.8-10 mm) before eventually entering the plastic stage, with a peak load of 33.2 kN. Ultimately, the structure and the battery experienced instability and failure together, with the temperature increasing to 59.5 °C. For the CCF/Epoxy

structure with embedded battery, with the highest peak load of 35.2 kN (the compressive strength of 45.88 MPa). The structure exhibited a petal-like failure mode, leading to ultimately failure, with the temperature reaching 41 °C....”

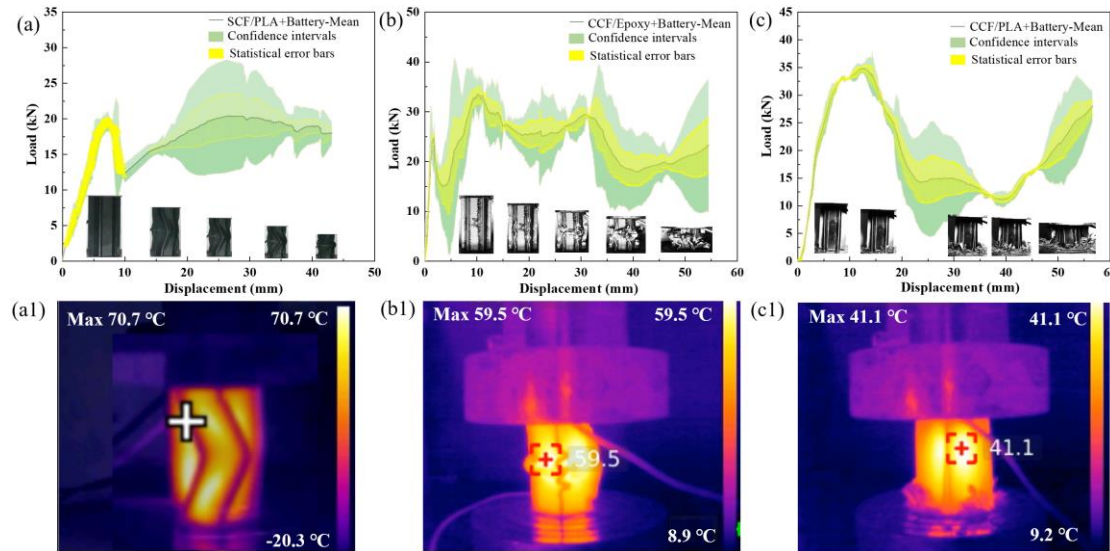


Figure 10. Quasi-static compression load–displacement responses, corresponding failure modes, and temperature evolution of 21700 cylindrical batteries embedded in honeycomb structures: (a, a1), SCF/PLA + battery, (b, b1) CCF/PLA + battery, (c, c1) CCF/epoxy + battery.

Comment 7

-- The "quality energy density" of the integrated structure is computed simply as battery energy / (battery mass + honeycomb mass). This is not physically meaningful for battery pack engineering:

Real packs include BMS, busbars, thermal management, frame, cooling plates, etc. SEA does not directly translate to pack safety. Energy density difference between 128 vs. 130 Wh/kg is negligible relative to measurement error.

Authors' Response to comment 7

Thank you for the valuable comments. It is fully agreed that the calculated energy density in this study should be regarded as a simplified estimate rather than a full battery pack-level metric.

The primary objective of this study is to focus on the analysis of structural deformation and battery mechanical failure mechanisms under crushing conditions. Here, the real-world BMS or full thermal management scenarios were not taken into account. Accordingly, the comparison among the three material configurations is based on which one provides more reasonable mechanical protection, rather than on an absolute energy density value. The discussion has been revised to emphasize that the simplified energy-density calculation is used only as an auxiliary indicator, whereas the main conclusions are drawn from failure modes, load response and structural protection behaviour.

“...Based on this value, the gravimetric energy density of the honeycomb-embedded battery system can be estimated for different material configurations. Specifically, the energy density is 128.28 Wh/kg for the SCF/PLA configuration, 101.64 Wh/kg for the CCF/PLA one, and 129.53 Wh/kg for the CCF/epoxy one. The gravimetric energy density reported here is a simplified structural-level estimate only based on the battery cell and the honeycomb protective structure, without including other pack components such as BMS, interconnects, thermal management devices, or supporting frame. It is therefore used only for conceptual comparison. In addition, the SEA values are derived under axial crushing conditions only and are intended for relative comparison within this specific loading scenario, rather than representing full battery pack performance under general service conditions.”

Comment 8

-- Despite repeated references to "mechanical-structural matching design," the paper does not perform an optimization (no objective function, no constraints, no parametric thoroughness). Only 72-80 mm heights are tested, and 72 mm is chosen empirically. The conclusions are thus not generalizable.

Authors' Response to comment 8

Thank you for the valuable comments. It should be clarified that the concept of "mechanical-structural matching design" mentioned in this study does not refer to a formal mathematical optimization process. Instead, it denotes an engineering-oriented structural design principle, under which the battery investigated is a 21700 cylindrical cell in a height of 70 mm, with all structural designs being carried out based on this dimension. The aim is to ensure that the deformation behaviour of the structure under static compression is mechanically matched to that of the cylindrical battery.

The present study intends to provide design methods and approaches for the mechanical matching of the 21700 cell with the honeycomb structures, thereby offering research insights for batteries and battery pack structures in different sizes. However, universally applicable conclusions cannot be drawn. The manuscript is carefully revised, and any wording implying universality is removed. To address the comments, the revised manuscript has clarified the definition of mechanical-structural matching and explicitly stated that the conclusions are valid within the investigated design range. Future work will focus on comprehensive parametric studies and establishing a more systematic optimization framework.

Page 30, line 451-453:

“...It should be noted that the present study does not aim to perform a global optimization of the battery-structure system. Instead, the novelty of this study is to identify the mechanical compatibility between the honeycomb structure and the cylindrical battery within the design range investigated....”

Comment 9

-- Many paragraphs contain repetitiveness, unclear phrasing, and grammatical mistakes (e.g., multiple run-on sentences, inconsistent terminology). The paper requires major English revision.

Authors' Response to comment 9

Thank you for pointing out this. In the revised manuscript, the entire paper has been carefully and systematically edited to improve language quality, sentence structure and technical terminology consistency. Long and run-on sentences as well as ambiguous phrases have been rewritten, with repetitive expressions being eliminated. All the changes in the manuscript have been marked in red.

Comment 10

-- Duplicate or nearly identical citations ([25] vs. [34]). Many references are conference reports, theses, or previous works by the same authors without broader context.

Authors' Response to comment 10

Thank you for pointing this out. In the revised manuscript, all duplicate and redundant references have been removed or consolidated. The relevant part of the introduction has been revised.

For example, Ref. [25] has been replaced as follows:

[25] Wang H, Wang Z, Fan Y, Gao Q, Wang H. Multi-objective lightweight design of automotive battery pack box for crashworthiness. *Int. J. Crashworthiness*. 2024.29(2), 292-307.

Many references, including conference papers by the same authors, theses, etc., have been replaced with new references:

[6] Asar A, Zaki W. A comprehensive review of the mechanisms and structure of interpenetrating phase composites with emphasis on metal-metal and polymer-metal variants, *Compos. Part B-Eng*. 275, 2024, 111314, 1359-8368.

[11] Lu P X, Wang J, Zhou J, Qiao Z, Cao J C, Li H, Du F P, Guan Z W. Investigate the in-plane and out-of-plane elastic properties of spiderweb-inspired hierarchical composite honeycombs using forward analytical and inverse numerical approaches, *Eng. Struct.* 346, Part A, 2026, 121600.

[13] Han S, Xiao G, Tan W, Mai P, Zhou A, Yu J, Ou J. Effect of tie parameters on strength and ductility of concrete columns reinforced with hybrid steel-fiber reinforced polymer (FRP) composite bars. *Eng. Struct.* 322, Part A, 2025, 119051.

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structure of battery. *Compos. Struct.* 2025, 352, 118586.

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[39] Alshahrani H, Ahmed A. Enhancing Impact Energy Absorption, Flexural and Crash Performance Properties of Automotive Composite Laminates by Adjusting the Stacking Sequences Layup. *Polymers (Basel)*. 2021 Oct 3;13(19):3404.

[40] Hu J F, Huang Y Y, Ge R Y, Wang J M, Lu W L. Out-of-plane compression performance of unidirectionally arrayed short fiber reinforced honeycomb structures: Experimental and numerical analysis. *Compos. Struct.* Volume 378, 2026. 119935.

Reviewer #3: The paper experimentally and numerically compares 3-D printed SCF/PLA, CCF/PLA and vacuum-bagged CCF/Epoxy honeycombs as energy-absorbing battery-pack shells. The work is timely, yet the manuscript is repetitive, omits key sample details, mis-reports statistics and lacks novelty justification. Major revision is required.

Comment 1

1. It is suggested to remove duplicated text/figures throughout your manuscript.

Authors' Response to comment 1

Thank you for this important comment. The manuscript has been carefully revised to remove duplicated descriptions and redundant figures. Repeated explanations of similar experimental observations and numerical results have been condensed to avoid repetition. In addition, several figures presenting similar information have been reorganised and combined where appropriate. As a result of these revisions, the conciseness and readability of the manuscript have been significantly improved, whilst the essential results and discussion have been retained.

Comment 2

2. Please add the fibre volume fraction, void %, wall-thickness tolerance and printing path images for every batch.

Authors' Response to comment 2

Thank you for the valuable suggestions. All the information has been added in the revised manuscript.

(1) Fibre volume fraction was determined based on material composition and verified through mass-volume measurements for each batch.

Page 6, line 83-84:

“...the fibre volume fraction ranging from 5.5% to 7.0% is exhibited by these composites....”

Page 6, line 95:

“...The fibre volume fraction of the structure in this study was determined as 17%....”

Page 7, line 110-111:

“...Under this process, the fibre content of the CCF/Epoxy composite material can reach approximately 50%.”

(2) The porosity results obtained from the CT test are as follows.

Page 9, line 132-136:

“Three-dimensional internal images were acquired via X-ray computed tomography (CT), allowing the void content (%) to be quantified. The measured void contents are 2.7% for SCF/PLA (Figure 2a), 3.46% for CCF/PLA (Figure 2b) and 1.32% for CCF/epoxy (Figure 2c). This quantitatively confirms that the filament-based additive manufacturing introduces a higher interfacial porosity compared with vacuum-assisted curing, primarily due to limited inter-bead fusion and fibre impregnation.”

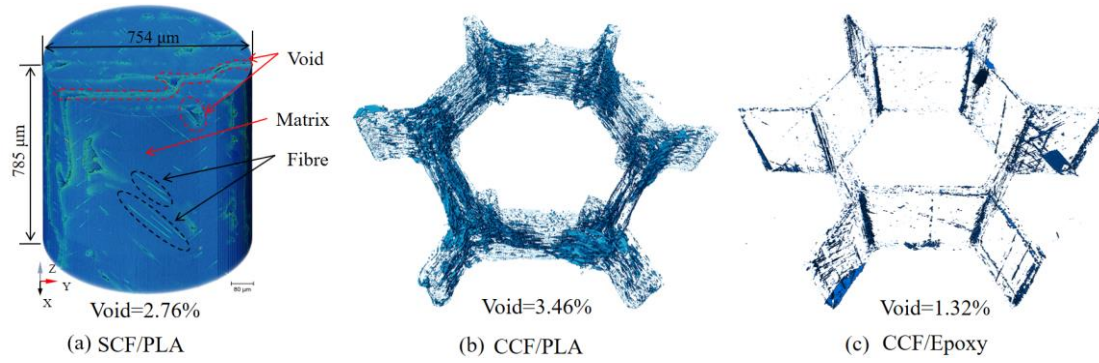


Figure 2. Comparison of void content (%) in honeycomb composite structures manufactured by different processes: (a) 3D-printed SCF/PLA, (b) 3D-printed CCF/PLA, and (c) vacuum bag cured CCF/epoxy, as determined by X-ray computed tomography (CT).

(3) Wall-thickness tolerance was measured at multiple locations using digital calipers, with the standard deviation being added in Table 2.

Table 2. Summary of geometries of the honeycomb structure investigated in this study.

Sample	Mass (g)	Length (mm)	Wide (mm)	Height (mm)	Thicknes s (mm)	Loading rate (mm·min ⁻¹)
SCF/PLA-1Cell-a	20.10	41.46	36.41	71.70	2.03±0.11	5.00
SCF/PLA-1Cell-b	20.40	41.51	36.62	71.75	2.04±0.08	5.00
SCF/PLA-1Cell-c	20.20	41.49	36.54	71.72	2.04±0.10	5.00
SCF/PLA-1Cell-d	20.15	41.48	36.63	71.74	2.02±0.08	5.00
SCF/PLA-1Cell-e	20.22	41.52	36.53	71.73	2.03±0.10	5.00
SCF/PLA-8Cell-a	77.30	73.52	71.38	73.50	2.09±0.09	5.00
SCF/PLA-8Cell-b	77.50	73.90	72.34	73.51	2.05±0.11	5.00
SCF/PLA-8Cell-c	77.40	73.87	72.21	73.45	2.07±0.12	5.00
CCF/PLA-1Cell-a	31.60	44.34	38.96	72.07	3.35±0.11	5.00
CCF/PLA-1Cell-b	31.70	44.35	38.89	72.10	3.36±0.13	5.00
CCF/PLA-1Cell-c	31.50	44.34	38.88	72.09	3.36±0.12	5.00
CCF/PLA-8Cell-a	115.00	77.19	73.49	71.72	3.02±0.11	5.00
CCF/PLA-8Cell-b	115.40	77.15	73.70	71.83	3.05±0.10	5.00
CCF/PLA-8Cell-c	115.20	77.16	73.74	71.80	3.03±0.09	5.00
CCF/Epoxy-1Cell-a	20.00	44.24	41.65	75.41	1.36±0.05	5.00
CCF/Epoxy-1Cell-b	20.00	43.04	42.96	73.85	1.55±0.03	5.00
CCF/Epoxy-1Cell-c	20.00	44.04	43.85	76.21	1.32±0.04	5.00
CCF/Epoxy-8Cell-a	77.50	72.29	68.75	75.16	1.75±0.04	5.00

CCF/Epoxy-8Cell-b	73.50	73.26	69.49	74.62	1.45±0.03	5.00
CCF/Epoxy-8Cell-c	77.20	73.16	68.48	76.01	1.55±0.02	5.00

(4) The printing path images (toolpath patterns) for 3D printed continuous fibre reinforced composite materials (CCF/PLA) have been included to clarify fibre deposition and manufacturing consistency.

Page 6-7 line 95-97:

“...The printing path of the 3D-printed CCF/PLA composite is presented in Figure 1(a). The as-printed material exhibits a good impregnation quality, with uniform resin distribution around the fibres and no significant fibre exposure....”

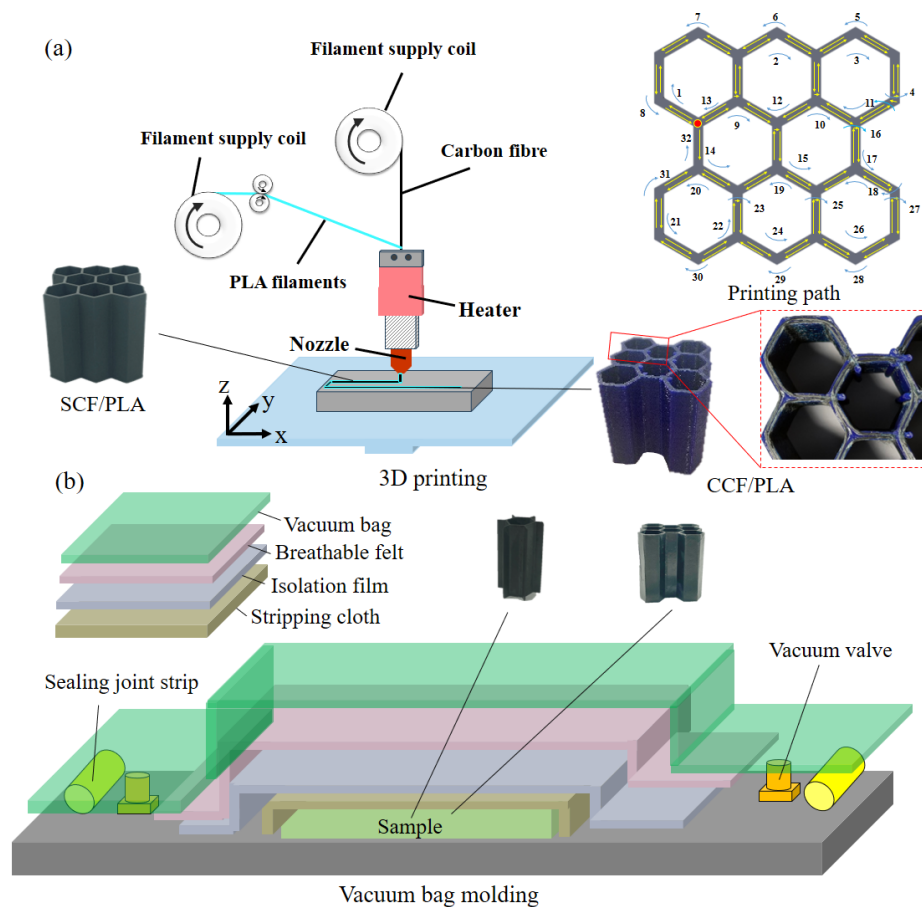


Figure 2. (a) Schematic of the 3D printing process of short/continuous carbon fibre thermoplastic composite, (b) preparation process of continuous carbon fibre thermosetting composite honeycomb structure by vacuum bag moulding method.

Comment 3

3. Why 5 mm min⁻¹ rate is relevant to crash?

Authors' Response to comment 3

Thank you for your comments. It is acknowledged that a loading rate of 5 mm·min⁻¹ corresponds to a quasi-static condition and does not represent a crash or impact event. The loading rate of 5 mm·min⁻¹ was not intended to

simulate crash conditions. Instead, it was employed to characterise typical deformation and energy absorption behaviour of the battery-structure system under controlled quasi-static loading.

Comment 4

4. State actual fibre mass % in PLA filaments; give tow spacing and consolidation pressure for VBM.

Authors' Response to comment 4

Thank you for your comments. The relevant information has been incorporated into the revised manuscript.

Page 6, line 82-83:

“...The actual fibre mass (%) in PLA filaments is 8-10%...”

Page 7, line 107-109:

“...a vacuum pressure of approximately 0.1 MPa was applied, and the initial tow spacing (0.2-1.0 mm) was reduced to approximately 0.05-0.5 mm after consolidation. Also, the curing cycle was performed with a temperature ramping rate of 2 °C/min to 80 °C...”

Comment 5

5. Please validate the SEA calculation. For example, force-displacement curves and exact integration limits are suggested to add.

Authors' Response to comment 5

Thank you for the valuable comments. The representative load-displacement curves have been added to the revised manuscript, with the integration limits being clearly indicated in Figure 6. The calculations regarding energy absorption and specific energy absorption have also been updated. The relevant part in the manuscript has been revised as follows.

Page 11-12, line 167-169:

“Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated up to the onset of densification to ensure a consistent comparison.”

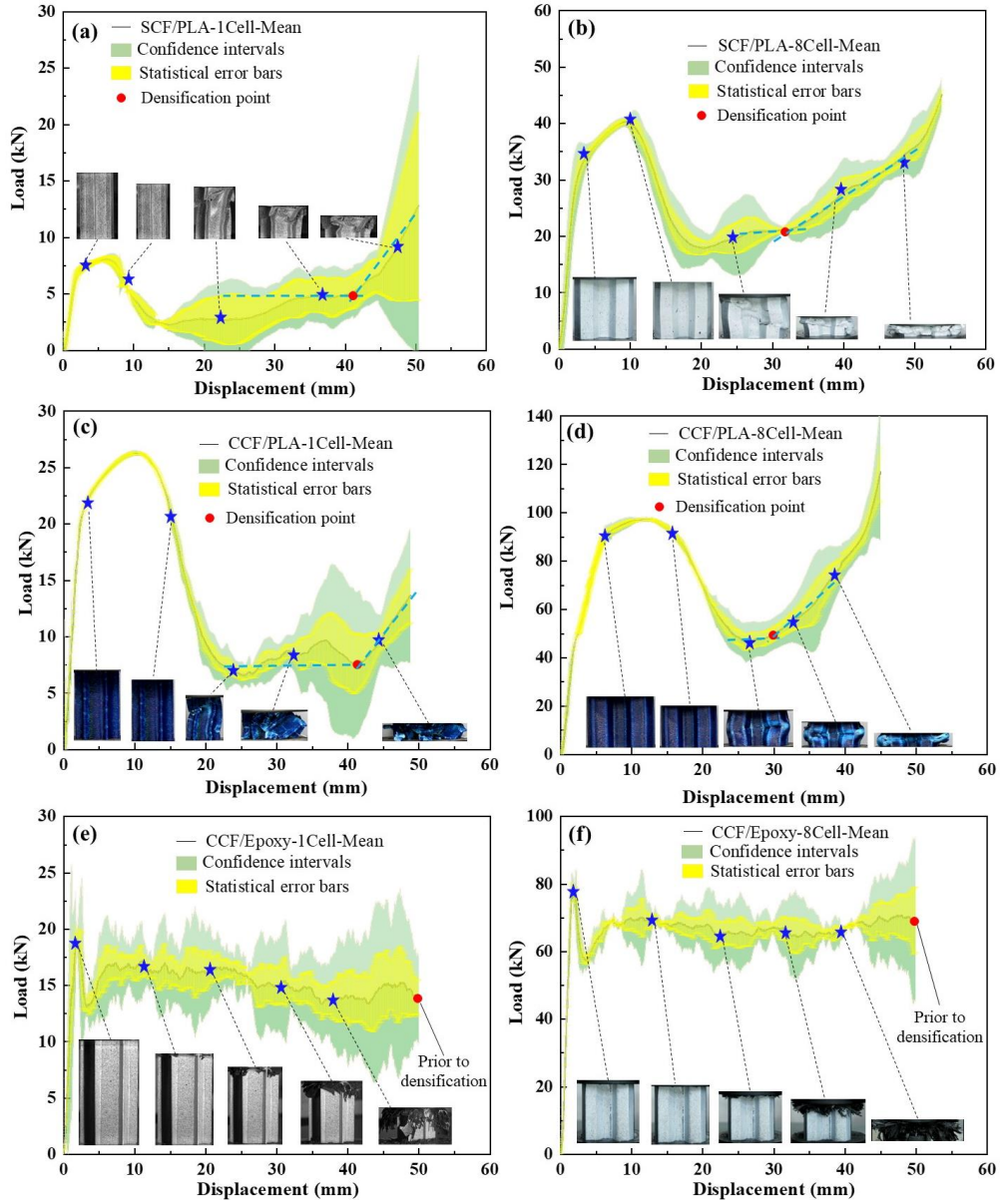


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static tests on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

Table 8. Energy absorption characteristics of honeycomb structure of 3D-printed short carbon fibre composites, 3D-printed continuous carbon fibre reinforced composite and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composite honeycomb structure.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
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SCF/PLA-1Cell-a	20.10	8.02		11.73		13.62	
SCF/PLA-1Cell-b	20.40	7.97		12.27		10.51	
SCF/PLA-1Cell-c	20.20	8.08	8.16	11.66	11.80	13.01	12.88
SCF/PLA-1Cell-d	20.15	8.10		11.70		13.58	
SCF/PLA-1Cell-e	20.22	7.98		11.50		13.67	
Variance	/	0.032		0.069		1.45	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	
CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

Comment 6

6. It is strongly suggested to discuss the micro-CT or SEM images in order to prove the failure modes

Authors' Response to comment 6

Thank you for the valuable suggestions. Additional discussion has been added in Section 4.2 to explicitly correlate the observed CT features with the proposed failure modes. These microstructural observations provide direct evidence supporting the identified deformation and failure mechanisms.

Page 22-23, line 319-332:

“Due to severe fragmentation after crush, the CCF/Epoxy specimens could not be examined by CT scanning. Therefore, post-crush CT analyses were performed on the 3D-printed SCF/PLA and CCF/PLA honeycomb structures. Figure 8(a) presents the overall 3D CT reconstruction of the crushed SCF/PLA honeycomb, which is consistent with the macroscopic observations and reveals interlayer delamination and matrix cracking. Figure 8(a1)

shows sectional views from different orientations, which more distinctly illustrate the internal damage features, including interfacial separation, local buckling and fracture.

For the CCF/PLA composite honeycomb, Figure 8(b) provides the 3D CT reconstruction after compression crush, exhibiting failure characteristics as observed during the physical test. The sectional views in Figure 8(b1) further reveal the layer-by-layer deposition features inherent to the 3D printing process, as well as the fibre deposition paths. In addition, internal damage modes such as delamination and fracture can be clearly identified within the structure. These CT results provide direct internal damage evidence, with supporting the observed macroscopic failure mechanisms.”

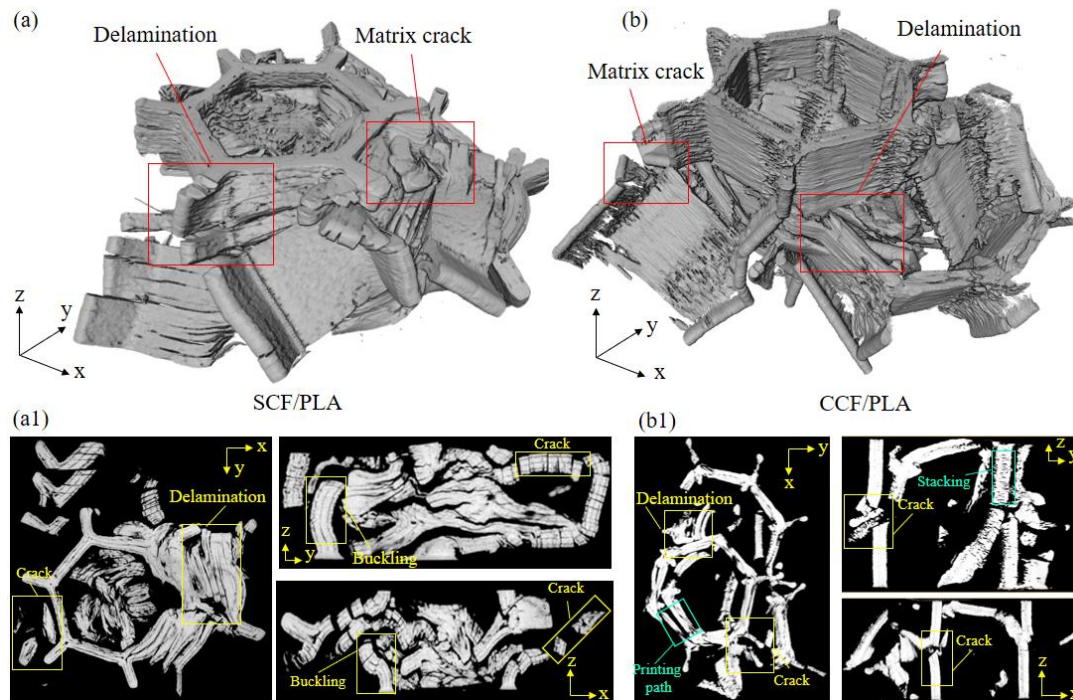


Figure 8. CT results of SCF/PLA (a, a1) and CCF/PLA (b, b1) after quasi-static compression.

Comment 7

7. Please shorten your introduction. Somer related studied and reviews are suggested to add. Besides, please compare fairly with recent CFRP honeycomb papers and state novelty clearly.

Authors' Response to comment 7

Thank you for your valuable comments. The introduction has been revised as follows in accordance with the suggestions.

Page 4-5, line 37-72:

“Batteries serve as the heart of electric aircraft and are essential for hybrid/electric propulsion. However,

current battery technology limits electric aircraft to small airframes with very limited range, making battery pack weight a critical constraint^[1]. Recent advances in electric aircraft largely rely on lightweight structural materials and improved power systems^[2-4]. Reducing battery pack weight lowers energy consumption and allows more batteries to be installed without increasing drone mass, which is vital for enhancing electric aircraft performance^[5, 6]. Despite its importance, research on aerospace battery systems remains limited, whereas significant progress in EV battery pack design has provided valuable insights into structural concepts and material selection^[7-10]. Commercial battery packs are primarily made of aluminum, steel, or carbon fibre reinforced plastic (CFRP)^[11-13]. Compared with aluminum, a CFRP cell shell is about 40 % lighter^[14-16] and the thermal conductivity is about 200 times lower, which can better protect the battery from cold and heat.

Numerous studies have focused on lightweight battery packs^[17-20]. For automotive applications, a high-strength, crash-resistant enclosure must address safety issues such as internal fire or explosion^[21, 22]. Over the past two decades, modelling of the battery pack has advanced toward more efficient management systems^[23, 24]. Pana et al.^[25] experimentally and numerically investigated a battery pack with a high-strength steel base and a CFRP outer shell, reporting a 6.6% weight reduction compared with an all-steel enclosure. Liang^[26] and Greve et al.^[27] used finite element simulations to evaluate the impact performance of battery and pack. Cellular and honeycomb structures have gained increasing attention to battery protection, due to their high specific strength, excellent energy absorption, and design flexibility^[28-31]. Unlich et al.^[32] proposed that the overall performance of hexagonal battery packs is superior to other test geometries. Sinz et al.^[33] designed a replaceable battery pack for electric buses and recommended adding beams beneath it to reduce intrusion and acceleration peaks for better protection. Other studies have explored composite materials in battery systems. Pan et al.^[34] studied the impact resistance of battery packs through computer simulations. Wang et al.^[35] investigated a novel battery pack system consisting of a non-modular battery pack (CTP) and two tubular structures with negative Poisson's ratio (NPR).

Han et al. ^[36] tested carbon/glass hybrid composite honeycomb enclosures under flexural and impact loads, with detailed damage analysis. Dong et al. ^[37] focused on structural battery composites, examining their out-of-plane compressive behaviour and electrochemical stability under a compressive stress of 10 MPa. Samir et al. ^[38] fabricated non-prismatic reinforced honeycomb (NPRH) sandwich structures via fused filament fabrication and conducted multi-directional quasi-static compression tests, reporting high specific energy absorption.

Limited research exists on the structural design of de-modular battery packs for improved protection. In electric aircraft, conventional wing fuel tanks could be replaced by battery packs integrated into honeycomb sandwich wing panels using unitized packaging technology (UPT). This enables both lightweight design and enhanced safety. To fill this gap, the current study carried out fabrication of thermoplastic short-carbon-fibre and continuous-carbon-fibre honeycomb structures via 3D printing and vacuum bag molding. The corresponding quasi-static crush energy absorption was evaluated through experimental and numerical investigations. Furthermore, the load-displacement responses were analysed to assess mechanical interaction and deformation compatibility between the battery and the protective structure.”

Reviewer #4: The manuscript presents an experimental and numerical investigation into the quasi-static energy-absorbing behavior of fibre-reinforced composite honeycomb structures intended for battery pack protection. Three manufacturing routes—3D-printed short carbon fibre PLA (SCF/PLA), 3D-printed continuous carbon fibre PLA (CCF/PLA), and vacuum-bag-moulded continuous carbon fibre epoxy (CCF/Epoxy)—are compared using compression tests and finite element simulations. The study further explores the mechanical interaction between honeycomb structures and cylindrical lithium-ion batteries, proposing an optimal "mechanical matching" between structural height and battery height. While the work is extensive and combines experiments with numerical modelling, several issues need to be addressed before further consideration. In particular:

Comment 1

1) The SCF/PLA, CCF/PLA, and CCF/Epoxy honeycombs differ substantially in wall thickness, mass, and density. As a result, comparisons of peak load and specific energy absorption are not fully normalized. It is unclear whether observed performance differences are due to material behavior, geometry, or mass effects. A comparison at equal relative density or equivalent geometry is needed.

Authors' Response to comment 1

Thank you for your valuable comments. In the present study, the comparison was intentionally conducted using compressive strength and specific energy absorption (SEA) to minimize the influence of geometric and mass differences. It is agreed that this normalization procedure was not sufficiently clarified in the original manuscript, which may have led to a misunderstanding. The manuscript has been revised to explicitly state the definitions and normalization rationale of compressive strength and SEA in Section 2.2.

Page 11-12, line 167-169:

“Compressive strength and specific energy absorption (SEA) were used for comparative evaluation, normalised by sample cross-sectional area and sample mass, respectively. Energy absorption was calculated up to the onset of densification in the load-displacement curve to ensure a consistent comparison.”

Comment 2

2) The mechanical properties of the 3D-printed composites are largely adopted from literature or treated as homogenized materials. No direct tensile, compressive, or interlaminar tests are reported for the printed specimens used in this study, despite the known sensitivity of 3D-printed composites to processing parameters.

Authors' Response to comment 2

Thank you for the valuable comments. The material parameters used in the numerical model were determined based on a combination of experimentally measured data and validated values from literature. Mechanical characterization tests, including tensile, bending and interlaminar evaluations, were performed on specimens fabricated using the same printing parameters as those adopted in this study. The measured experimental data are provided in appendix.

Page 36-37, line 636-629:

“Mechanical characterisation tests are carried out on SCF/PLA composite specimens manufactured under identical printing conditions. The composites exhibited a tensile strength of 32.40 MPa and a tensile modulus of 2.94 GPa. Under compressive loading, the measured strength and modulus are 47.63 MPa and 1.51 GPa, respectively. The corresponding flexural strength and flexural modulus are found to be 84.54 MPa and 6.32 GPa, whilst the shear strength and shear modulus reached 20.74 MPa and 0.30 GPa, respectively (Figure A1).

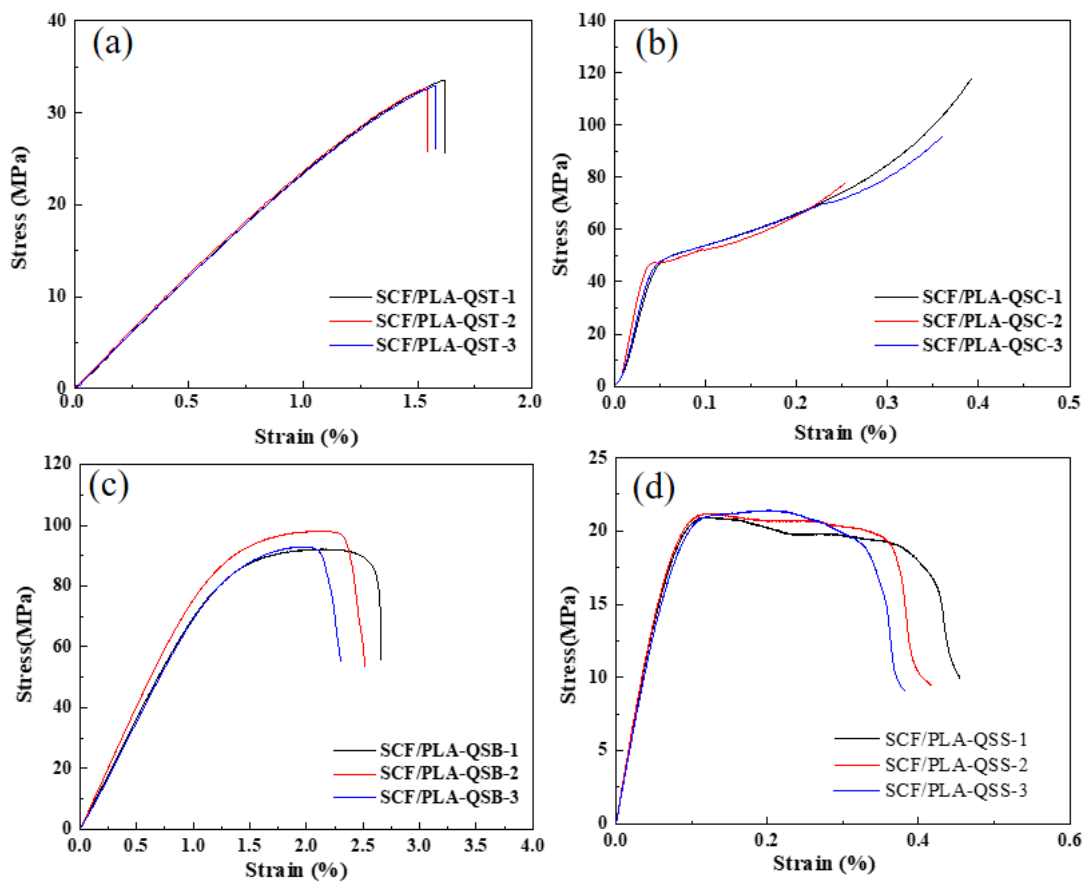


Figure A1. Mechanical test results of SCF/PLA in (a) tension, (b) compression, (c) bending and (d) shearing.

According to experimental measurements conducted on specimens fabricated in the identical printing conditions, the tensile strength and modulus of the CCF/PLA composites are determined as 333.46 MPa and 21.97 GPa, respectively. The measured flexural strength and modulus are 273.37 MPa and 16 GPa, respectively, whilst the shear strength is 32.47 MPa (Figure A1), shown in Appendix.”

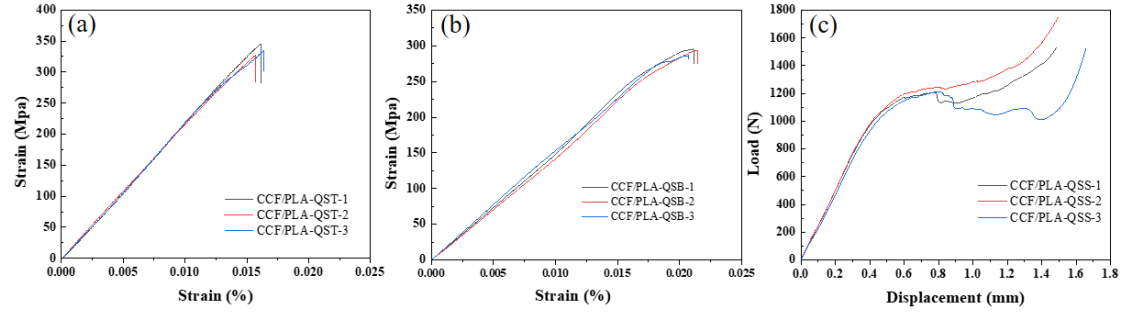


Figure A1. Mechanical test results of CCF/PLA in (a) tension, (b) bending, and (c) shearing.

Comment 3

3) The material properties of Table 3 are poorly defined. Different values of the fracture strain and failure displacement are defined, with the corresponding values being contradictory. Corrections and explanations are required.

Authors' Response to comment 3

Thank you for your valuable comments. The parameters listed in Table 3 correspond to different damage mechanisms implemented in the numerical model and therefore do not represent contradictory values.

Specifically, two damage criteria were employed to describe the failure behaviour of the SCF/PLA composite honeycomb: ductile damage and shear damage. Fracture strain values are given in percentages and indicate the damage initiation under different stress states. The different failure displacement values stem from the two damage modes and are calibrated via inverse modeling based on reference [40] to best fit experimental data. After careful review, an error in the second line of the shear damage definition has been corrected, as shown in Table 3 updated.

Table 3. Properties of short carbon fibre composite honeycomb ^[40].

Parameter		Value	Units
Density		1.35	g/cm ³
Young's Modulus		2294	MPa
Poisson's Ratio		0.3	/
Ductile damage (Damage evolution)	Fracture strain	1.45	%
	Stress triaxiality	-0.33	/
	Strain rate	0.000656	s ⁻¹
	Failure displacement	0.14143	mm
Shear damage (Damage evolution)	Fracture strain	1.35	%
	Shear stress ratio	1.6	/
	Strain rate	0.00416	s ⁻¹
	Failure displacement	1.1343	mm

Comment 4

4) A model height of 72 mm is defined and a stress triaxiality of -0.33 (Table 3). How are the values defined? To which cases do they correspond to? Moreover, the use of the FE models is more descriptive than predictive.

Sensitivity analysis and mechanistic explanations are largely missing.

Authors' Response to comment 4

Thank you for your valuable comments. Firstly, the height of the cylindrical battery is 70 mm, which is 2 mm higher than the set height of the battery. The actual height of the honeycomb structure used in the quasi-static compression experiment, which is 72mm, is matched accordingly. The same geometric configuration is adopted in the finite element model to ensure consistency between the experimental tests and the numerical simulations. In the ductile damage model available in the commercial code Abaqus, stress triaxiality is defined as the ratio between the hydrostatic stress and the equivalent von Mises stress:

$$\eta = \sigma_m / \sigma_{eq}$$

where σ_m is the mean (hydrostatic) stress and σ_{eq} is the equivalent von Mises stress.

In the present study, the honeycomb structure is subjected to quasi-static compression, where the deformation of the cell walls is mainly governed by shear-dominated stress states during progressive buckling and collapse. For such stress conditions, the stress triaxiality is typically close to -0.33, which corresponds to a compressive-shear dominated state in a ductile damage formulation.

Page 13, line 187-189:

“Among them, the stress triaxiality of -0.33 was calculated to represent the compressive shear dominated stress state during honeycomb wall buckling under quasi-static compression.”

The mesh sensitivity has been re-analyzed, with the modifications as follows.

Page 37-38, line 637-649:

“In order to prove the mesh independence of the proposed damage model, three mesh sizes (0.5, 1 and 1.5 mm) were used to investigate the mesh sensitivity in the honeycomb structure. As illustrated in Figure A2(b) (Appendix), the simulations with mesh sizes of 0.5 mm and 1 mm show good agreement with the experimental peak load, indicating that the numerical predictions are relatively insensitive to further mesh refinement. In contrast, the model with a mesh size of 1.5 mm significantly underestimates the peak load, as the coarse mesh cannot adequately capture the local deformation and damage evolution of the honeycomb cell walls.

In addition, the computational efficiency was evaluated. The CPU time for mesh sizes of 0.5, 1 and 1.5 mm was approximately 280, 165 and 27 min, respectively. Considering both prediction accuracy and computational

efficiency, a mesh size of 1 mm was selected for the subsequent simulations. This sensitivity analysis demonstrates that the selected mesh size provides a reasonable balance between the computational cost and accuracy.”

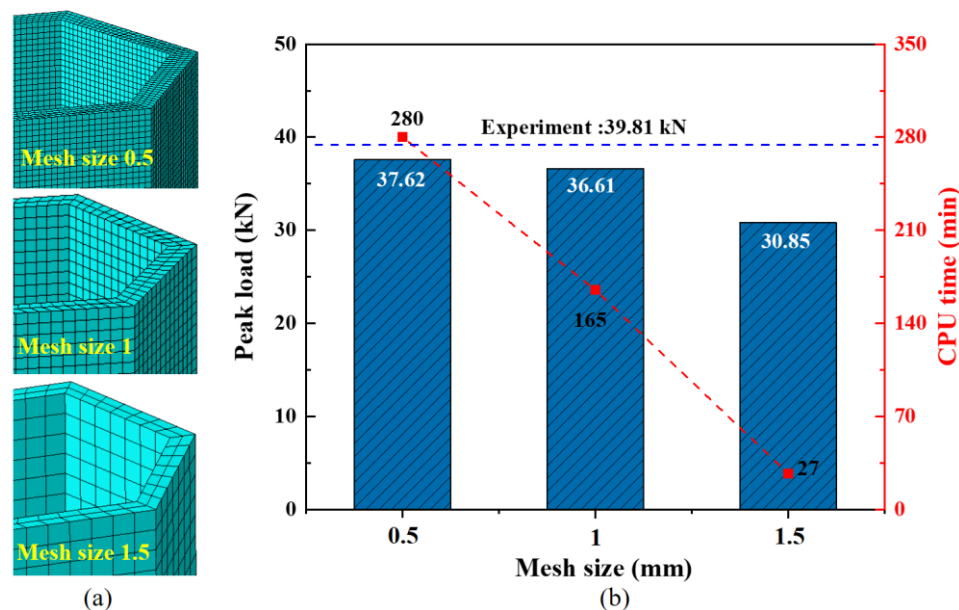


Figure A2. Mesh geometries in mesh sensitivity study: (a) specimens for mesh geometries and (b) corresponding peak load histories obtained from quasi-static compress test.

Comment 5

5) Nearly all results are lacking statistical analysis. Indicatively, what is the variance of peak load, energy absorption and specific energy absorption in Table 8? What is their statistical robustness and reproducibility?

Authors’ Response to comment 5

Thank you for the valuable comments. For each configuration, repeated quasi-static compression tests were conducted on 3 or more specimens fabricated under identical processing conditions to evaluate repeatability and statistical robustness (for example, 5 sets of experiments were conducted for SCF/PLA 1 cell). The variance of the peak load, energy absorption and specific energy absorption has now been calculated and is summarised in the revised Figure 5 and Table 8.

Page 18, line 275-278:

“The compressive response of the three honeycomb structures is illustrated in Figure 6, which depicts both the load-displacement curves and the corresponding failure progression. The densification points are also marked on the curves. To ensure statistical robustness, the curves were analyzed to derive the mean, variance and confidence intervals, as well as the boundary intervals for the energy absorption....”

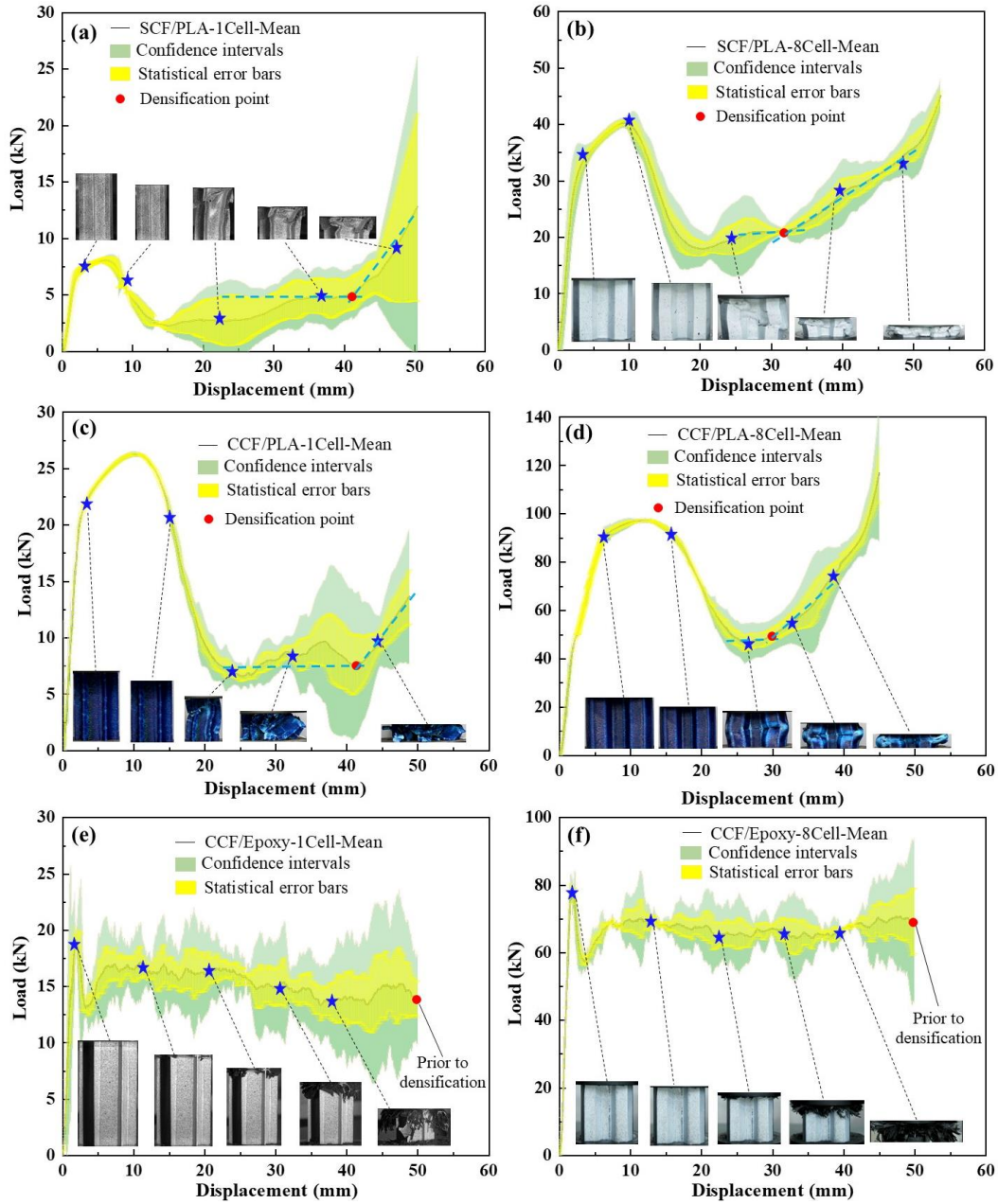


Figure 6. The load-displacement curves and failure modes at different stages following quasi-static crush tests on (a) 1-cell and (b) 8-cell honeycomb structures of 3D-printed short carbon fibre composites (SCF/PLA), (c) 1-cell and (d) 8-cell honeycomb structures of 3D-printed continuous carbon fibre composites (CCF/PLA), (e) 1-cell and (f) 8-cell honeycomb structures of vacuum bag moulding continuous carbon fibre composites (CCF/Epoxy).

Page 21-22, line 315-318:

“All data on the peak load, compressive strength, energy absorption and specific energy absorption of the honeycomb structures are listed in Table 8. For 8-cell honeycomb structures made of different materials, it is

observed that the 3D-printed CCF/PLA composites provide the highest compressive strength (18.61 MPa), surpassing those of CCF/Epoxy (15.95 MPa) and SCF/PLA (7.58 MPa).”

Table 8. Energy absorption characteristics of honeycomb structures made of 3D-printed short carbon fibre, 3D-printed continuous carbon fibre and Vacuum Bag Moulding (VBM) continuous carbon fibre reinforced composites, respectively.

Sample codes	Sample mass (g)	Peak load (kN)	Average value (kN)	Compressive strength (MPa)	Average value (MPa)	Specific energy absorption (J/g)	Average value (J/g)
SCF/PLA-1Cell-a	20.10	8.02		11.73		8.03	
SCF/PLA-1Cell-b	20.40	7.97		12.27		6.50	
SCF/PLA-1Cell-c	20.20	8.08	8.16	11.66	11.80	7.50	7.58
SCF/PLA-1Cell-d	20.15	8.10		11.70		7.86	
SCF/PLA-1Cell-e	20.22	7.98		11.50		8.01	
Variance	/	0.032		0.069		0.33	
SCF/PLA-8Cell-a	77.30	40.23		7.77		10.60	
SCF/PLA-8Cell-b	77.50	39.07	39.81	7.66	7.578	10.19	10.32
SCF/PLA-8Cell-c	77.40	40.12		7.44		10.17	
Variance	/	0.273		0.09		0.04	
CCF/PLA-1Cell-a	31.60	26.39		38.11		18.91	
CCF/PLA-1Cell-b	31.70	24.23	25.29	34.99	36.53	18.07	18.48
CCF/PLA-1Cell-c	31.50	25.25		36.28		18.46	
Variance	/	0.778		1.62		0.12	
CCF/PLA-8Cell-a	115.00	97.40		18.55		20.18	
CCF/PLA-8Cell-b	115.40	97.64	97.68	18.61	18.61	19.93	20.07
CCF/PLA-8Cell-c	115.20	98.01		19.20		20.10	
Variance	/	0.063		0.0018		0.26	
CCF/Epoxy-1Cell-a	20.20	19.59		25.78		43.34	
CCF/Epoxy-1Cell-b	20.10	20.26	20.33	26.67	26.75	43.41	43.27
CCF/Epoxy-1Cell-c	20.30	21.13		27.81		43.07	
Variance	/	0.397		0.692		0.02	
CCF/Epoxy-8Cell-a	77.50	80.46		16.01		37.46	
CCF/Epoxy-8Cell-b	73.50	79.83	80.18	15.88	15.95	38.55	38.25
CCF/Epoxy-8Cell-c	77.20	80.24		15.97		38.75	
Variance	/	0.068		0.0027		0.32	

Comment 6

6) The empirical formula provided in Equations (4)-(7) is not validated against independent testing sets, while it is not written in a reinforcement phase density-independent manner. What is its range of validity in terms of height and phase volume fractions? Why has a multi-segment approach followed, instead of unified formulations?

Physically-motivated formulations need to be considered (TWS 217 113855)

Authors' Response to comment 6

Thank you for the valuable comments. First, it needs to be noted that the empirical expressions were established based on the combined experimental and numerical results obtained in the present study. It aims to describe the relationship between structural height and the compressive response of the integrated honeycomb–battery system. These formulations are therefore intended to capture the observed trends within the investigated parameter space, rather than to provide a universally applicable predictive model. The applicability of the proposed equations has now been clarified in the revised manuscript.

Due to the limited number of experimental configurations available in this study, a completely independent experimental dataset is not available currently. However, during the development of the empirical relationships in Equations (4)-(7), the fitting procedure was conducted using the results from four structural height configurations, whilst the remaining height configurations were instead used as validation cases that were not involved in the fitting process.

This approach provides a hold-out validation, allowing the predictive capability of the proposed empirical relationships to be preliminarily evaluated using the data that were not used during model calibration. The comparison between the predicted and experimental responses for these configurations (Four structure height) shows good agreement, indicating that the proposed relationships can reasonably capture the influence of structural height within the parameter range investigated.

To clarify this point, an explanation has been added in the revised manuscript to describe the fitting dataset and the validation dataset used for evaluating the empirical formulation.

Page 30, line 451-455:

"It should be noted that the present study does not aim to perform a global optimization for the battery-structure system. Instead, the novelty of this study is to identify the mechanical compatibility between the honeycomb structure and the cylindrical battery within the design range investigated. The empirical relationships were calibrated using the results from four height configurations, whilst the remaining configurations were used as validation cases to assess the predictive capability of the proposed expressions."

Second, the proposed empirical relationships are limited to the specific geometries and material systems investigated (honeycomb height: 72-80 mm; SCF/PLA and CCF/PLA composites with fixed reinforcement architecture). This limitation has been stated in the revised manuscript.

Page 29-30, line 449-451:

“...With the fibre volume fraction and reinforcement distribution being kept constant across all specimens, the reinforcement phase density is not explicitly included as a variable....”

Thank you for pointing out the relevant study published in Thin-Walled Structures (217, 113855). The work focuses on constitutive modeling of TPMS-based composites, whereas the present study investigates the mechanical-structural matching design of honeycomb protective structures for battery systems, where the key design parameter is the honeycomb height. Consequently, the physically-based formulations from the cited study are not directly applicable to our system-level design problem.

Third, a multi-segment formulation was adopted to account for the different deformation regimes observed during crushing compression. As shown in the load-displacement responses, the mechanical behaviour evolves through a few stages (elastic deformation of the honeycomb, progressive cell wall buckling, and coupled deformation between the battery and the protective structure). These distinct mechanisms cannot be accurately captured by a single unified expression. Hence, the piecewise formulation provides a better approximation of the structural response in each deformation stage.

Comment 7

7) The novelty of the work needs to be properly defined. The "mechanical synergy" between the battery and its height largely reflects an expected geometric alignment effect and does not constitute on its own a novel design principle.

Authors' Response to comment 7

Thank you for the valuable comments. The novelty of this study lies not in the geometric matching itself, but in the identification of the underlying structural mechanical mechanism responsible for the observed mechanical synergy. It is demonstrated that the interaction between the battery height and the surrounding composite structure leads to a redistribution of stress and stiffness participation, thereby significantly altering the load transfer paths and the global mechanical response. Through combined experimental work and validated finite element analyses, a quantitative relationship of structural height, stiffness contribution and deformation compatibility has been established. This provides a mechanics-based explanatory and predictive framework for structural design, rather than a purely geometric observation. The manuscript has been revised to clarify this point and to better emphasize the structural mechanics contribution of the study.

Page 30, line 451-453:

“It should be noted that the present study does not aim to perform a global optimization for the battery-structure system. Instead, the novelty of this study is to identify the mechanical compatibility between the honeycomb structure and the cylindrical battery within the design range investigated.”

Comment 8

8) The introduction section needs to be extended to properly place the current work in the context of interpenetrating phase composites: Comp Part C 17 100596, Virtual Physic. Prototyp. 20 e2505992, Comp. Part B. 275 111314

Authors' Response to comment 8

Thank you for your valuable suggestions. We have revised the introduction accordingly.

Page 4-5, line 37-64:

“Batteries serve as the heart of electric aircraft and are essential for hybrid/electric propulsion. However, current battery technology limits electric aircraft to small airframes with very limited range, making battery pack weight a critical constraint ^[1]. Recent advances in electric aircraft largely rely on lightweight structural materials and improved power systems ^[2-4]. Reducing battery pack weight lowers energy consumption and allows more batteries to be installed without increasing drone mass, which is vital for enhancing electric aircraft performance. ^[5, 6]. Despite its importance, research on aerospace battery systems remains limited, whereas significant progress in EV battery pack design has provided valuable insights into structural concepts and material selection ^[7-10]. Commercial battery packs are primarily made of aluminum, steel, or carbon fibre reinforced plastic (CFRP) ^[11-13]. Compared with aluminum, a CFRP cell shell is about 40 % lighter ^[14-16] and the thermal conductivity is about 200 times lower, which can better protect the battery from cold and heat.

Numerous studies have focused on lightweight battery packs ^[17-20]. For automotive applications, a high-strength, crash-resistant enclosure must address safety issues such as internal fire or explosion ^[21, 22]. Over the past two decades, modelling of the battery pack has advanced toward more efficient management systems ^[23, 24]. Pana et al. ^[25] experimentally and numerically investigated a battery pack with a high-strength steel base and a CFRP outer shell, reporting a 6.6% weight reduction compared with an all-steel enclosure. Liang ^[26] and Greve et al. ^[27] used finite element simulations to evaluate the impact performance of battery and pack. Cellular and honeycomb structures have gained increasing attention for battery protection due to their high specific strength, excellent energy absorption, and design flexibility ^[28-31]. Unlich et al. ^[32] proposed that the overall performance of

hexagonal battery packs is superior to other test geometries. Sinz et al. [33] designed a replaceable battery pack for electric buses and recommended adding beams beneath it to reduce intrusion and acceleration peaks for better protection. Other studies have explored composite materials in battery systems. Pan et al. [34] studied the impact resistance of battery packs through computer simulations. Wang et al. [35] investigated a novel battery pack system consisting of a non-modular battery pack (CTP) and two tubular structures with negative Poisson's ratio (NPR). Han et al. [36] tested carbon/glass hybrid composite honeycomb enclosures under flexural and impact loads, with detailed damage analysis. Dong et al. [37] focused on structural battery composites, examining their out-of-plane compressive behaviour and electrochemical stability under a compressive stress of 10 MPa. Samir et al. [38] fabricated non-prismatic reinforced honeycomb (NPRH) sandwich structures via fused filament fabrication and conducted multi-directional quasi-static compression tests, reporting high specific energy absorption.”

[6] Asar A, Zaki W. A comprehensive review of the mechanisms and structure of interpenetrating phase composites with emphasis on metal-metal and polymer-metal variants, *Compos. Part B-Eng.* 275, 2024, 111314, 1359-8368.

[13] Shingare K B, Alkunte S, Li BS, Schiffer A, Kinloch I, Liao K. Multifunctional TPMS-based interpenetrating phase composites: A comprehensive review of structure, properties, piezoelectricity and applications. *Compos. Part C: Open Access*, 17, 2025, 100596.

Comment 9

9) The language needs to be proof-read. There are frequent grammatical errors and inconsistent terminology (e.g., "continue" vs. "continuous" carbon fibre).

Authors' Response to comment 9

Thank you for pointing out this. The manuscript has now been carefully proofread and professionally edited to correct grammatical errors and improve overall readability.

Comment 10

10) The manuscript is overly long containing repetitive descriptions of similar results, condensation is required. That should be combined with figure restructuring and proper labelling.

Authors' Response to comment 10

Thank you for this valuable comment. The manuscript has now been carefully revised to improve conciseness and readability.

Page 25-28, line 380-409:

“As shown in Figure 11, the finite element predictions are compared with the experimental results to evaluate the predictive capability of the proposed damage model. Overall, the simulations reproduce the load-displacement responses and the main failure characteristics observed in the experiments. For the SCF/PLA honeycomb structures (Figures 11a and b), the numerical results show good agreement with the experimental curves, with the error in peak load within 6 %. The model also captures the characteristic deformation stages, including the elastic response (initial stiffness), peak load, progressive collapse and final densification. Also, micro-buckling is observed in both the experimental and simulation results, as shown in Figure 11(a). These phenomena indicate that the progressive buckling and damage accumulation of the cell walls are reasonably simulated.

For the CCF/PLA honeycomb structures (Figures 11c and d), the predicted responses also agree well with the experimental results. The discrepancies in peak load are between 5 and 9 %, whilst the differences on plateau load remain below 10 %. The micro-buckling phenomenon of the structure is captured in Figure 11(c), as well.

However, in the present model the honeycomb walls are represented using homogenised material properties. Fibre-matrix micromechanical interactions, discrete fibre fracture and layer-wise additive manufacturing defects are neglected. Consequently, the simulations exhibit fewer fibre fragments and more continuous damage patterns than those observed experimentally. Nevertheless, the model accurately reproduces the global structural response, including the initial stiffness, peak load, post-peak softening and overall crushing behaviour of the CCF/PLA honeycombs (Figures 11c and d). This indicates that the model is suitable for capturing the macroscopic deformation and progressive collapse behaviour of the integrated structure ^[57, 58].

For the CCF/epoxy honeycomb structures (Figures 11e and f), the simulations also reproduce the key deformation stages, including the elastic response, sudden load drop, and subsequent plateau region. The

discrepancies between the predicted and experimental mechanical responses remain approximately 10-11 %.

Figures 11(e) and (f) show the ply splitting and fibre fragmentation phenomenon of the structure, which are reasonably captured using the Abq_Ply_Fabric model. Overall, the FE models provide reliable predictions of the global mechanical response and dominant failure mechanisms of the battery protective system under crush compression.”

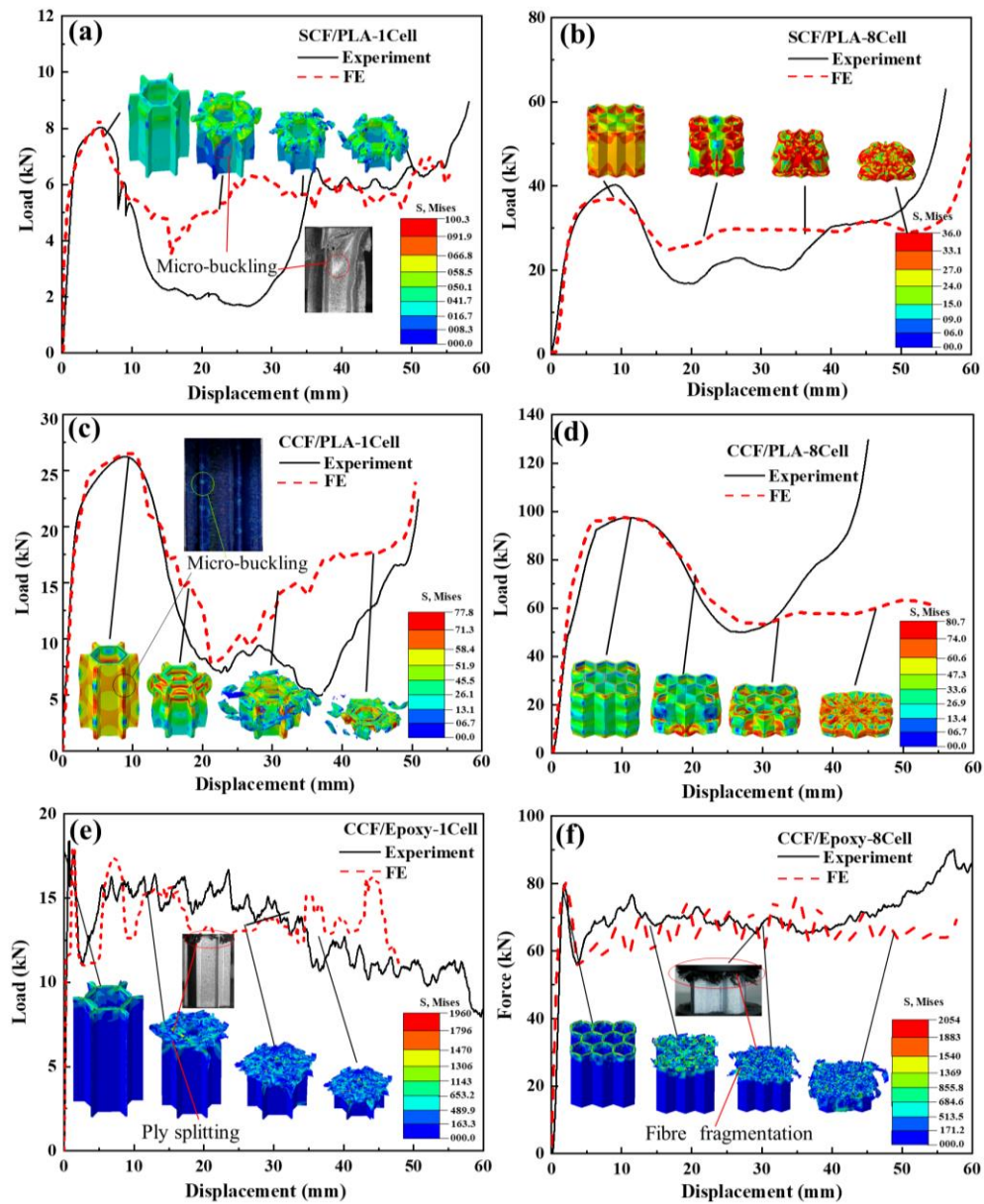


Figure 11. The load-displacement curves and predicted failure modes following crush tests on 1-cell (a) and 8-cell (b) honeycomb of 3D-printed SCF/PLA composite, 1-cell (c) and 8-cell (d) honeycomb of 3D-printed CCF/PLA composite and 1-cell (e) and 8-cell (f) honeycomb of vacuum bag moulding CCF/Epoxy composite. (the solid lines correspond to test data and the dashed lines to the FE predictions).

Second, the structure of several figures has been re-organised. Related results are now grouped into combined figures to facilitate direct comparison between different material systems and structural configurations. Correspondingly, the figure captions have been revised and clarified to improve readability and ensure consistent labelling.

Page 28-29, line 426-438:

“Finite element simulations were conducted to evaluate the mechanical interaction between the honeycomb structure and the cylindrical battery at different structural heights (72-80 mm), as shown in Figure 13. For the 3D-printed SCF/PLA honeycombs (Figure 13a), when the structural height of 72 mm is close to the battery height of 70 mm, the structure and battery deform cooperatively, resulting in the highest peak load of 23.08 kN. As the structural height increases, the protective structure collapses earlier than the battery, leading to a gradual reduction in peak load and energy absorption. Figure 13(c) further shows that the optimal structural matching height lies between 72-74 mm, in terms of peak load. Although the peak load at 74 mm is similar to that at 72 mm, the energy absorption at 72 mm (Figure 13d) is about 3.2% higher, indicating a more effective structural matching condition.

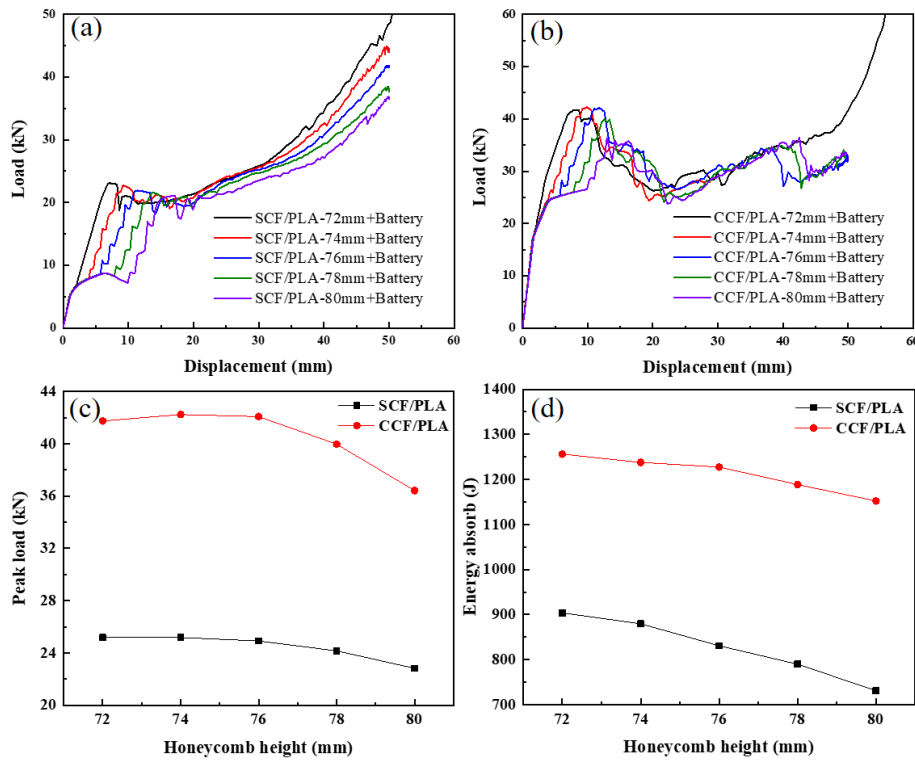


Figure 13. Mechanical responses of integrated honeycomb-battery systems with varying honeycomb heights: (a-b) load-displacement curves of 3D-printed SCF/PLA and CCF/PLA honeycombs, and (c-d) corresponding peak

load and energy absorption.

A similar trend is observed for the CCF/PLA honeycombs (Figure 13b). At larger structural heights (e.g., 78-80 mm), premature structural failure reduces the load-sharing capability between the honeycomb and the battery, resulting in decreased peak load and energy absorption. Overall, the results indicate that a structural height of 72 mm, slightly higher than the battery height, provides the optimal mechanical compatibility and maximizes the load-bearing capacity and energy absorption of the integrated system.”

Third, redundant explanations of similar trends in load-displacement responses, energy absorption behaviour and failure modes have been avoided, whilst the main quantitative results and mechanistic interpretations are retained.

Page 17-18, line 259-269:

“Based on the load-, voltage- and temperature-displacement curves in Figure 5, the battery response can be divided into three stages (Figure 5a). At Stage I (Mechanical deformation), the battery undergoes elastic and plastic deformations until structural buckling occurs, whilst voltage and temperature remain nearly constant. The increase on load contributes to the accumulation of progressive strain energy. At Stage II (Separator deformation), after buckling, the separator with a high failure strain begins to carry the load. The applied load gradually decreases as the separator deforms under continuous platen displacement, whilst the voltage remains around 3.4 V. This stage highlights stress redistribution and progressive damage within the cell. At Stage III (Short circuit and thermal response), once the separator reaches its mechanical limit, a cathode-anode short circuit occurs, leading to a rapid voltage drop and temperature rise to 129 °C. The accelerated changes are attributed to localised shorting and Joule heating. Further crush results in the ultimate mechanical failure and complete voltage collapse. This stage illustrates the coupling among the structural degradation, electrochemical response and thermal effect.”

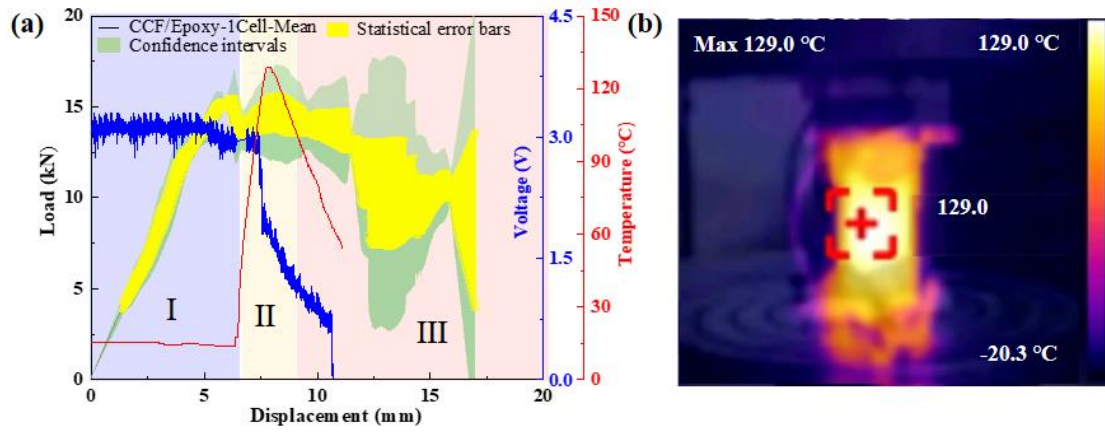


Figure 5. (a) The curves of load, voltage, and battery temperature versus time under quasi-static compression, and (b) the temperature results of the failure center.